The University of Manchester

*Department of Earth & Environmental Sciences*

# *Easter Field Course*

# *EART25102*

## *Second year, Second semester*

## *Fort William part: 7th – 14th April 2026*


***Course Leaders:***

***Steve Covey-Crump, Russell Garwood,***

***Margaret Hartley, Carson Richardson***

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# Course introduction

The EART25102 field course to Scotland comprises two parts:

(a) further training in mapping geological structures – in this case multiply-deformed, moderately-metamorphosed units at Kinlochleven and Cuil Bay on the north-western coast of Scotland;

(b) development of field data collection techniques used for sedimentary and igneous units – in this case, Caledonian igneous units at Glen Coe and Kentallen, and Carboniferous sedimentary and Permian igneous units exposed on the coast of Fife, eastern Scotland.

This guide covers the Kinlochleven, Cuil Bay, and Glen Coe/Kentallen part of the field course. The *Geological Background* section is simply for those who wish to place what they are mapping in geological context but is entirely optional reading; the rest of the field guide may be useful in completing the work.

Our primary aims in this part of the field course (6-field days) are focused on (not in this order):

(1) producing a geological map of an area 2-3 $km^2$ in size located on the coast at Cuil Bay – the mapping area is primarily on one limb of a significant fold, has an intrusive igneous body, and has minor structures associated with a later deformation event (3 days);

(2) illustrating the geometry of folds that crop out along the coast near Kinlochleven by means of a profile plane cross-section with accompanying stereogram constructed from map data (2 days);

(3) producing a compass-and-pace map (on a blank piece of paper) of a small area (~ 100 $m^2$) in Glen Coe that contains numerous dykes intruding andesitic and rhyolitic country rock (½ day)

(4) recording notebook observations of the igneous and metamorphic units seen at Kentallen – these will include both hand specimen descriptions and descriptions of field relations (½ day).

This is an area that is structurally rather complex, but almost all this complexity lies in the interpretation of the structures – completing the exercises is relatively straightforward provided the task of collecting and recording observations is undertaken systematically.

Some of the days will be staff-led – we will show you key localities and provide training in the mapping skills. At other times, however, you will be mapping in pairs or threes – when you are doing this the staff will be doing their own mapping within the mapping area but will be on-hand to provide some guidance if needed. Throughout we will be emphasizing observational / recording skills not skills of interpretation.

# Assessment

**1.** Field slips of the Cuil Bay mapping area (40% of field course). **One map per mapping pair**

- Quantity and quality of all types of orientation data (bedding, cleavage, lineations, younging)
- Accurately located exposures
- Well-located and formation boundaries and labelled fold traces
- General presentation: quality of inking and shading, presence of a comprehensive key

**2.** Profile plane cross-section with supporting stereogram and A3 field slip from the coastal part of the Kinlochleven mapping area (20% of field course). **One section per mapping pair**

- Quantity and quality of bedding and cleavage (or axial plane) and younging data
- Well-located and oriented fold traces and well-drawn form-lines
- Consistency of section with stereogram and A3 field-slip
- General presentation: quality of inking and shading, presence of a comprehensive key

## Itinerary (may be reordered in response to weather conditions)

**Tue 7 Apr** Departure from Manchester at 9:00 am. ***We will stop for a short (~ 30 min) shop at a supermarket in Fort William*** ~ 5:15 pm. Arrival at Linnhe Lochside Holidays at ~ 6 pm

**Wed 8 Apr** Kinlochleven shore-side: staff-led day on taking planar and linear measurements, and recording those measurements and the location of exposures on a map
***Shop at supermarket, Fort William***
*Evening*: tidy up your data on the A3 field-slip, and plot a stereogram of the data collected

**Thur 9 Apr** Kinlochleven shore-side: mapping in twos or threes
*Evening*: tidy your data on the A3 field-slip and plot a stereogram of the data collected today. Use the stereogram to select a profile plane for your cross-section

**Fri 10 Apr** **am:** compass-and-pace mapping exercise in the Pass of Glen Coe
**pm:** excursion to the contact aureole of the Ballachulish pluton at Kentallen Pier, and to the contact of the Kentallen intrusion a few hundred metres to the south
*Evening*: tidy notebook observations collected today; ink-in and colour your compass-and-pace map.
***Note that this day's work is primarily practice for assessed exercises that you will do in Week 2***

**Sat 11 Apr** Cuil Bay: staff-led day introducing the mapping units at Cuil Bay, and key points of interest, including a short exercise examining second phase folds that refold an earlier cleavage
*Evening*: ink-in data collected today; continue Kinlochleven cross-section construction
***Shop at supermarket, Fort William***

**Sun 12 Apr** Cuil Bay: mapping in pairs or threes at Cuil Bay
*Evening*: ink-in data collected today; complete Kinlochleven cross-section (if it is not already completed)

**Mon 13 Apr** Cuil Bay: mapping in pairs or threes at Cuil Bay
*Evening*: ink-in data collected today, and complete your field maps

**Tue 14 Apr** **9:00- 9:30 am:** inspection of accommodation
**9:45 am:** departure from the accommodation, and journey to Fife – we will arrive in Fife in early afternoon but may need to wait until 3 pm to check-in to the hotel
*Evening*: if not done already, complete field maps and cross-sections/stereograms for submission in the morning

**Wed 15 Apr** **8:45 am:** submit map and cross-section

# Health and Safety / Behaviour

As staff members we draw your attention to the advice on health and safety and on behaviour at the back of this book. You must read this advice and should see a member of staff immediately if there is any part of it that you do not understand.

**Key points: in the field**

The weather in Scotland can be very pleasant but may change rapidly to being cold, wet, and windy. In such circumstances there is a very real risk of exposure. *No-one should go into the field unless they are fully equipped with adequate footwear, waterproof clothing, gloves, and spare warm clothes*. It is easy to slip and sprain an ankle or incur other fall-type injuries – *take care not to become separated from your mapping partners so that there is always someone on hand to help*, and make sure that you have exchanged mobile phone numbers with your colleagues (reception will probably be OK but be aware that it is not always guaranteed). *Make sure that you are at pre-arranged meeting places at the pre-arranged time.* Almost certainly there will be locals keeping an eye on our activities – don't drop litter and generally treat the land with respect.

*****

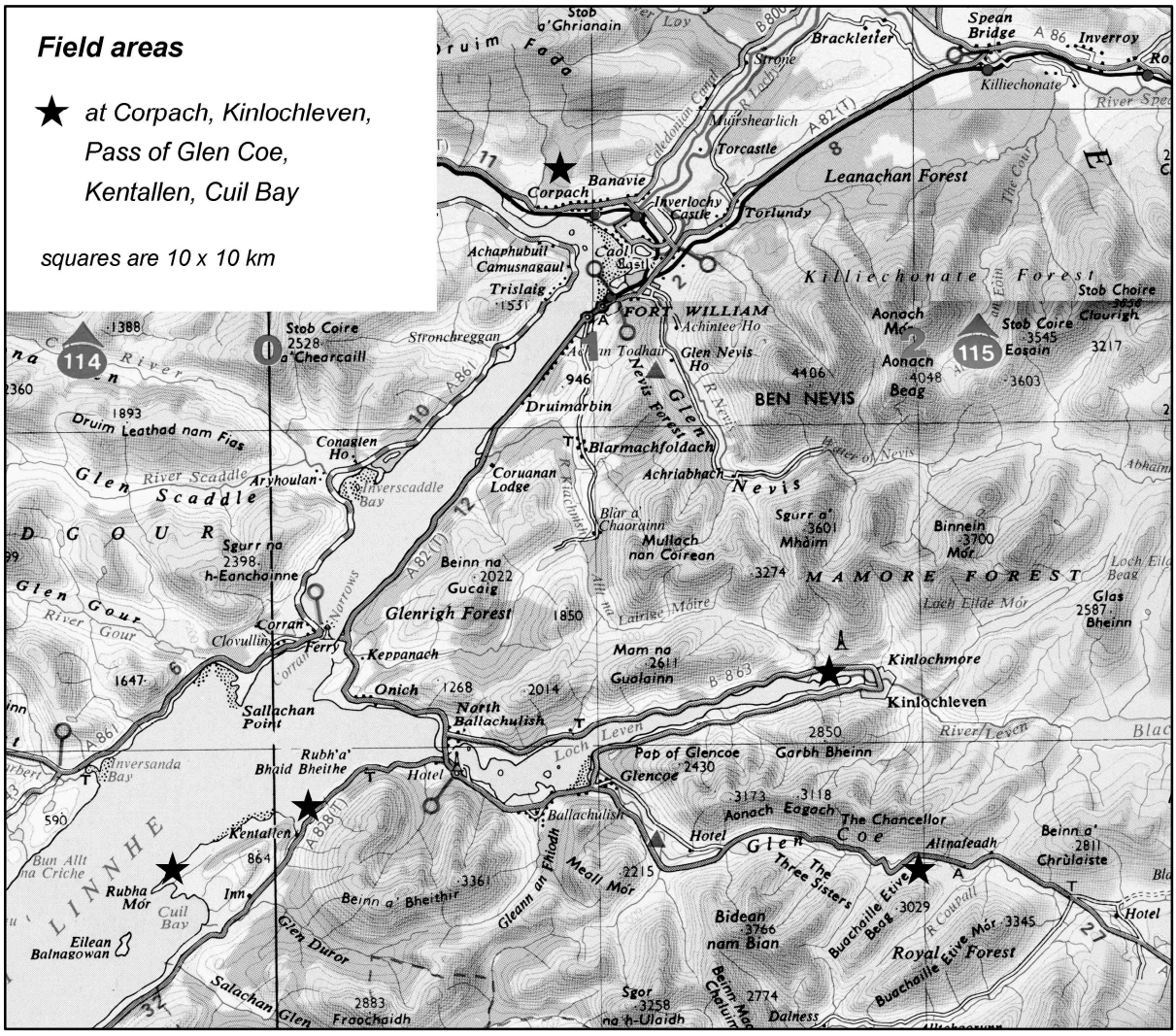
The local mountain rescue team is: ***Glencoe Mountain Rescue Team*** (www.glencoemountainrescure.org.uk). They operate out of Glencoe village. In the event of injury, attempt to contact the teaching staff first but if for some reason this is unsuccessful, the procedure recommended by the mountain rescue team is to ***call 999 or 101 and ask for 'Police' then 'Mountain Rescue'***. 999 is the number to use for urgent, life-threatening situations; use 101 for non-life-threatening injuries but ones which nevertheless require fast medical help. Asking for 'police' first ensures that there are no delays in assessing which emergency service is required.

*****

**Key points: in the accommodation (Linnhe Lochside)**

There will be other members of the public (including small children) staying at Linnhe Lochside Holidays at the same time as you. The owners of the accommodation will therefore be sensitive to noise. They will also be sensitive to general untidiness inside the caravans. The usual standards of behaviour apply. *At all times the word of the owners is to be followed without argument*; if you have any issues arising from that, bring the matter to the attention of one of the teaching staff. The accommodation should be kept clean and tidy; the owners may make inspections from time to time if they think they have reason to. Any damage/breakages should be reported. *Take care to switch off all electrical appliances (especially heating and cookers) when you are not in the caravans.*

# Field locations




The accommodation is at Corpach, a few miles north-west of Fort William (northernmost star). The four field locations we will visit are all indicated.

# Grids / Magnetic North

The coastal part of the Kinlochleven mapping area is at

NB. 1° = 60′ ; 1′ = 60′′

British grid reference: NN 173 622

Latitude/longitude: 56° 42′ 56′′ N , 4° 59′ 13′′ W (= 56.715556°N, 4.986944°W)

The British grid (Ordnance Survey grid) is what is on your map. Converting between the British grid and latitude / longitude is computationally complicated but there are several on-line calculators that do the job. These give very slightly different answers depending upon how they have done the maths. A good one is at

www.movable-type.co.uk/scripts/latlong-os-gridref.html

but the one I have used is a spreadsheet available from the web page

www.ordnancesurvey.co.uk/business-government/tools-support/os-net/coordinates

or directly from

www.ordnancesurvey.co.uk/documents/resources/projection-and-transformation-calculations.xls

**Now you need to consider three norths:**

1. Geographical north
2. Magnetic north (where your compass points)
3. Grid north (where the grid lines on your map point)

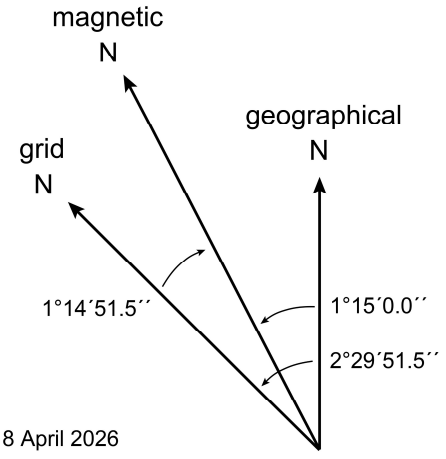
Magnetic north varies with time and place (see https://maps.ngdc.noaa.gov/viewers/historical_declination). Two web sites that give magnetic declination (***magnetic north with respect to geographical north***) for any place in the world on a given date are:

www.geomag.bgs.ac.uk/data_source/models_compass/wmm_calc.html

www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml

The NOAA site allows a choice of models of the Earth's magnetic field. For the World Magnetic Model (WMM-2025), the magnetic declination at Kinlochleven on 8 April 2026 is 1° 15′ 0.0′′ W changing at 12.6′ E per year. [= - 1.25° changing at + 0.21° per year.]

However, you want ***magnetic north with respect to grid north***. To obtain this you need to know the grid convergence (the angle between grid north and geographic north). The spreadsheet noted above calculates this as well. On the coastal part of the Kinlochleven mapping area, grid convergence is 2° 29′ 51.5′′ W (= - 2.4976°). Using this with the magnetic declination (1° 15′ 0.0′′ W) indicates that magnetic north is 1° 14′ 51.5′′ E (= 1.2476°E) of grid north on your field slips on 8 April 2026.




**2026 magnetic north on your compass ~ 1°E**

A handy website that calculates magnetic north with respect to British grid north at any place in Great Britain without you needing to worry about grid convergence is

www.geomag.bgs.ac.uk/data_service/models_compass/gma_calc.html

# GEOLOGICAL BACKGROUND

***Additional Reading:***

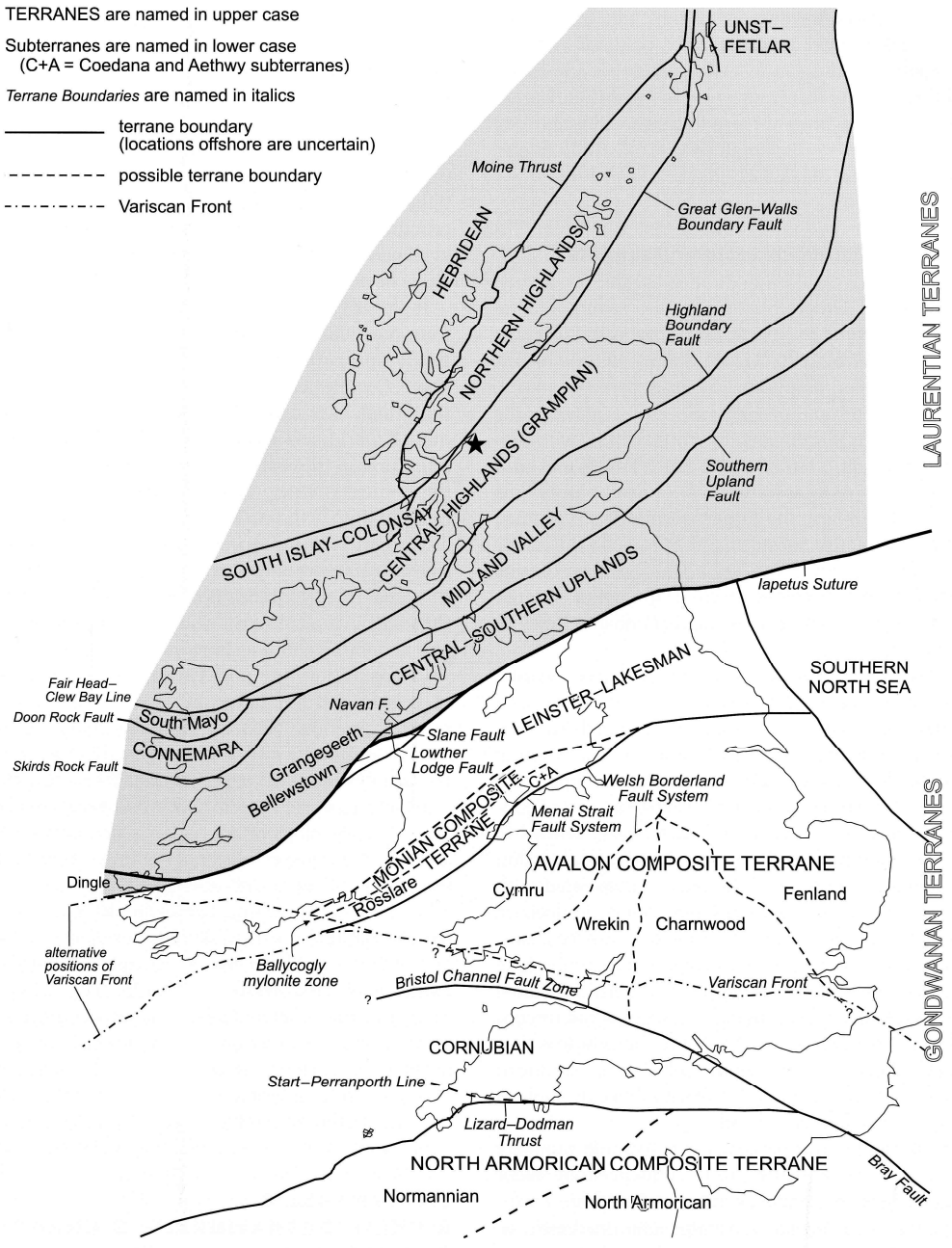
The notes on the Neoproterozoic and Palaeozoic geological evolution of Scotland over the next few pages focus on the parts of that story that are most relevant for the present field course. For a more comprehensive account see

- Woodcock N, Strachan R (editors), 2012, *Geological History of Britain and Ireland* (2nd edition). Wiley-Blackwell
- Smith M, Strachan R (editors), 2024, *The Geology of Scotland* (5th edition). Geological Society, London [This is a new edition of Trewin NH (editor), 2002, *The Geology of Scotland* (4th edition). Geological Society, London]
- Chew DM, Strachan R, 2014, The Laurentian Caledonides of Scotland and Ireland. *Geological Society, London, Special Publications* 390: 45-91

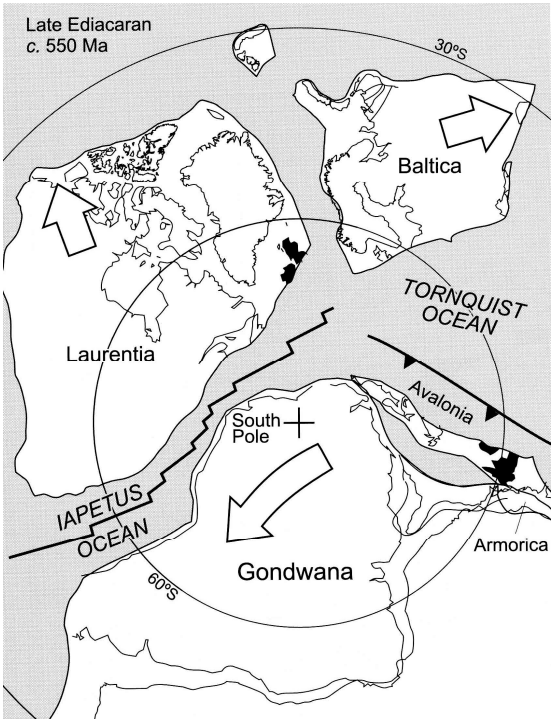
# Geological Background

The basement geology of Scotland comprises an assemblage of fault-bound crustal blocks. These blocks were juxtaposed (with subsequent readjustments) during a series of Ordovician to Devonian orogenic events that were associated with the formation and closure of oceans that lay between the continents of Laurentia (to the north-west), Baltica (to the east) and Avalonia (to the south-east). These orogenic events are known collectively as the Caledonian Orogeny.

The rocks that we will be examining on this field course belong to the Central Highlands (Grampian) terrane, which is bounded to the north-west by the Great Glen Fault, and to the south-east by the Highland Boundary Fault.




*The terranes of the British Isles (after Woodcock & Strachan, 2012). The star shows our location*

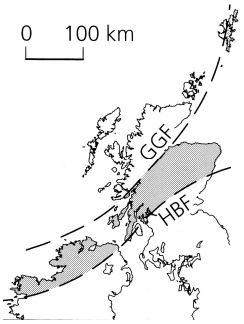


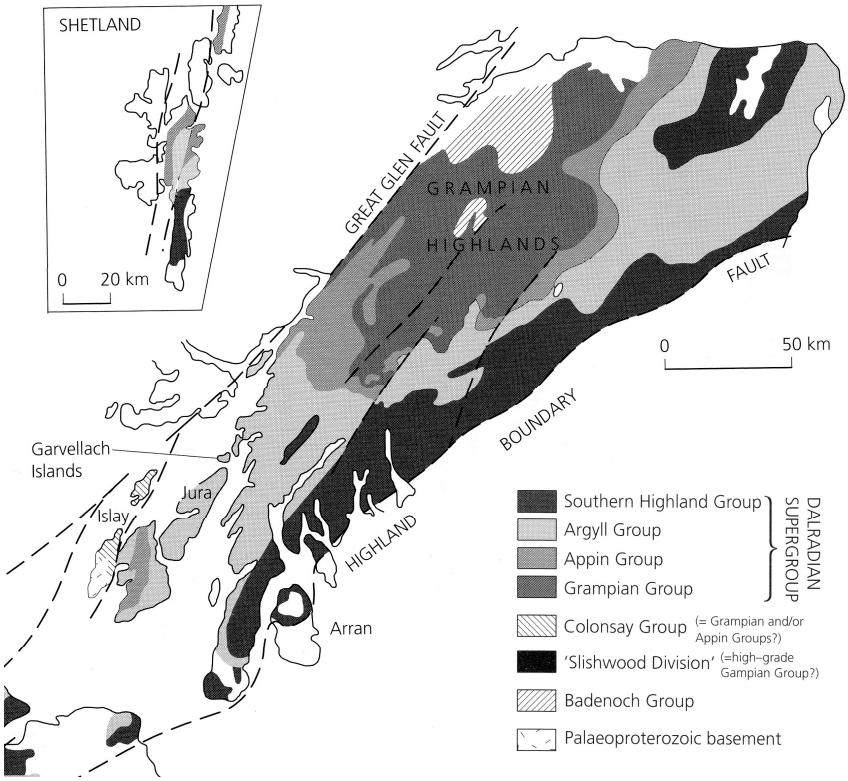

*Palaeocontinental positions at the end of the Neoproterozoic (after Woodcock & Strachan, 2012).*

The Central Highlands terrane is dominated by the Dalradian Supergroup, a thick sequence of Neoproterozoic to Early Ordovician sedimentary rocks that were deposited on the southern margin of Laurentia during the rifting that culminated in the opening of the Iapetus Ocean between Laurentia and Gondwana.

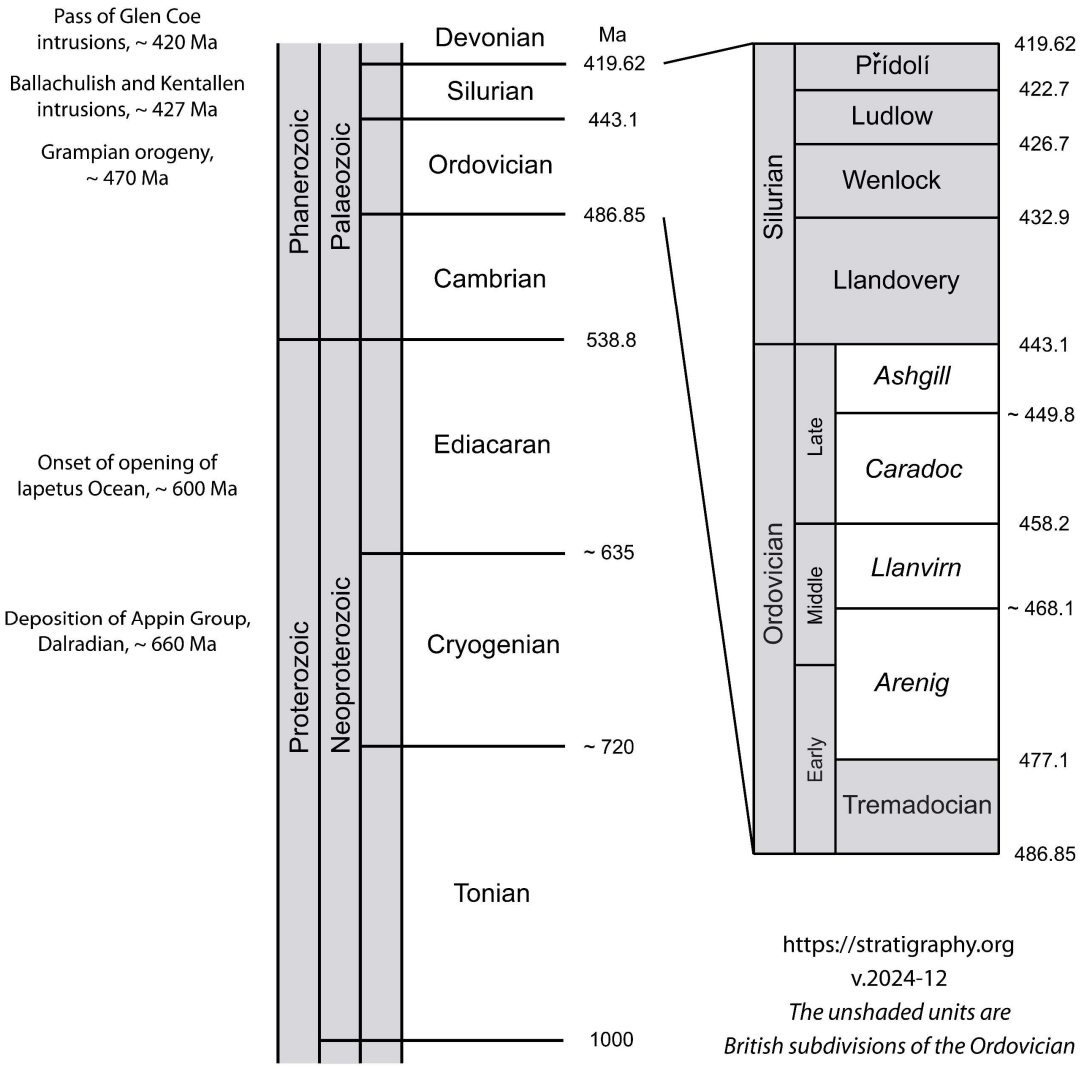
In the Early Ordovician (~470 Ma), as Iapetus began to close, the Dalradian units were strongly deformed and metamorphosed when a volcanic arc (now underlying the Midland Valley) collided with Laurentia. We will be mapping structures that formed during this deformation event within the Caledonian Orogeny.

In the Late Silurian to Early Devonian (~430 - 400 Ma), during the late stages of the Caledonian Orogeny, the Central Highlands terrane was intruded by the 'Newer Granites'. The igneous and contact metamorphic rocks that we will be examining on this field course belong to this phase of igneous activity.




*Central Highlands Terrane and Dalradian outcrop in Scotland (after Woodcock & Strachan, 2012).*




| | |
|---|---|
| oldest land plant macrofossil | Wenlock |
| Late Ordovician mass extinction | 445 Ma |
| oldest land plants, terrestrial arthropods | mid-Ordovician |
| Great Ordovician Biodiversification Event (two pulses) | mid-Arenig, late Llanvirn - early Caradoc |
| origin of crown groups of most modern phyla | end-Ediacaran - Cambrian |
| advent of biomineralization (and worm-world) | end-Ediacaran |
| first trace fossils of burrows | ~555 Ma |
| Shuram-Wonoka carbon isotope excursion (carbon cycle disruption) | 560 Ma |
| Gaskiers glaciation | 579.2 Ma |
| appearance of large, soft-bodied, multicellular organisms | Ediacaran |
| Marinoan ‘Snowball Earth’ glaciation | ~645 - 635.2 Ma |
| Sturtian ‘Snowball Earth’ glaciation | 717 - 660 Ma |
| Neoproterozoic Oxygenation Event (10% → 50%+ present atm $O_2$) | 800 - 650 Ma |
| appearance of metazoans | early Tonian |

**Dalradian Supergroup**

***Plate tectonic context***

At ~900 Ma most of the Earth's land masses were amalgamated in a supercontinent (Rodinia). Subsequently, subduction zones and accretionary orogens developed around the periphery of this supercontinent, while at the same time the continental interior underwent multistage rifting events. The rifting ultimately led to the break-up of Rodinia during the second half of the Neoproterozoic.

The protoliths of the older parts of the metasedimentary Dalradian Supergroup were deposited in extensional basins formed within Rodinia as it underwent rifting. At ~600 Ma this rifting developed into the opening of the Iapetus Ocean, that separated Laurentia to the north-west from Gondwana to the south-east. The younger parts of the Dalradian sequence were deposited on the Laurentian passive margin during the latest Neoproterozoic and early Palaeozoic as Iapetus Ocean opened and then began to close.

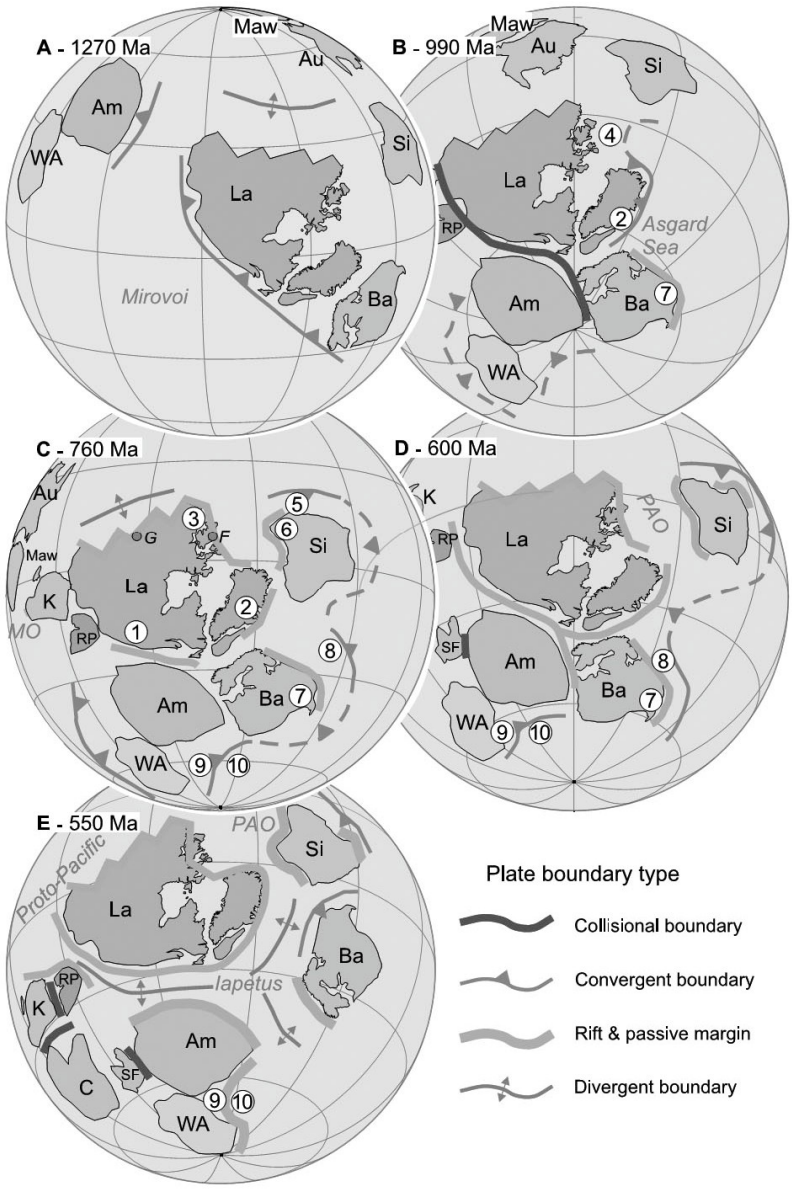



*Plate reconstructions during the formation of Rodinia (1000 - 900 Ma) and its subsequent break-up (760 - 530 Ma) (after Cawood et al., 2016, Earth and Planetary Science Letters 449: 118-126).*

*Abbreviations:* Am – Amazonia, Au – Australia, Ba – Baltica, C – Congo, K – Kalahari, La – Laurentia, Maw – Mawson, RP – Rio de la Plata, SF – Sao Francisco, Si – Siberia, WA – West Africa, MO – Mozambique Ocean, PAO – Palaeo-Asian Ocean.

*The location of the Dalradian units is just below ② on these figures. England and Wales are at ⑩*

***Stratigraphy***

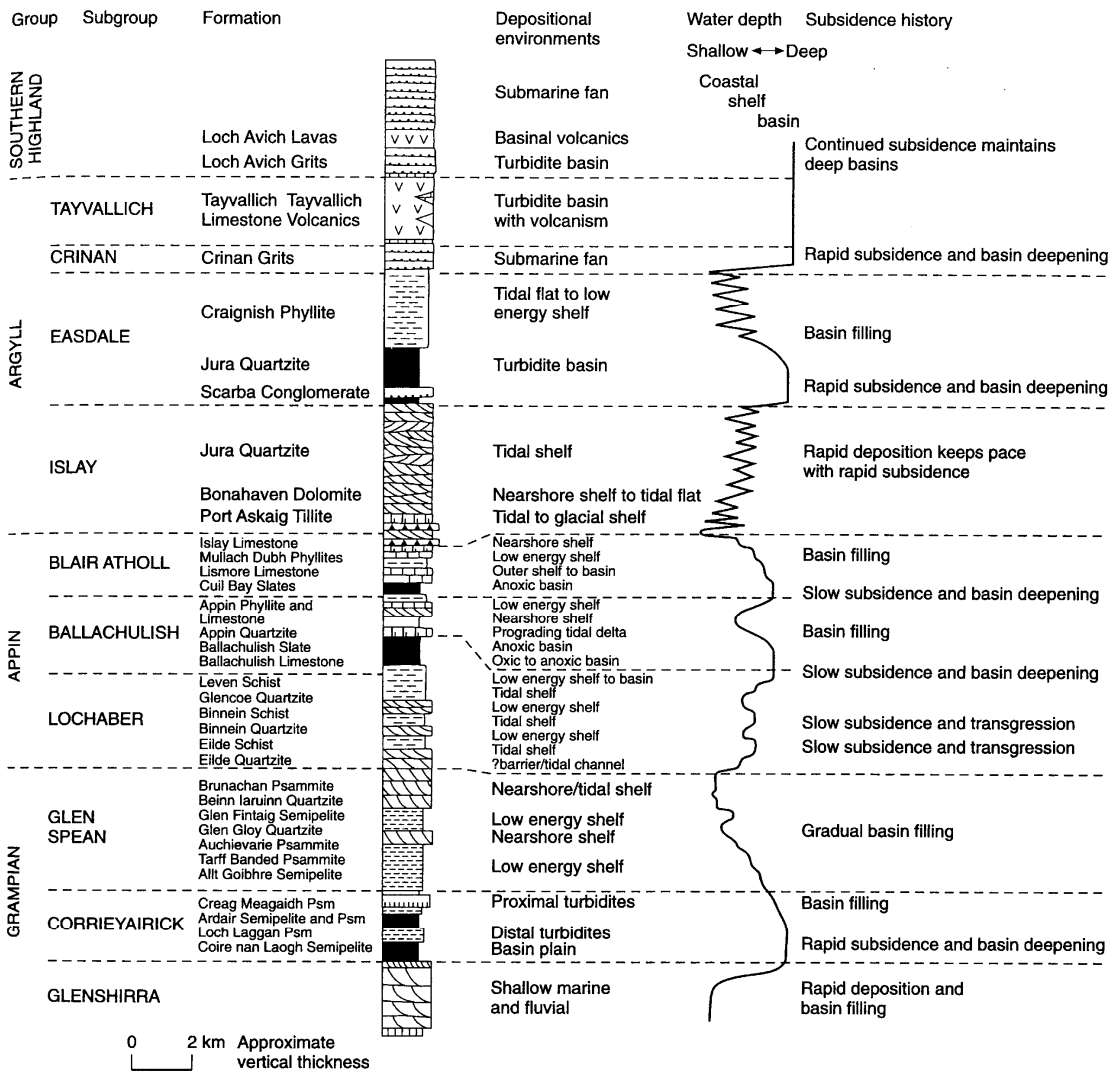
The Dalradian metasediments are a sequence of marine sands, silts, muds, and limestones that were deposited on a basement of older metamorphic rocks (the Badenoch Group, formerly known as the 'Central Highland Migmatite Complex'). Dalradian sedimentation post-dates the deformation and mid-amphibolite facies metamorphism of the Badenoch Group that occurred at ~ 840 Ma, and probably began ~ 730 - 700 Ma, at about the same time as the onset of the Sturtian 'Snowball Earth' glaciation. The total thickness of the succession was at least ~ 25 km but the complete vertical thickness was never deposited at one place – rather the depocentre migrated south-eastwards with time as the basin deepened with prolonged crustal stretching.

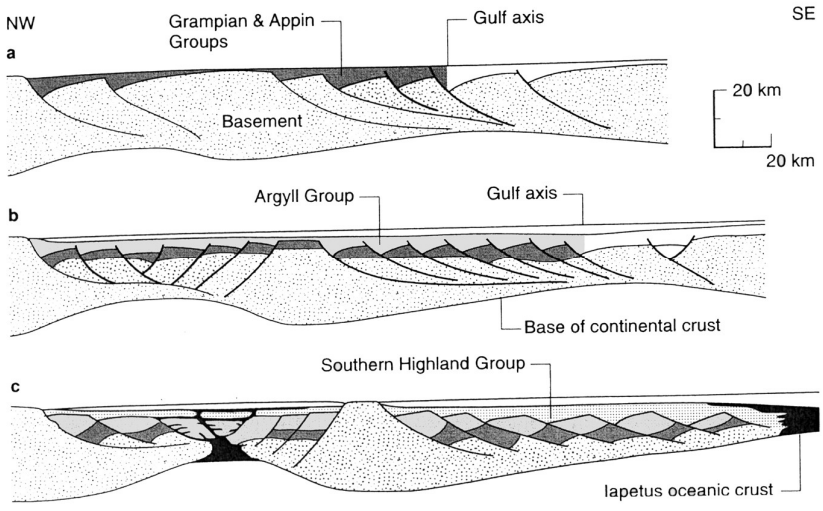
The Dalradian sequence is subdivided into the Grampian, Appin, Argyll, and Southern Highland Groups. The ***Grampian Group*** is composed primarily of micaceous to quartzose psammites and semi-pelites. These record the initiation of extension, followed by a period of rapid rifting and basin deepening (associated with turbidites), and then shallowing as the basin filled during the transition from active rifting to a more gradual thermally-induced basin subsidence. The ***Appin Group*** comprises quartzites, limestones, semi-pelites and pelites which were deposited on a stable marine shelf. Cyclical alternations of quartzites and muddy sediments are thought to correspond with repeated episodes of rifting and thermal subsidence. Towards the top of the group, limestones and dark mudstones become more prevalent and this is interpreted to indicate a period of low sediment supply during a prolonged phase of thermal subsidence and basin widening. Thickness variations of the units suggest that the Appin Group sediments were deposited in a set of half-grabens. The base of the succeeding ***Argyll Group*** is marked by a tillite (the Port Askaig Tillite). The units immediately above this tillite are shallow water, intertidal, stromatolitic dolomites suggestive of a warm, arid climate. Hence the Port Askaig Tillite has been linked with one of the 'Snowball Earth' glaciations (probably the Marinoan glaciation, ~ 645 - 635 Ma). The rest of the Argyll Group succession comprises quartzites (originally thick tidal shelf sands), pebbly psammites (originally fan turbidites), graphitic pelites (originally deep-water organic-rich mudstones), and calc-pelites (originally tidal flat sediments), and appears to record fault-controlled sedimentation. Towards the top of the Argyll Group there are abundant metabasite sills and volcanics (Tayvallich Volcanic Formation) indicative of rift-related basic magmatism. Contemporaneously, the rift-related 'Vuirich Suite' of granites were intruded. Together the Tayvallich Volcanics and Vuirich Suite granites mark the onset of the opening of Iapetus Ocean at 605 - 590 Ma. The ***Southern Highland Group*** is dominated by feldspathic psammites and hemipelagic mudstones representing steady basin deepening. Near the base of the group there are beds containing drop-stones and diamictites (the Macduff Slates) that date to the Gaskiers glaciation (579.8 Ma). At the very top of the Southern Highland Group are deep water sandstones, limestones, slates, cherts and slump breccias, some of which units contain Cambrian trilobites and Early Ordovician conodonts.

*Dalradian units in the mapping areas at Kinlochleven and Cuil Bay (within the Appin Group)*

The Dalradian units at Kinlochleven (Binnein Quartzite, and Binnein Schist) are from the lower part of the Appin Group. The quartzites contain cross-bedding (commonly herring-bone), ripple marks, and erosional features, while occasional mudcracks indicate intermittent emergence. They become less feldspathic towards the north-east, with palaeocurrent evidence indicating NE-SW flow. These units are interpreted as shelf deposits with sediment being supplied from the south-west, the quartzites representing shoaling events in which the deposits were tidally reworked.

At Cuil Bay the units between the Ballachulish Slates and Cuil Bay Slates are from the middle of the Appin Group, and are also interpreted as shelf deposits formed during a period of basin filling between two phases of basin-deepening rifting – the slates are from mudstones deposited when the basin was at its deepest.




*Top: Dalradian stratigraphy and sedimentary evolution. Bottom: schematic sections illustrating the evolution of this part of the Laurentian margin during deposition of the Dalradian. (After Trewin, 2002.)*

## Grampian Orogeny

### *Plate tectonic context*

In the Cambrian and Early Ordovician the Laurentian margin in northern Scotland was a shallow-water carbonate shelf that passed south-eastwards into the deep marine turbidite basins of the Southern Highland Group. Sedimentation was terminated in the Mid-Ordovician by the Grampian Orogeny (478 - 460 Ma). The Grampian orogenic event was caused by the collision of a volcanic arc with the Laurentian margin. This arc, commonly known as the Taconic Arc and including units now underlying the Midland Valley of Scotland at its eastern end, had developed over an intra-oceanic subduction zone within the Iapetus Ocean. The collision led to the obduction of an ophiolite nappe, together with regional deformation and metamorphism.

To the far south during the Early Ordovician (~475 Ma), Avalonia, a continental fragment that included southern Britain and Ireland, rifted away from the margins of Gondwana and began migrating northwards. This marked the start of the closure of the Iapetus Ocean and the opening of the Rheic Ocean between Avalonia and Gondwana. At this time, subduction was also occurring along the western margin of Baltica which was rotating anticlockwise and beginning to converge with Laurentia.

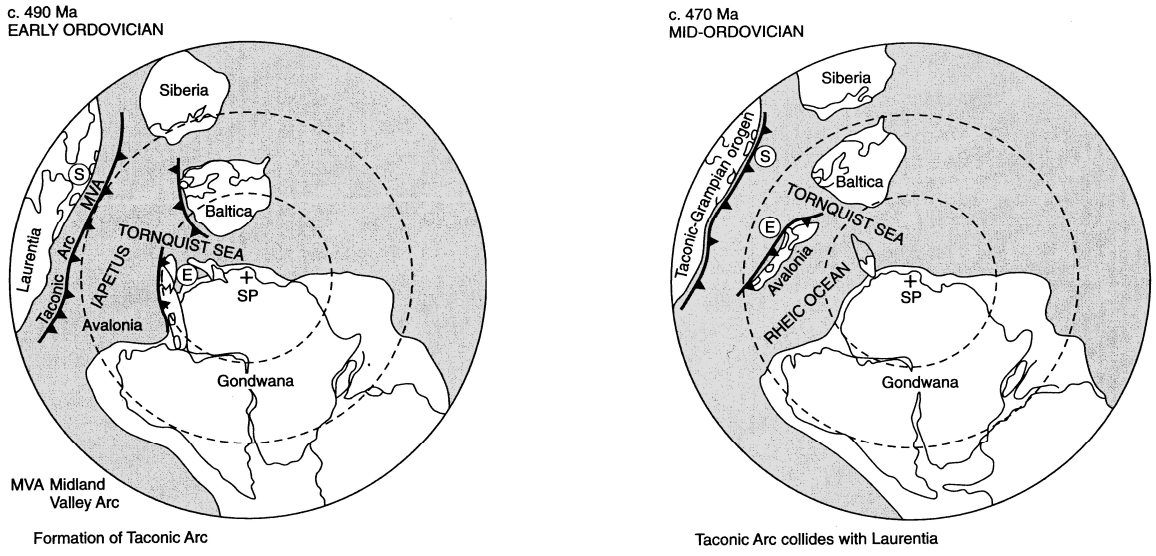
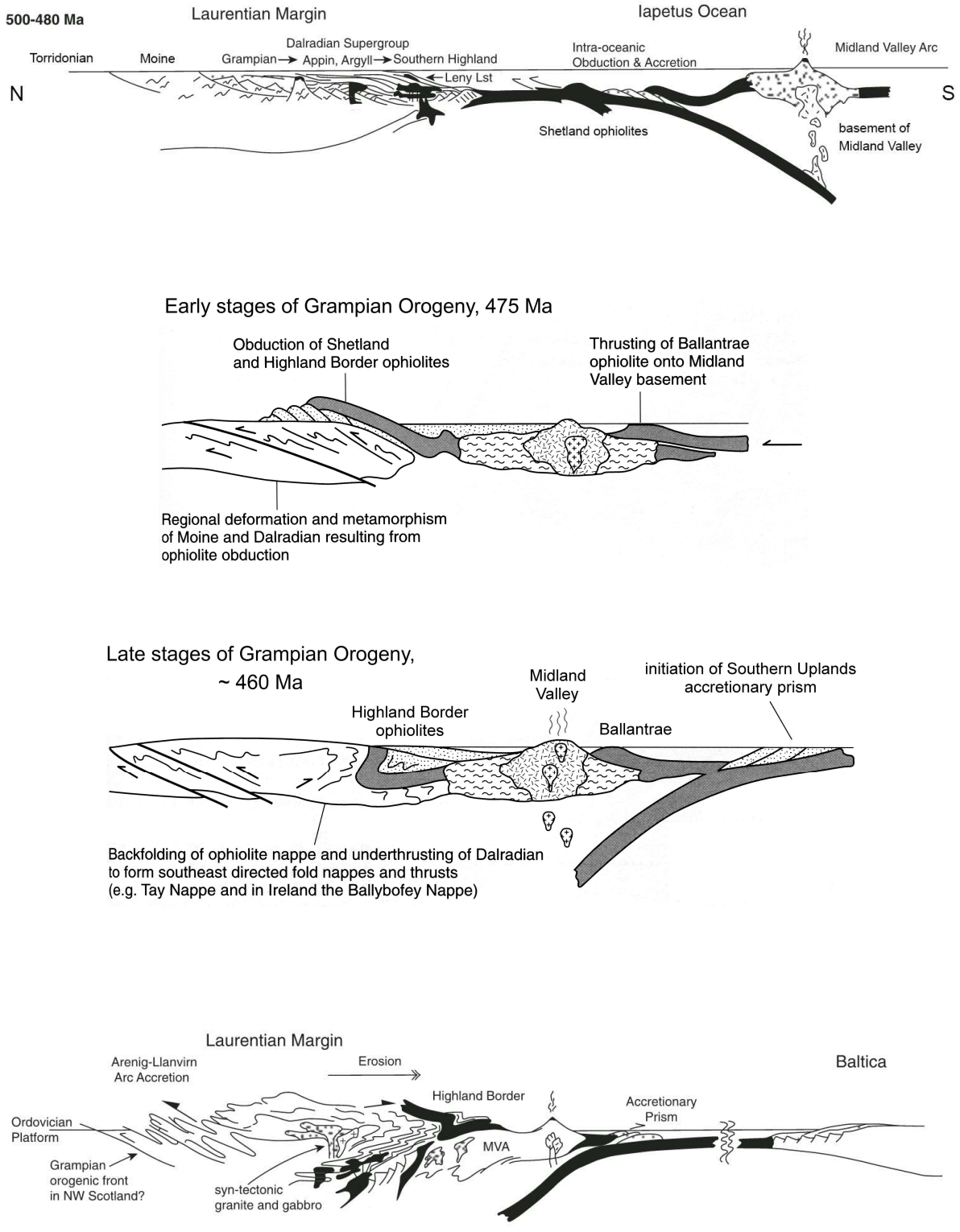



*Plate reconstructions showing the plate tectonic context of the Scottish Highlands during the Grampian Orogeny (after Trewin, 2002).*




*Schematic sections across Scotland before (top) and after (bottom) the Grampian Orogeny, with two intermediate steps (middle two). Note the switch in direction of subduction (southward to northward) after collision of the arc with Laurentia. In the Kinlochleven and Cuil Bay mapping areas, $D_1$ corresponds with the deformation illustrated for the 475 Ma section, $D_2$ with the deformation illustrated for the 460 Ma section, and $D_3$ with some late readjustments in the terminal stages of the Grampian Orogeny. (Top and bottom sections after Leslie et al., 2008, Geological Society of America Memoir 202: 307-343; middle two sections after Woodcock & Strachan, 2012.)*

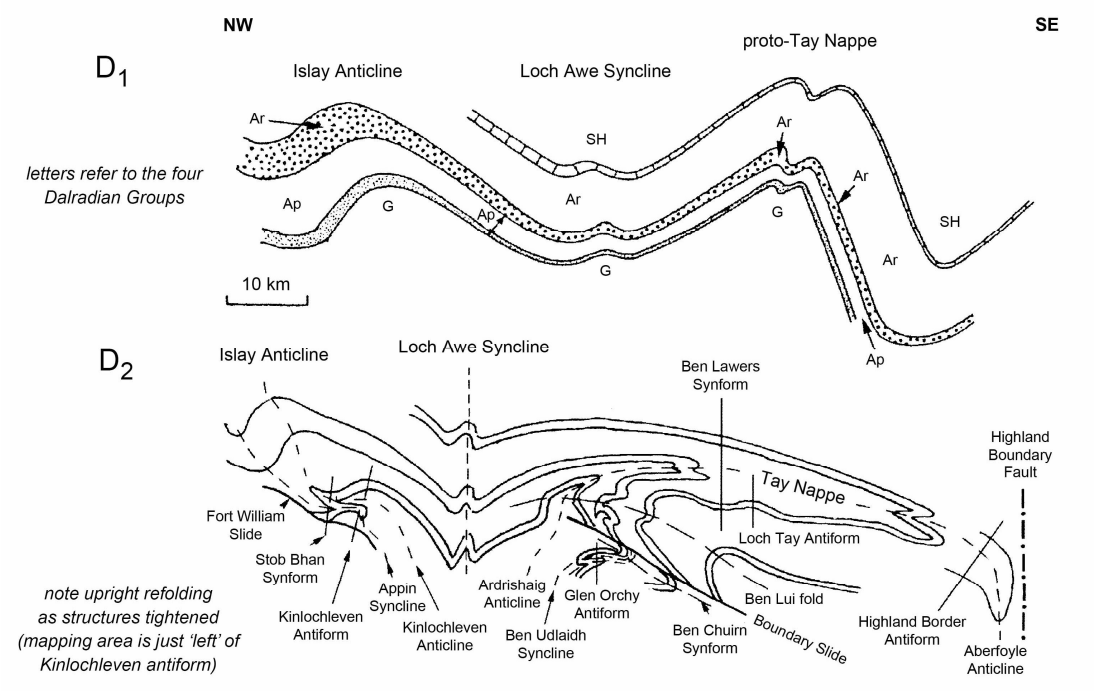
***Deformation***

The Grampian Orogeny affected units in the Northern Highland terrane to the north of the Great Glen Fault as well as units in the Central Highlands (Grampian) terrane. North of the Great Glen Fault it led to widespread NW-directed, ductile thrusting and folding under amphibolite facies metamorphic conditions. South of the Great Glen Fault in the Central Highlands terrane, the Dalradian Supergroup experienced three major ductile deformation events, associated with initial crustal thickening ($D_1$), gravitational spreading and back-thrusting of the thickened pile as it heated up ($D_2$), and differential uplift and exhumation ($D_3$).

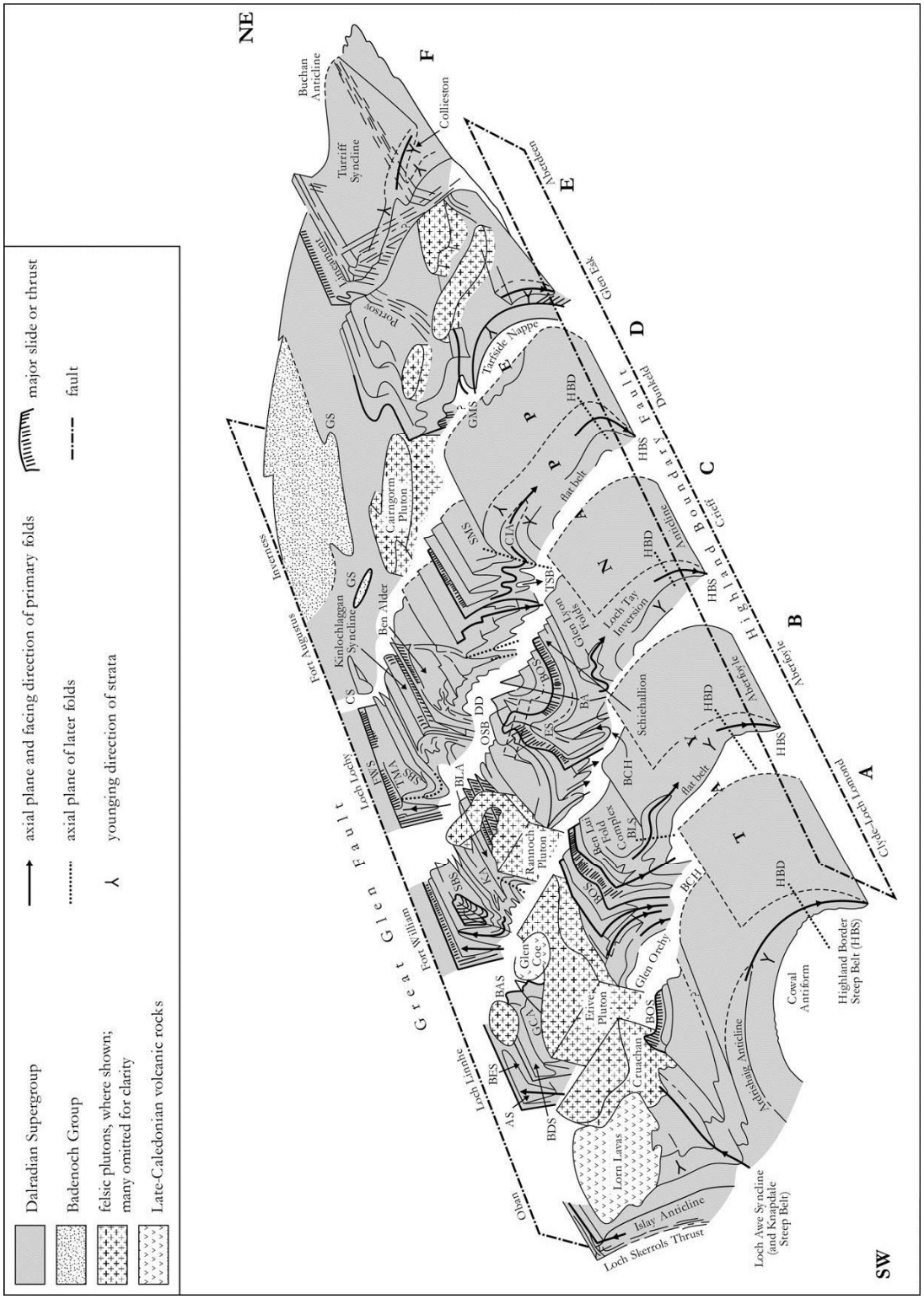
$D_1$ involved NW-directed ductile thrusting over distances of at least tens of kilometres, accompanied by NE-SW trending upright folding ($F_1$) on scales of up to tens of kilometres. In micaceous lithologies the folding was associated with the development of an axial-planar, penetrative slaty cleavage ($S_1$) that fans across the Dalradian outcrop from NW facing in the north-west to SE facing in the south-east. These areas are separated by a central zone of upright folding represented by the Loch Awe Syncline and the Geal Charn - Ossian Steep Belt. Associated with $D_1$ is a stretching lineation ($L_1$) which appears as stretched pebble clasts or as a mineral elongation lineation.

$D_2$ involved vertical shortening and SE-directed shearing (back-thrusting). This refolded the earlier folds, forming upright $F_2$ folds in the NW and, through a combination of back-thrusting and extensional collapse of the orogenic belt, a major recumbent fold (the Tay Nappe) to the SE. The associated foliation is typically a crenulation cleavage ($S_2$), but in places this has completely transposed the earlier $S_1$ fabric. The usual lineation is a crenulation ($L_2$) formed by the hinges of $F_2$ micro-folds of the $S_1$ cleavage.

$D_3$ was a series of late, low-strain events, the net large-scale effect of which was to rotate $D_1$ and $D_2$ structures locally into steep attitudes – the present-day distribution of 'flat belts' and 'steep belts' in the Dalradian is largely the result of $D_3$. $D_3$ produced large-scale upright folds but did not generally produce penetrative fabrics. Gentle crenulations are the most common small-scale $D_3$ structures.




*General NW-SE cross-section across the Grampian orogenic structures of south-western part of the Grampian Highlands terrane (from Treagus, 2009, The Dalradian of Scotland, Geologists' Association Guide 67). Top: reconstruction of the $D_1$ fold geometry; bottom: present (post-$D_3$) geometry.*



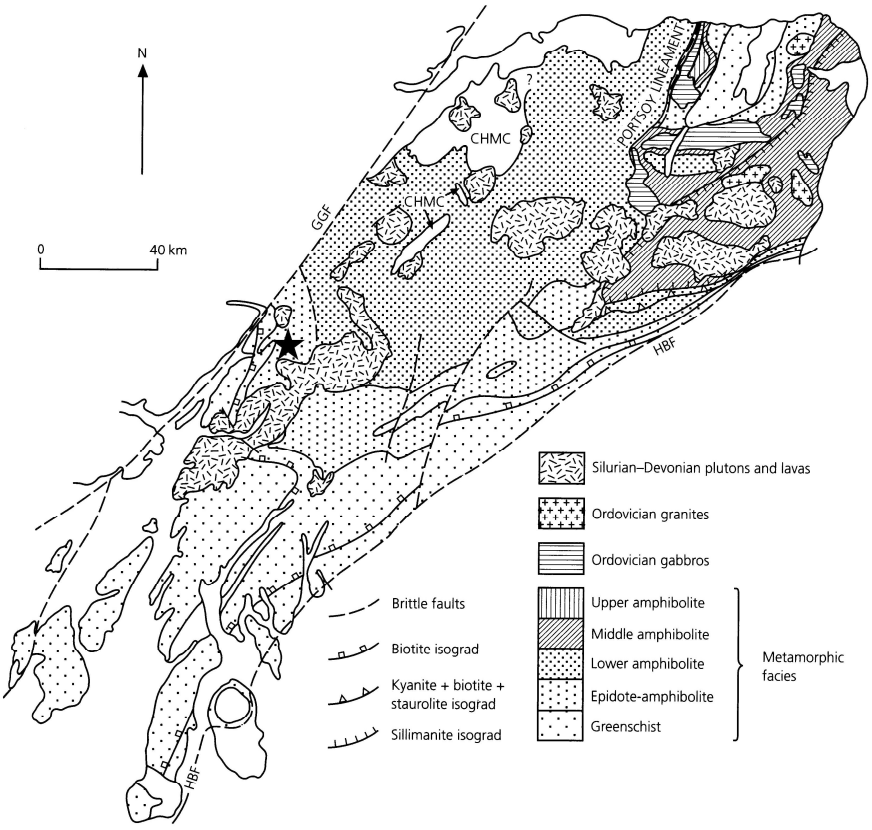

*Block diagram of the major structures in the Central Highlands terrane (after Stephenson et al., 2013, Proceedings of the Geologists' Association 124: 3-82). Abbreviations: AS = Appin Syncline; BA = Bohespic Antiform; BAS = Ballachulish Slide; BCH = Beinn a'Chuallich folds; BDS = Beinn Don Syncline; BES = Benderloch Slide; BLA = Beinn na Lap Antiform; BLS = Ben Lawers Synform; BOS = Boundary Slide; CIA = Creag na h'Iolaire Anticline; CS = Corrieyairack Syncline; DD = Drumochter Dome; ES = Errochty Synform; FWS = Fort William Slide; GCA = Glen Creran Anticline; GMS =Glen Mark Slide; GS = Grampian Slide; HBD = Highland Border Downbend; HBS = Highland Border Steep Belt; KA = Kinlochleven Anticline; OSB = Geal-charn–Ossian Steep Belt; SBS = Stob Ban Synform; SMS = Sron Mhor Synform; TMA = Tom Meadhoin Anticline; TSB = Tummel Steep Belt.*

***Metamorphism and Grampian Orogeny intrusions***

Over much of the Dalradian outcrop the regional metamorphism during the Grampian Orogeny is represented by the classical Barrovian zonal sequence: chlorite, biotite, garnet, staurolite, kyanite, sillimanite and sillimanite + K-feldspar. The metamorphic grade increases from greenschist facies in the south-west to mid-amphibolite facies in the north-east. Pressures of 900 - 1200 MPa corresponding with depths of burial of 25 - 35 km are widely found over the South-Western and Central Highlands but these decrease markedly to ~ 300 MPa (10 km) in the north-east.

The Dalradian units of the north-east were intruded, ~ 470 Ma, by a suite of gabbros (the 'Newer Gabbros'). Locally the contact aureoles of these overprint the peak regional metamorphic assemblages prior to their involvement in $D_3$. At about the same time, or very shortly afterwards, but before $D_3$, S-type granites (the 'Grampian Granites') derived from partial melting of the Dalradian metasediments were intruded into the north-eastern part of the Dalradian outcrop.

The thermal peak of the regional metamorphism associated with the Grampian Orogeny occurred after $D_1$ since porphyroblasts overgrow $S_1$. At many locations the porphyroblasts exhibit S-shaped inclusion trails which imply that they grew during $D_2$, and hence that the thermal peak was during or shortly after $D_2$. The chemical zonation of the garnets suggests that they grew as pressure was decreasing, which is consistent with the suggestion that $D_2$, at least in part, represents extensional collapse of an over-thickened (during $D_1$) crust as it heated up. $D_3$ is associated with the development of retrograde minerals and so occurred after the metamorphic peak but before the intrusion of an undeformed pegmatite that has been dated at 462 Ma.



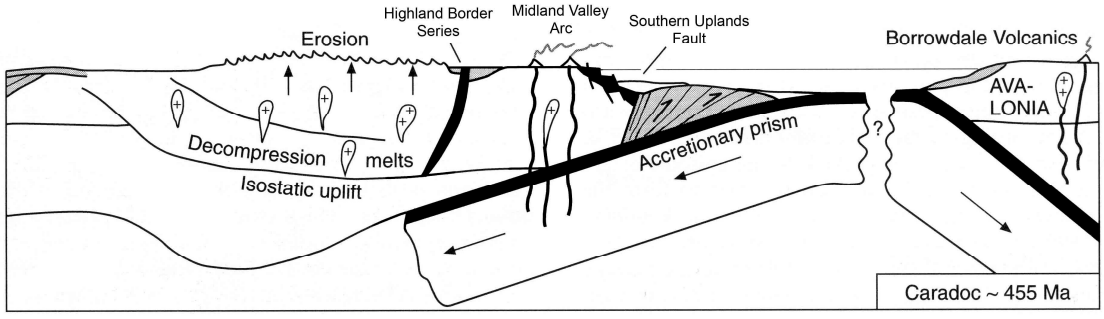

*Metamorphic zones within the Dalradian rocks. Also shown are the pre-Dalradian Badenoch Group (CHMC) and the later (Silurian-Devonian) 'Newer Granite' igneous bodies that crop out in the Central Highlands terrane. Kinlochleven (star) is at epidote-amphibolite facies grade (just above the garnet-in isograd); Cuil Bay just to the west is at greenschist facies grade (just below the biotite-in isograd). (After Woodcock & Strachan, 2012.)*

**Late to post-Grampian Orogeny events**

The presence at the surface today of Dalradian rocks that were metamorphosed at depths of 35 km, coupled with the observation that the crust in the Grampian Highlands is currently ~35 km thick, implies that at peak metamorphic conditions (~470 Ma) the thickness of the Grampian crust was ~70 km. Barrovian high-grade garnet detritus is found in mid-Llanvirn (465 Ma) conglomerates to the south of Glasgow, which implies that within 5 million years of peak metamorphism, crust metamorphosed at 35 km depth had reached the surface. This requires local erosion rates of about 7 mm $yr^{-1}$, that is, erosion-rates similar to those in the Nanga Parbat region of the modern Himalaya. The implication is that during the Grampian Orogeny, the $D_1$ crustal thickening phase generated high mountains, and that fast erosion, possibly assisted by extensional collapse during $D_2$, led to rapid exhumation.

By ~455 Ma, in the Southern Uplands terrane to the south, an accretionary prism had formed indicating that by then northward subduction under Laurentia had been initiated. In the Central Highlands terrane at ~455 Ma a suite of post-tectonic S-type granitic intrusions were emplaced. These are interpreted as the products of decompression melting of the over-thickened Grampian crust as it continued to be eroded and exhumed.

There is no record of igneous activity in the Central Highlands terrane in period 455 - 430 Ma. When igneous activity began again ~430 Ma it was as subduction was ceasing. The lack of evidence for active subduction-related igneous activity in this 25 myr interval is attributed to shallow angle subduction under Laurentia, or to the removal of the evidence by the extreme rates of uplift and erosion at this time.

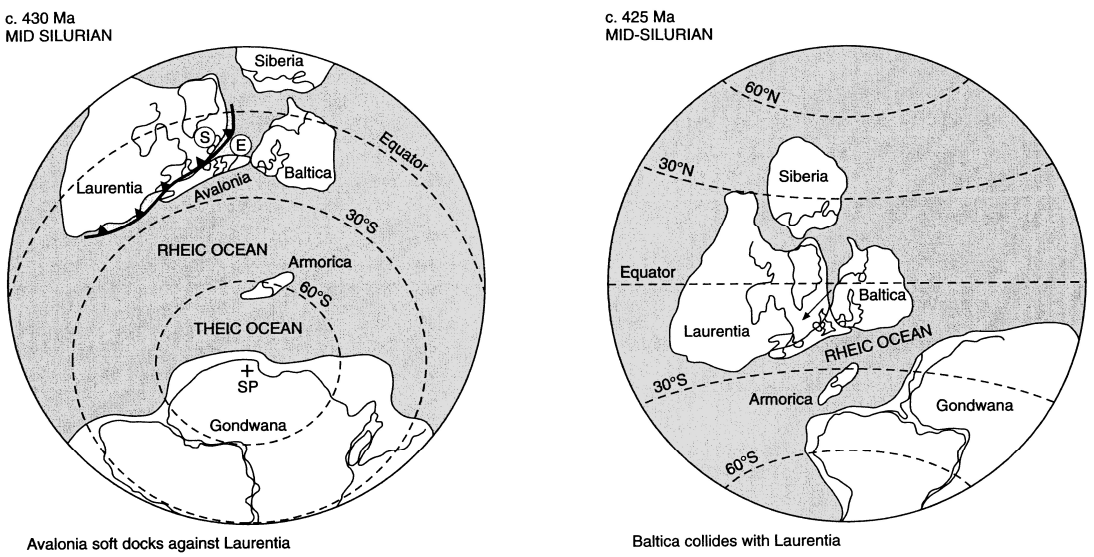



*Schematic section showing the tectonic context of the Scottish Highlands during the emplacement of the post-tectonic granites (after Trewin, 2002).*

**Convergence of Laurentia with Baltica-Avalonia and the final closure of Iapetus**

***Plate tectonic context***

Southwards subduction of Iapetus Ocean under Avalonia ceased ~450 Ma and from then on Avalonia was brought closer to Laurentia (*i.e.*, the Iapetus Ocean was narrowed) by continued northwards subduction under the Southern Uplands accretionary prism. By ~430 Ma the ocean was effectively closed and the Lake District on the northern edge of Avalonia had become the site of a foreland basin. Further subduction of Avalonia under Laurentia was resisted by buoyancy constraints: by ~426 Ma subduction had slowed and by 423 Ma orthogonal shortening of the Southern Uplands accretionary prism had switched to sinistral transpression because the collision between Laurentia and Avalonia was oblique and not 'head-on'.



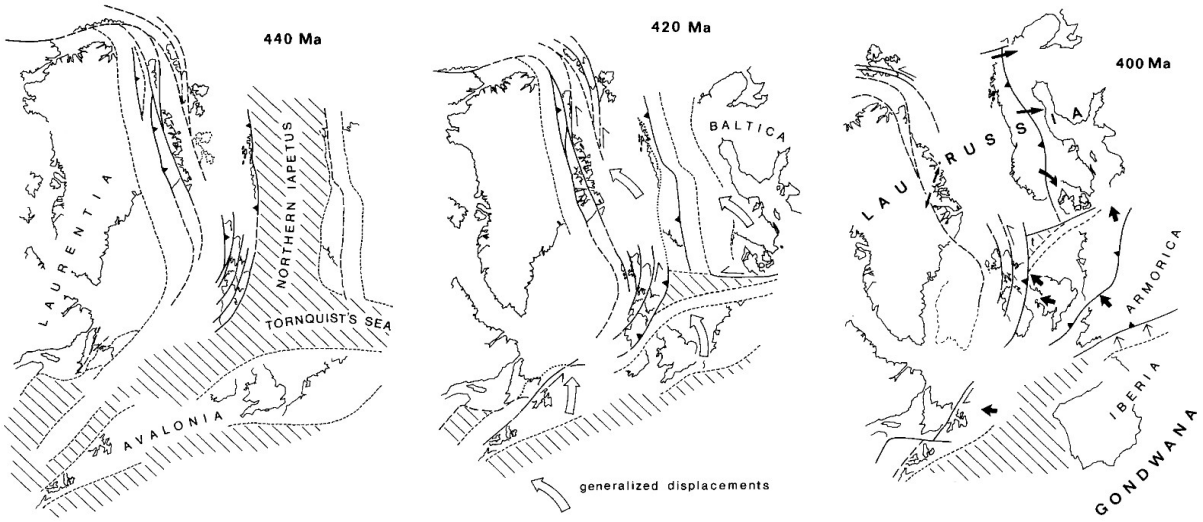
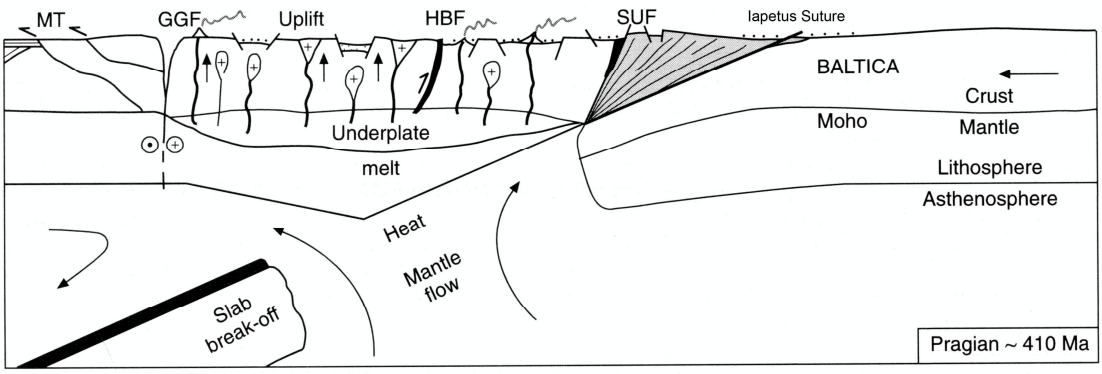




*Plate reconstructions during the Scandian Orogeny and final closure of the oceans between Laurentia, Avalonia and Baltica. Note that the oblique collision of Avalonia with Laurentia set up sinistral strike-slip displacement on the faults cross-cutting Scotland, and that this sense of offset was reinforced when Baltica joined the party. (Top two figures after Trewin, 2002; bottom three figures after Soper et al., 1992, Journal of the Geological Society, London 149: 871-880.)*

As the Iapetus Ocean between Laurentia and Avalonia closed, Laurentia also collided with Baltica. This collision was underway by 435 Ma and led to the Scandian Orogeny (~435 - 425 Ma) during which episode large scale thrust and nappe deformation occurred in Scandinavia, Greenland, and in north-western Scotland (it was during this event that the major thrust systems in the Northern Highlands terrane, namely the Sgurr Beag, Navier and Moine thrusts, were activated). The outcome of the mutual collision of these three plates was the initiation of substantial sinistral strike-slip movement on the Great Glen Fault. The movement on the Great Glen Fault, which was perhaps 700 km+, occurred over the period ~428 - 410 Ma and resulted in the juxtaposition of the Scandian-deformed Northern Highlands terrane with the Central Highlands terrane which shows no Scandian tectonic overprint.

***Silurian to Early Devonian igneous activity***

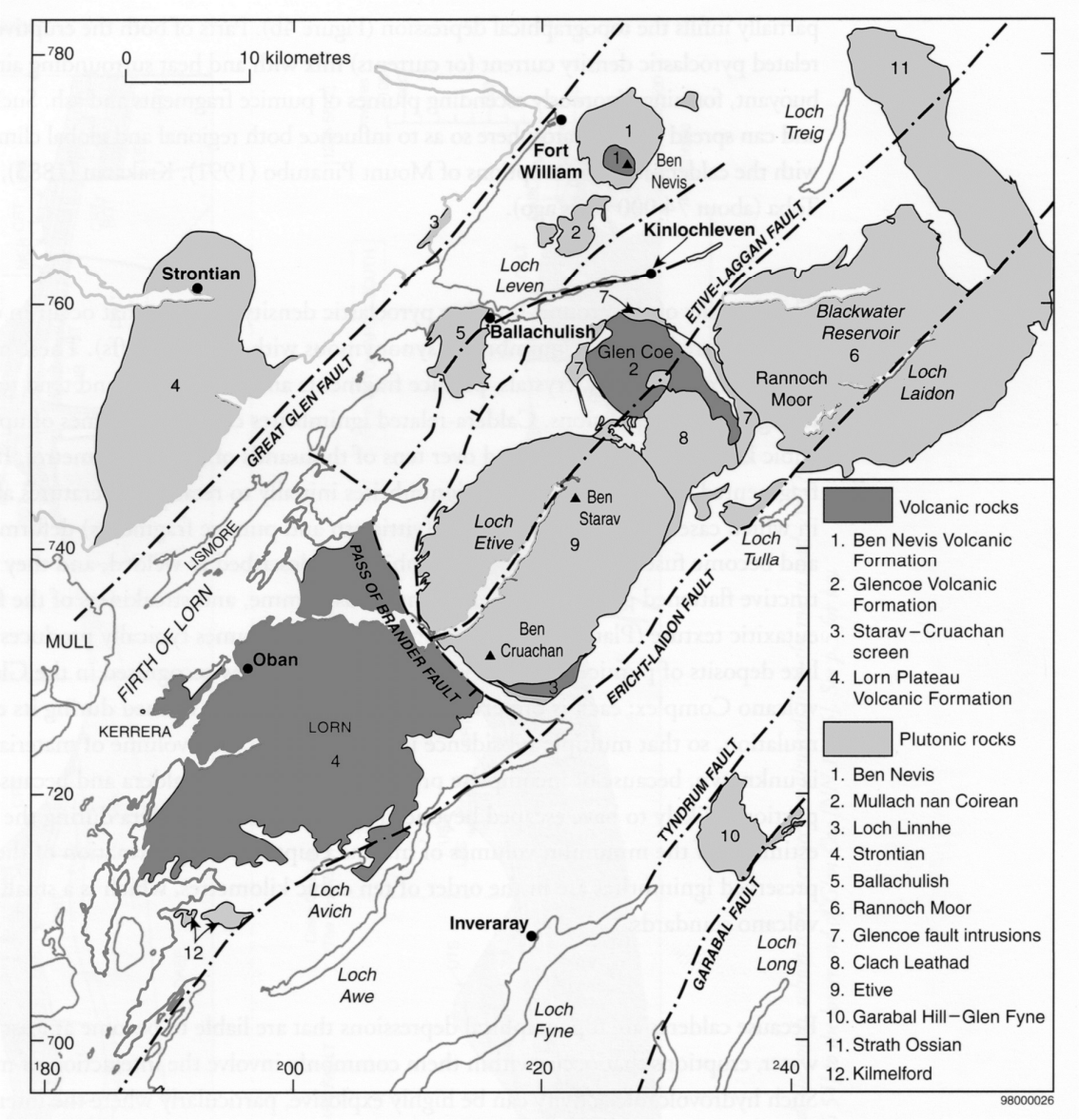
From ~430 Ma, as subduction of Iapetus under Laurentia slowed and ceased, there began a period of I-type granite emplacement in the Grampian and Northern Highland terranes which lasted at least 22 million years. These 'Newer Granites' were closely associated with hydrous alkali-rich rocks (appinites and lamprophyres). It is presently considered that this period of igneous activity followed failure (break-off) of the subducted slab. The idea is that slab break-off led to a linear domain of mantle upwelling through the slab break. The heating induced by the upwelling hot asthenopheric mantle led to partial melting (at depths of 100 - 200 km) of previously metasomatized subcontinental lithospheric mantle that lay above the slab break, thereby generating alkali-rich, hydrous appinite magmas. Rapid uplift, thermal weakening, and focusing of deformation along this linear domain may have helped to fix the location of the lithosphere-penetrating Great Glen Fault itself, and this fault then provided an easy upward pathway for the magmas. During their ascent, some of the magmas underwent fractional crystallization to produce alkali-rich intermediate and silicic magmas that were emplaced in the upper crust north-west of the Great Glen Fault. However, the chemistry of most of the Newer Granites cannot be explained by fractional crystallization from appinitic magmas but rather it is inferred that the volatiles and heat that accompanied the rising appinitic magmas promoted partial melting of mafic and intermediate lower crust, thereby generating large volumes of dioritic magma. These dioritic magmas then ascended to their level of buoyancy (~ 10 km depth) where they were modified by *in situ* fractionation and country rock assimilation to produce the Newer Granites that crop out today. Large volcanic edifices, mostly since eroded away, were formed above these intrusions, with the plumbing systems for these volcanic edifices being represented today by swarms of dykes.




*Schematic section showing the tectonic context of the Scottish Highlands during the emplacement of the Newer Granites (after Trewin, 2002).*

Rapid uplift caused by high heat flow meant that later igneous complexes were emplaced into the deeper levels of older ones. The resulting cross-cutting relationships between the volcanics, hyperabyssal intrusions, and plutons are particularly well displayed immediately to the east of the Great Glen Fault.

Many of these igneous features (the appinites, Newer Granites, volcanic edifices) are exposed close to the mapping area. We will not visit them on this field course but we see several NE-SW trending dykes that belong to the Etive dyke swarm. These form part of the plumbing system of one or more of these volcanic edifices, now eroded away.



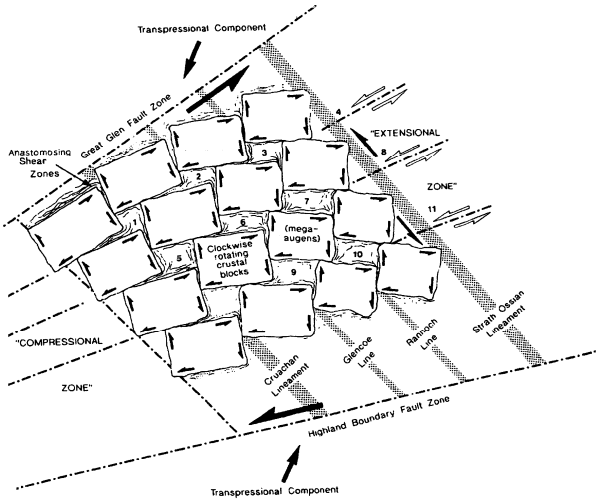

*Silurian-Devonian plutons and volcanics in the south-west Grampian Highlands (after Neilson et al., 2009, Journal of the Geological Society of London 166: 545-561)*

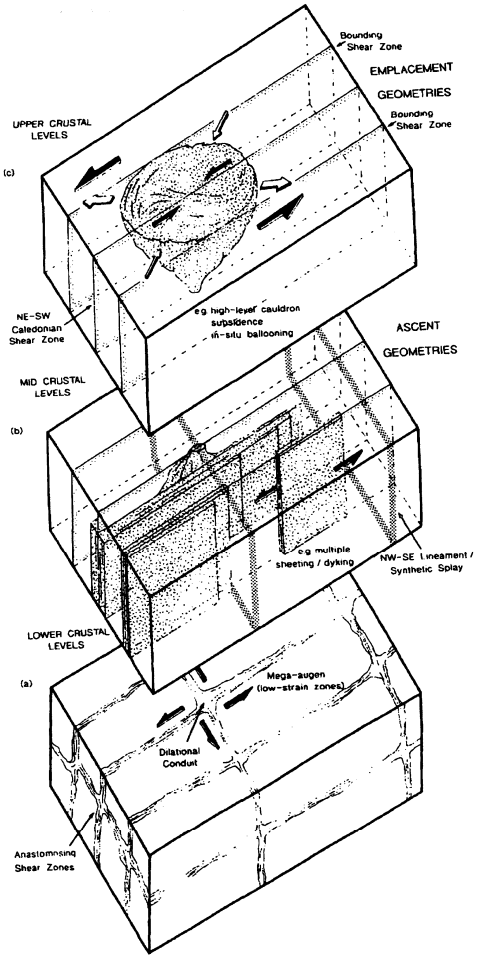
| *Dates* (quoted in Neilson *et al*., 2009) | |
|---|---|
| Garabal Hill appinitic diorite | 428 ± 9.8 Ma |
| Ballachulish pluton | 427 ± 1 Ma |
| Rubha Mor appinite (Cuil Bay) | 426.5 ± 4.2 Ma |
| Top of Lorn Plateau Volcanic Formation | 425.0 ± 0.7 Ma |
| Rannoch Moor pluton | 422.5 ± 0.5 Ma |
| Glen Coe Fault intrusion | 419.6 ± 5.4 Ma |
| Clach Leathad pluton | 417.9 ± 0.9 Ma |
| Cruachan intrusion (Etive pluton) | 414.9 ± 0.7 Ma |
| Inner Starav intrusion (Etive pluton) | 408.0 ± 0.5 Ma |

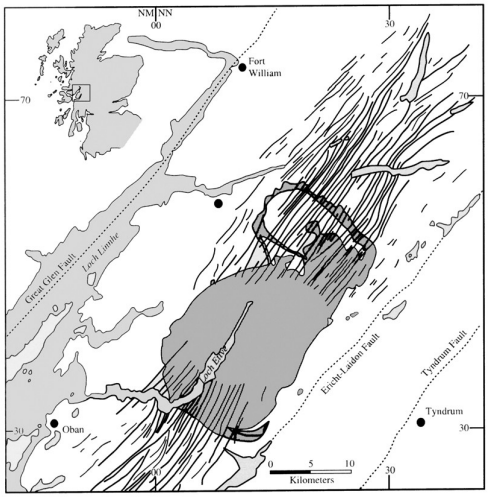
*Emplacement of the Silurian-Devonian plutons*

Many of the Newer Granite plutons have well developed thermal aureoles, with contact metamorphism clearly overprinting regional metamorphism. The inner aureole rocks contain mineral parageneses of pyroxene-hornfels facies (*e.g.*, cordierite + andalusite + biotite + K-feldspar + quartz / corundum in pelites), indicative of emplacement in the upper crust (~ 8 - 10 km).

The plutons are approximately aligned in NE-trending arrays along major strike-slip faults or lineaments. It has been suggested that high-angle NE-SW trending Caledonian shear zones and high-angle NW-SE trending pre-Caledonian lineaments effectively compartmentalized the lower crust into a set of blocks that rotated relative to one another during sinistral strike-slip movement on the Great Glen and Highland Boundary faults. As these blocks rotated, transtensional zones were set up at the intersections of the two sets of lineaments, locally facilitating the ascent of magma in a series of steep sheets. Assembly of the plutons in the upper crust then resulted from rotational ballooning.



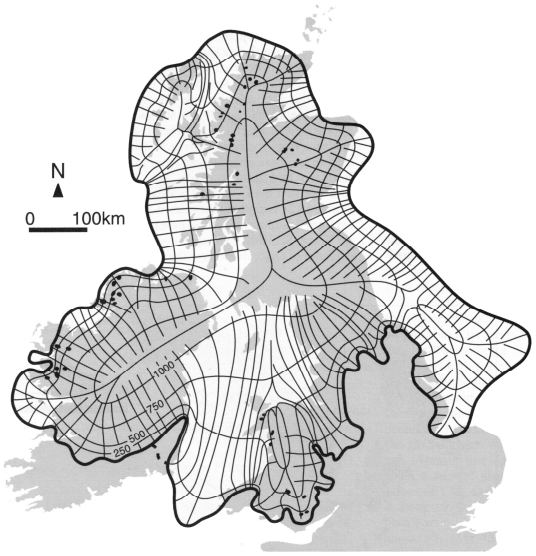



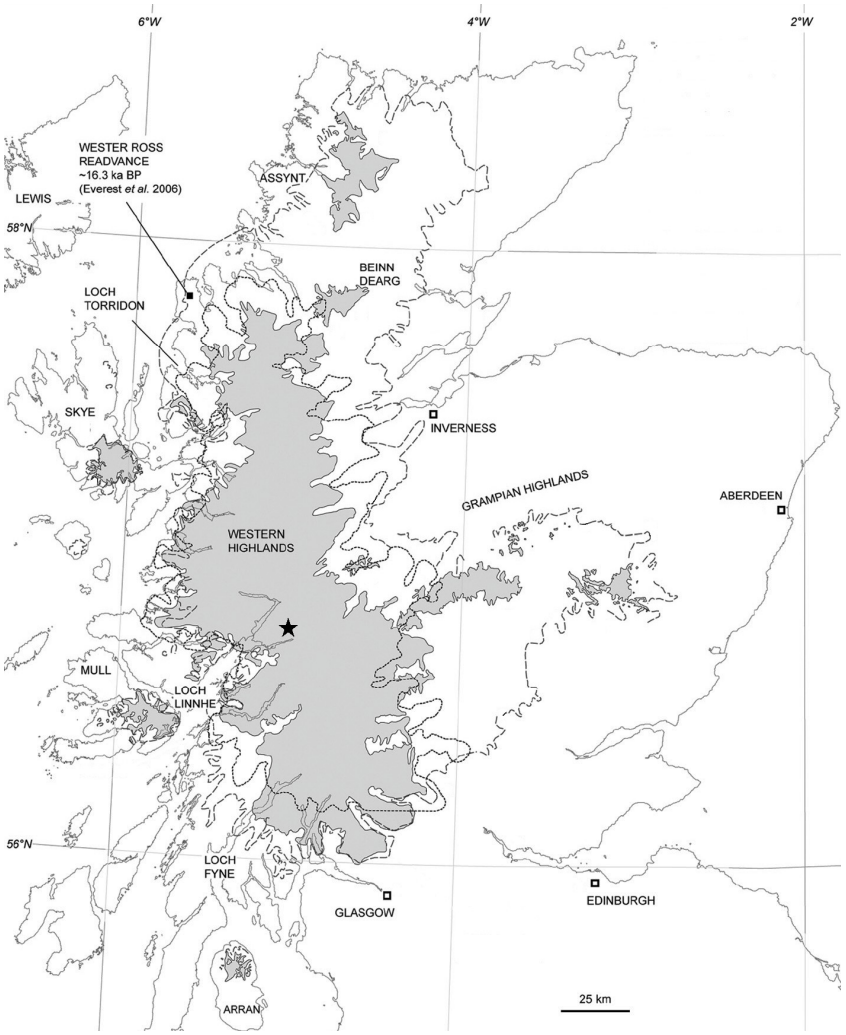

*Top left and right: model for emplacement of the south-west Grampian Highlands Newer Granite plutons (after Jacques & Reavy, 1994, Journal of the Geological Society of London 151:955). Bottom left: the Etive dyke swarm considered to be part of the plumbing of a large volcano that became intruded by the Etive pluton and was subsequently removed by erosion (after Morris & Hutton, 1993, Scottish Journal of Geology 29: 69-72). Note that the dykes are parallel to the Great Glen Fault.*

**Quaternary**

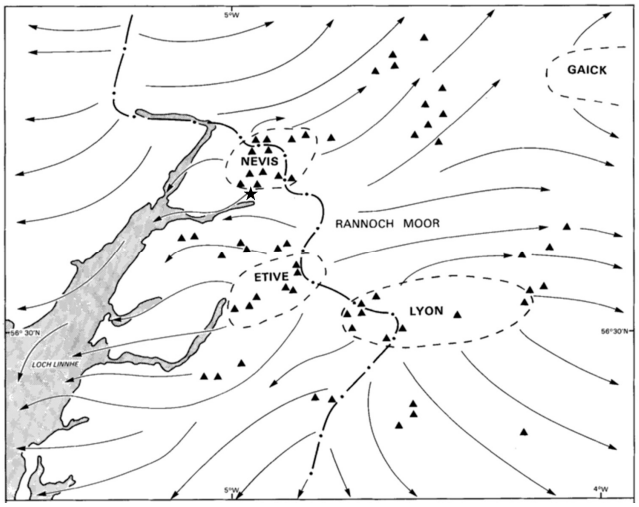
Scotland was heavily glaciated in the Quaternary but little evidence remains on land for the numerous glacial episodes that occurred before the last one. From ~26 ka glaciers formed in the Scottish mountains and amalgamated to form a large ice sheet. Glen Coe and Loch Leven formed major east to west routes for the ice. After 18 ka the ice was in retreat, albeit a retreat punctuated by several re-advances of varying magnitude. There was a sustained warm period (the Windermere interstadial), 14.7 - 12.9 ka, although the ice may not have completely disappeared at this time. This was terminated by a major climatic reverse (the *Loch Lomond stadial* in Scotland but known to the rest of the world as the *Younger Dryas*) when a thick ice cap formed just east of Glen Coe on Rannoch Moor. Ice flowed westwards down Glen Coe and Loch Leven but did not overtop the peaks lining these valleys. This cold period ended abruptly 11.7 ka. Rannoch Moor was ice-free within a few hundred years, and since then slope processes have shaped the landscape.




*Top: maximum extent of the last glacial ice sheet, 22.8 ka, with ice flow lines (after Trewin, 2002).*



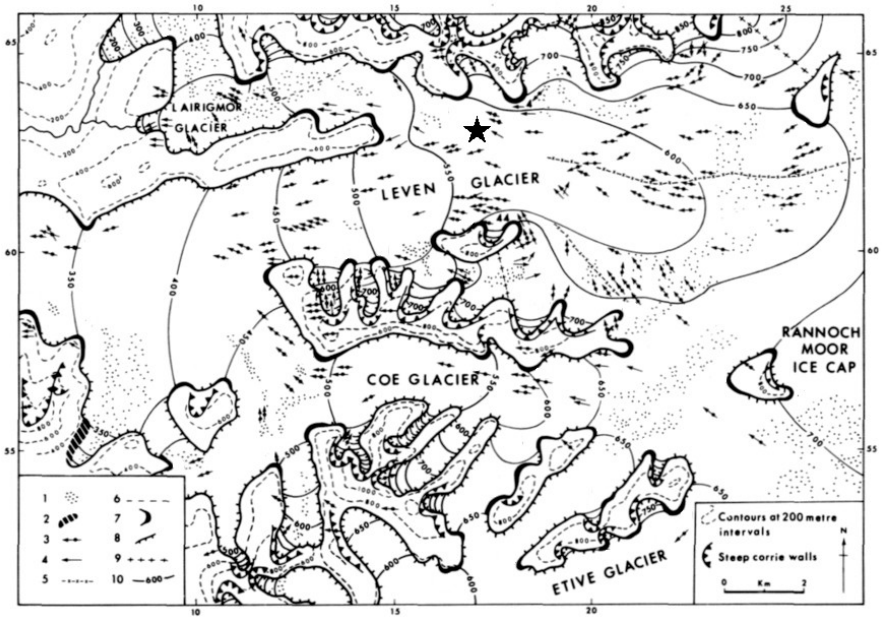

*Left: extent of the Younger Dryas ice sheet, ~ 12 ka. The shaded area is the current best estimate based on field observations; the long-dashed lines give an earlier estimate; the short-dashed lines show the ice limits predicted from an ice flow / climate model. (After Golledge et al., 2008, Quaternary Science Reviews 27: 888-904)*

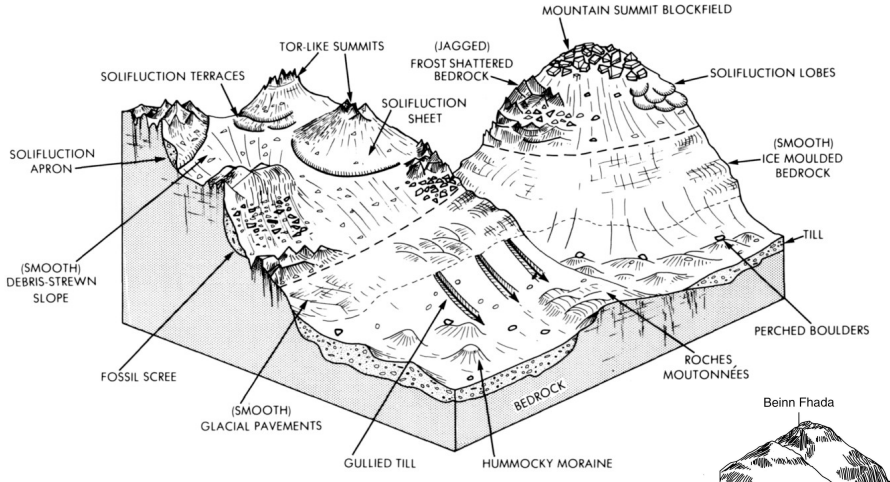



*Left: ice flow directions during the Last Glacial Maximum as estimated primarily from erratics. Note the ice divide running Etive to Nevis. The dashed lines indicate ice domes, the triangles are present-day summits over 1000 m. (After Thorp, 1987, Journal of Quaternary Science 2: 103-112)*

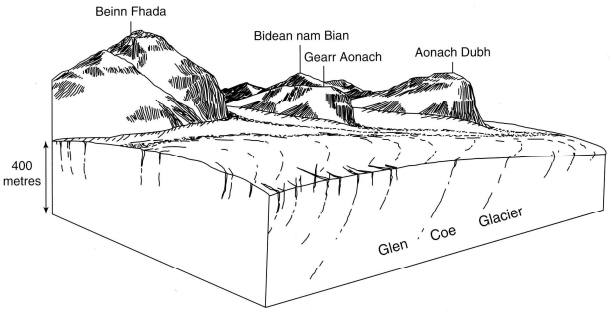
*Below: map showing the Leven and Coe glaciers during the Younger Dryas as mapped from present day landforms (after Thorp, 1981, Scottish Journal of Geology 17: 49-64). The star indicates the location of the Kinlochleven mapping area*

1 *hummocky moraines*
2 *terminal moraines*
3 *striae*
4 *friction cracks*
5 *N and W limits to Rannoch Moor granite (see p.22) erratics*
6 *E limit of Mullach nan Coirean granite (see p.22) erratics*
7 *trim-line*
8 *ice limit*
9 *ice shed*
10 *height of ice surface (50 m intervals)*




*Surface features used to evaluate the position of trim-lines (after Thorp, 1981, Scottish Journal of Geology 17: 49-64)*




*Artist's conception of the Coe Glacier during the Younger Dryas (after Kokelaar & Moore, 2006, Classical Areas of British Geology: Glencoe Caldera Volcano, Scotland, British Geological Survey)*

# SITE-SPECIFIC INFORMATION

**Further reading**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***Kinlochleven***


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- Roberts JL, Treagus J, 1977, The Dalradian rocks of the Loch Leven area. *Scottish Journal of Geology* 13: 165-184.


***Cuil Bay***


- Treagus JE, Treagus SH, 1971, The structures of the Ardsheal peninsula, Argyll, their age and regional significance. *Geological Journal* 7: 335-346.


***Kentallen***


- Pattison DRM, Harte B, 1997, The geology and evolution of the Ballachulish Igneous Complex and aureole. *Scottish Journal of Geology* 33: 1-29.
- Bowes DR, Wright AE, 1967, The explosion-breccia pipes near Kentallen, Scotland, and their geological setting. *Transactions of the Royal Society of Edinburgh* 67: 109-143.

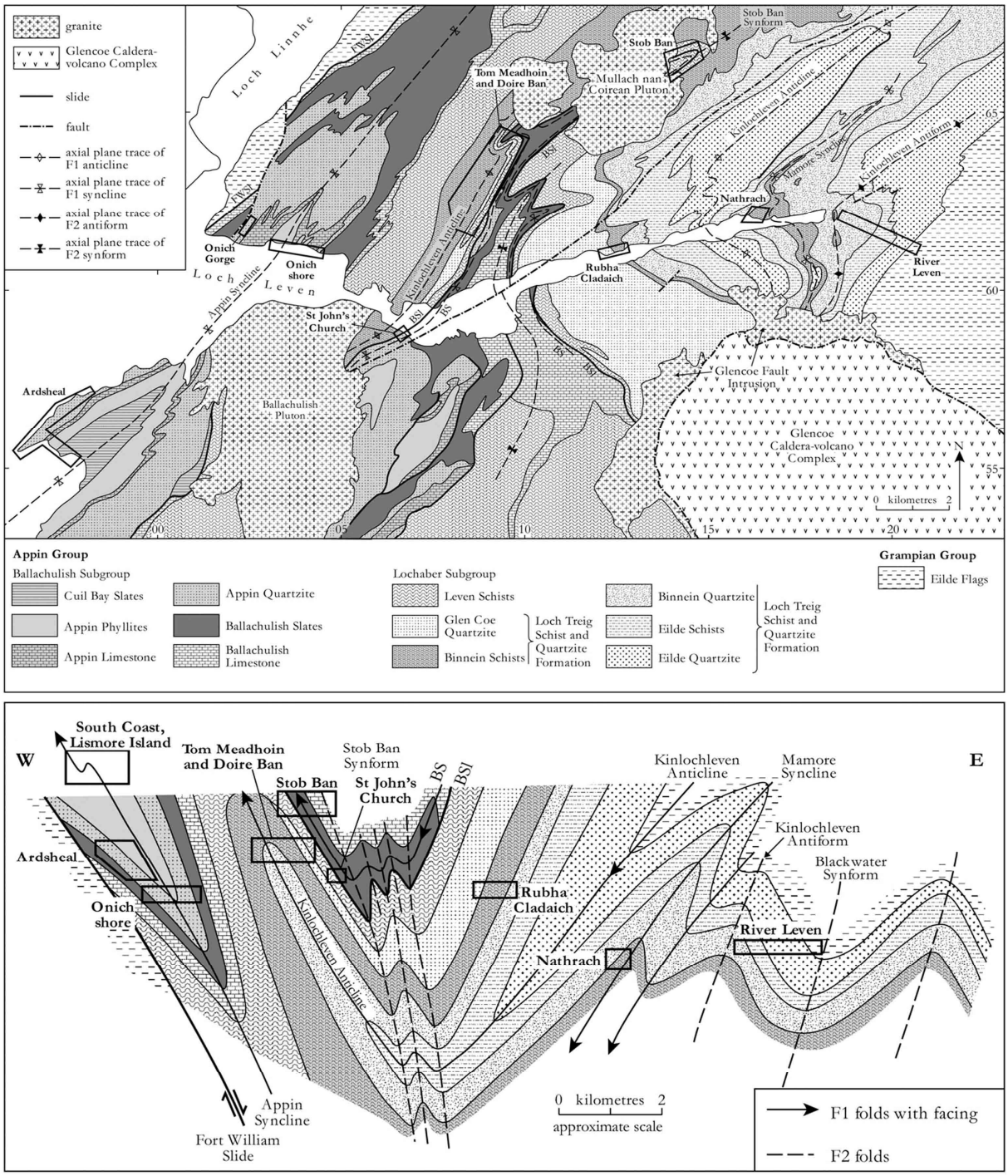

***Glen Coe***


- Kokelaar BP, Moore ID, 2006, *Classical Areas of British Geology: Glencoe Caldera Volcano, Scotland.* British Geological Survey, 127pp.

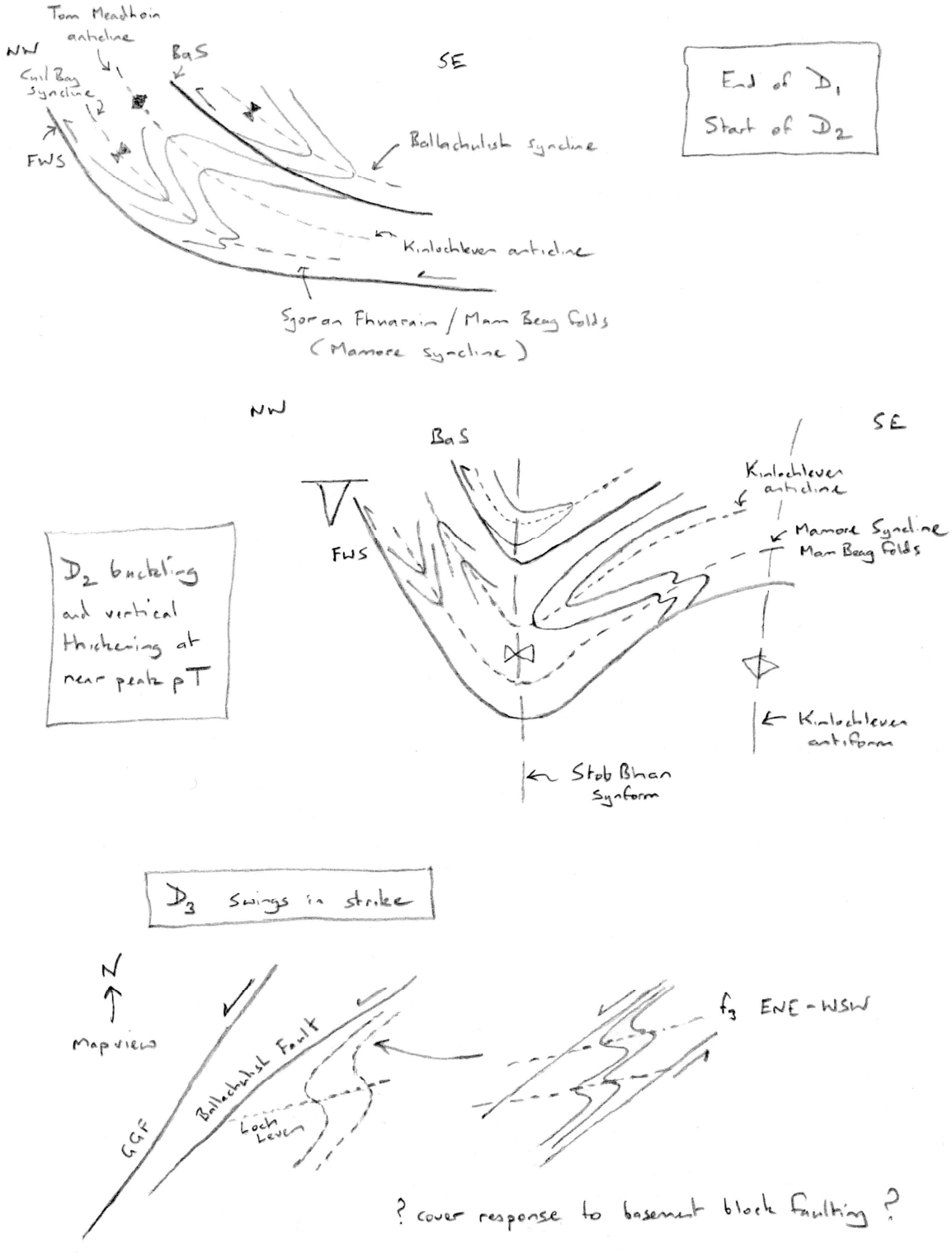
# LOCH LEVEN

The Loch Leven area shows a series of $D_1$ folds that have been folded around upright, approximately NE-SW trending, $D_2$ folds. Superimposed on these structures are two locally developed phases ($D_3$ and $D_4$) of spaced crenulation cleavage and minor fold formation. $D_1$ to $D_4$ all date to the Grampian Orogeny.

At Cuil Bay (Ardsheal) and on the shore section at Kinlocheven we will be examining structures on one limb of essentially the same $D_1$ syncline, but while at Cuil Bay it is a synform, at Kinlochleven it is an antiform because the units are turned upside down by the $D_2$ Stob Bhan synform in between the two localities.




*Geological map and simplified east-west cross-section (along the shores of Loch Leven) through the Dalradian stratigraphy (after Treagus et al., 2013, Proceedings of the Geologists' Association 124: 148-214).*



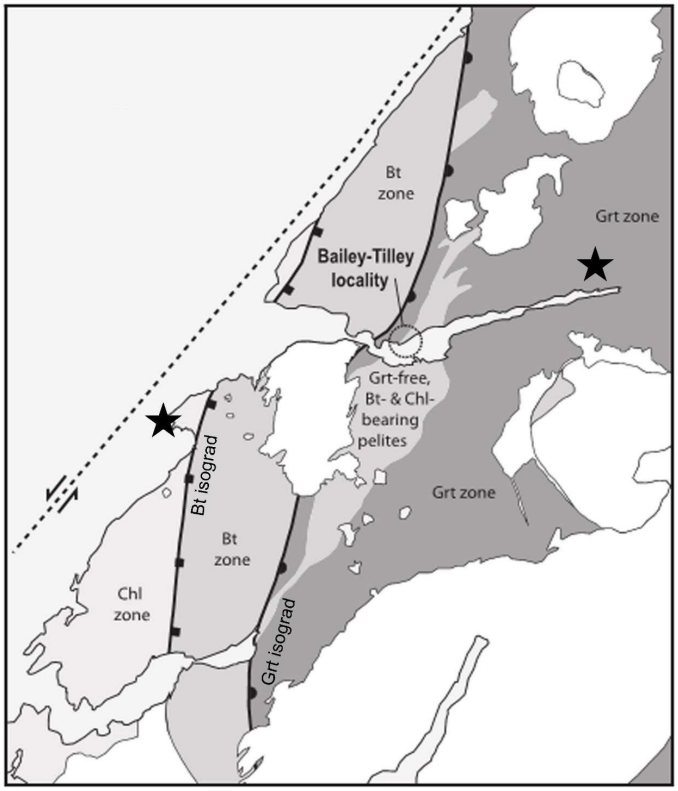

*Notebook sketches of the structures along the full length of Loch Leven as drawn by a student who was attempting to get to grips with the relationship between the $D_1$ and $D_2$ structures. Top two sketches are schematic sections; bottom sketch a schematic map representation. The Kinlochleven mapping area is at the SE end of the sections; the Cuil Bay syncline is at the NW end. Note that at the SE end of the section the $D_2$ folding has turned the Dalradian stratigraphy upside down.*

***Stratigraphy***

The formations in the Kinlochleven and Cuil Bay mapping areas belong to the *Appin Group* of the *Dalradian Supergroup*.

| *Group* | *Subgroup* | *Formation* | |
|---|---|---|---|
| Appin Group | Blair Atholl (base of) | ***Cuil Bay Slates*** | ***Centimetre-scale graded beds of graphitic semi-pelites and pelites with minor dark grey limestones and some more psammitic beds*** |
| | Ballachulish | ***Appin Phyllite*** | ***Centimetre-scale graded and cross-laminated semi-pelites interbedded with pelites, and locally with thin meta-limestones and flaggy quartzites*** |
| | | ***Appin Limestone*** | ***Interbedded dolomitic limestones (up to 10s cm thick), phyllitic semi-pelites, quartzites and psammites*** |
| | | ***Appin Quartzite*** | ***Thick-bedded, massive white quartzites, which locally contain pebble-sized clasts of quartz and feldspar; sometimes poorly graded, and generally cross-bedded*** |
| | | ***Appin Transition*** | ***Sequence of psammites, semi-pelites and graphitic pelites that is transitional between the Ballachulish Slates below and the Appin Quartzite above, and which displays abundant small-scale graded bedding, cross-lamination, and soft-sediment deformation structures*** |
| | | Ballachulish Slate | Pyrite-bearing, black slates and graphitic phyllites with thin semi-pelite beds that exhibit graded bedding and cross-lamination |
| | | Ballachulish Limestone | Grey-green calcareous phyllites, cream and grey dolostones, dark bluish grey limestones, and intercalations of slaty pelite; the phyllites are commonly amphibolitic |
| | Lochaber | Leven Schist | Thick homogeneous sequence of greenish grey phyllites or schists; thin quartzites are present at the base where there is a transitional contact with the underlying Glencoe Quartzite |
| | | Glencoe Quartzite | Very feldspathic; cross-bedded units up to 4 m thick, strongly slumped; lower contact characterized by K-feldspar pebble beds and mud slithers |
| | | ***Binnein Schist*** | ***Occasional carbonate, calc-silicate, and graphitic horizons; quartzite ribs at both upper and lower contacts*** |
| | | ***Binnein Quartzite*** | ***Less feldspar but more muscovite than the other quartzites; characteristic white weathering and rusty ellipsoidal spots*** |
| | | Eilde Schist | Graphitic seams throughout; good graded bedding; quartzite ribs at both upper and lower contacts |
| | | Eilde Quartzite | Similar to Glencoe Quartzite but thinner cross-bedded units and feldspar pebbles far less conspicuous |

***Regional Metamorphism (Grampian orogeny)***




*Map of the Loch Leven area (mapping areas indicated by the stars – Kinlochleven to the east and Cuil Bay to the west) showing the location of the regional metamorphic isograds (after Pattison, 2013, Journal of the Geological Society, London 170: 417-434).*

The phyllites and slates of the Cuil Bay field area contain the chlorite zone assemblage of muscovite + chlorite + quartz + albite + graphite. Just to the east, biotite appears with the biotite forming due to reactions of the kind:

*dolomitic units*

$$musc + dol + qtz + H_2O = bt + chl + cc + CO_2$$

*carbonate-free units* (Tschermak Exchange between coexisting chl, musc, bt)

$$musc + chl = bt + (Fe,Mg)_{-1}Si_{-1}Al_2 + qtz + H_2O$$

The temperatures and pressures of the biotite-in isograd are estimated at ~475°C and ~700 MPa.

The Kinlochleven field area is in the garnet zone with the garnet isograd being given by either of the following reactions:

$$musc + chl + epi + qtz + ilm = gt + bt + H_2O$$
$$musc + chl + epi + ab + ilm = gt + bt + qtz + H_2O$$

The peak metamorphic conditions are estimated to be $T \sim 500°C$, $p \sim 700$ MPa.

A pressure of 700 MPa equates to a depth of ~25 km. Peak metamorphism was at approximately the same time as $D_2$.

***Deformation events recognizable in the Kinlochleven area***

*$D_1$ structures*

$D_1$ is responsible for the principal younging confrontations observed across the mapping area – these help to locate a number of mappable $F_1$ axial traces. $S_1$ is a penetrative micaceous cleavage that is everywhere nearly bedding-parallel. It is invariably present in relatively thick pelitic units, and is easily seen in the cores of $F_2$ folds where $S_2$ clearly crenulates it. However, in thin pelitic units it is overprinted by $S_2$, and in localities where $S_2$ is also nearly bedding-parallel (which is common in the mapping area), careful observation is required to distinguish it from $S_2$. $S_1$ is not generally visible in the more quartzitic lithologies. In thin section $S_1$ appears as a strong preferred orientation of muscovite and biotite and of the enclosed quartz and feldspar grains. Minor $F_1$ folds are recognizable as $F_1$ where they are clearly refolded folds, where the dominant cleavage ($S_2$) transects them, or where $S_2/S_0$ has a different asymmetry to that of "its" fold. It is challenging to make reliable observations of $S_1/S_0$ asymmetry.

*$D_2$ structures*

The mapping area lies between two major $D_2$ folds – the Stob Bhan synform to the west and the Kinlochleven antiform to the east. There are no mappable $F_2$ folds in the main part of the mapping area – indeed, in the mapping area, $D_2$ is essentially just a near-bedding-parallel shearing deformation. This shearing is locally intense, particularly in the pelitic lithologies close to the contacts with more quartzitic units, where it is responsible for small scale $F_2$ folds that have a strongly developed axial planar crenulation cleavage and variable plunge.

$D_2$ is responsible for the dominant cleavage ($S_2$) in the sandy pelites to more quartzitic lithologies in the mapping area. $S_2$ is a closely spaced (typically ~ 1 mm) crenulation or composite fabric (a transposition fabric in zones of high $D_2$ strain) that has the appearance of being formed at peak metamorphic conditions, and which is usually steeply dipping and trending NE-SW to N-S. In thin sections of semi-pelitic lithologies it clearly deforms $S_1$ in intense crenulations with recrystallization of the mica on the $S_2$ planes. In quartzitic lithologies it is a strong grain shape fabric. $S_2$ locally contains a mineral lineation (rodding of quartz and feldspar, elongate rusty spots). Minor $F_2$ folds are recognizable as $F_2$ because $S_2$ is axial planar to them. It is difficult to do much with observations of $S_2/S_0$ asymmetry but it is worth noting whether the strike of $S_2$ is clockwise or anticlockwise to that of $S_1$.

The growth of biotite on $S_2$ and the appearance of porphyroblasts around which $S_2$ is deflected, both indicate that $D_2$ occurred at about the same time as the metamorphic peak. Garnet porphyroblasts occur locally in all pelitic lithologies, and microcline porphyroblasts in the Binnein and Eilde Schists. The porphyroblasts contain inclusion trails (quartz, feldspar, mica) representing the $S_1$ fabric; in the garnet these are strongly curved while in the microcline they are less so, the implication being that the microcline porphyroblasts grew early in $D_2$ while the garnet porphyroblasts continued to grow well into $D_2$.

*$D_3$ structures*

$D_3$ is responsible for ENE-WSW trending, upright folds that cause the main swing in $S_0$ (and $S_1$ and $S_2$) across the mapping area. The cores of $F_3$ folds have a well-developed, spaced (5 - 10 mm) axial planar crenulation ($S_3$) but $S_3$ is generally weakly developed elsewhere. In thin section, $S_3$ is seen to result from the alignment of mica on the limbs of microfolds of $S_1$ or $S_2$ without new mica growth, that is, it postdates peak metamorphic conditions.

*$D_4$ structures*

$D_4$ is represented by a sporadically developed, N-S trending, steeply dipping crenulation ($S_4$). It is spaced with no new mineral growth. There are no major $D_4$ structures.

***Rule of thumb:*** *relatively high metamorphic grade, tightly spaced fabrics and small-scale structures are $D_2$; lower grade, spaced crenulations and more open folds are $D_3$. $D_1$ minor structures are only recognizable locally, although $F_1$ fold axial traces are mappable from younging information.*

## *Deformation events recognizable at Cuil Bay*

As an example of an account of the structural geology of an area, here is one extracted from Treagus *et al.*, 2013, *Proceedings of the Geologists' Association* 124: 148-214 (http://dx.doi.org/10.1016/j.pgeola.2012.07.009) describing the structural geology of the Cuil Bay mapping area.

### 9. Ardsheal Peninsula (NM 961 555–NN 007 579) (J.E. Treagus)

#### 9.1. Introduction

The Ardsheal Peninsula GCR site, on the south-east side of Loch Linnhe, gives a superbly-exposed section across five stratigraphical units in the Appin Group, from the Appin Transition Formation of the Ballachulish Subgroup, up into the Cuil Bay Slates of the Blair Atholl Subgroup. Although the whole outcrop width of the F1 Appin Syncline is represented on the peninsula, the almost continuous clean coast section described here is the on the north-west limb. Minor structures related to the major F1 fold are exceptionally well exposed, and minor structures belonging to the regional D2 phase are superimposed on the early structures in the eastern part of the GCR site.

The area of the GCR site was described in general by Bailey (1960), who was the first to recognize the major structure as an upward-, NW-facing primary syncline. It was subsequently described in detail by Treagus and Treagus (1971), after Bowes and Wright (1967) had given a very different account of the structure in their examination of the setting of the igneous rocks of the area. The latter are discussed in site reports in the *Caledonian igneous rocks of Great Britain* GCR volume (Stephenson et al., 1999).

#### 9.2. Description

The five most instructive areas of exposed Dalradian rocks in this GCR site are situated along the south-west and north-west coasts of the peninsula (Fig. 20, localities 1–5). The youngest formation, the Cuil Bay Slates, is exposed at locality 1, the type locality for this formation on the north-west side of Cuil Bay, between NM 9763 5539 and NM 9724 5544. At the east end of these exposures, on a small peninsula below a fishing croft, centimetre-thick graded beds of grey silt to black mud (now slaty semipelite and pelite) are particularly well seen. These beds are folded into 3 m-wavelength, tight to open, upright folds, which plunge to the south-west, face upwards and verge to the north-west (Fig. 21b). They have an axial-planar slaty cleavage dipping at 50–60° to the south-east and a stretching lineation of elongate chlorite-muscovite stacks, which pitches steeply to the north-east on the cleavage. Elsewhere, the slaty cleavage is strongly affected by a second generation of open folds, particularly well seen on the south-east side of this small peninsula, Rubha Beag. These folds also plunge to the south-west and have a north-west vergence, but they are related to a crenulation cleavage, which dips at 60–70° to the north-west and cross-cuts the earlier folds. The effect of this folding is to rotate the earlier slaty cleavage into flat-lying attitudes and, locally, to refold the earlier fold-set. The first cleavage surfaces are strongly crenulated and kinked; some kinks are due to original refraction through the semipelite and pelite beds, others result from the second deformation and yet others are related to later deformation associated with the intrusion of dykes.

The junction with the next oldest stratigraphical unit, the Appin Phyllite, is not exposed; but this unit occupies the whole of the next small peninsula to the west, Rubha Meadhonach (locality 2). Bedding, which is less easy to see in these rather uniform grey phyllitic semipelites, dips steeply to the south-east and locally exhibits upward-younging graded units and ripple-drift cross-bedding, particularly on the north-west side of the outcrop. The first cleavage dips steeper to the south-east than the bedding, and intersections plunge gently to the south-west. The second cleavage and folds are not strongly developed, but the intersection of the second cleavage with the first plunges at varying angles to the north-east. Kink bands, commonly conjugate, are well developed.

The westernmost subsidiary peninsula of the GCR site, Rubha Mor, exposes a further three stratigraphical units, the Appin Limestone and the Appin Quartzite, as well as the distinctive transitional beds to the Ballachulish Slates (the Appin Transition Formation). The Appin Limestone and its steep SE-dipping junction with the Appin Quartzite are exceptionally well displayed on the south-east coast of Rubha Mor between NM 966 557 and NM 963 556 (locality 3). The youngest beds, on the seaward side, are mixed metacarbonate rocks, quartzites, phyllitic semipelites and psammites (Fig. 21b), followed downwards by a clear sequence of white quartzite, grey metacarbonate rock, cream dolomitic metalimestone and phyllitic semipelite, each a metre or two thick, before the well-exposed junction with the Appin Quartzite is reached high on the cliff. These latter beds are seen in a spectacular fold-pair in a 15 m-long section at the south-west end of their outcrop. Here, disharmonic minor folds and refracting penetrative axial-planar cleavage are beautifully displayed; the vergence of minor folds and cleavage/bedding is to the north-west and the plunge is 20–40° to the south-west (Fig. 21b).

The Appin Quartzite consists of thick-bedded (up to 2 m) white quartzite with quartz and feldspar pebbles up to 20 mm across in a gritty matrix; the clasts are only weakly elongated in a poorly developed cleavage in the finer grained beds. The quartzite commonly displays cross-beds younging to the south-east, which are especially well-seen near the top of the formation at NM 9640 5565 (locality 3) and near the base on the north-west side of the promontory at about NM 968 557 (locality 4). The Appin Transition Formation is seen on the shore of Loch Linnhe from about NM 963 558 to NM 971 565 (locality 5). These steeply SE-dipping beds of striped psammites, semipelites and graphitic pelites display a variety of SE-younging sedimentary structures including small-scale grading, ripple-drift cross-bedding, load structures, slump-folds and sedimentary dykes. Cleavage in the pelitic beds is penetrative. The vergence of minor folds and cleavage/bedding intersection is to the north-west and the plunge is 10–30° to the south-west.

#### 9.3. Interpretation

All of the stratigraphical units that crop out at localities 1–5 exhibit the same north-west vergence of both the earliest folds of bedding and of the cleavage/bedding intersection related to those folds. According to Bailey (1960), Voll (1964) and Treagus and Treagus (1971), they lie on the north-west limb of the regional fold, the Appin Syncline, which all these authors agree is one of the early (D1) fold-set. The hinge-zone and south-east limb of this fold, affecting the Appin Phyllite and Limestone Formation, is best seen, up plunge to the north-east, in the *Onich* GCR sites. A generalized structural profile of the fold as it affects the rocks of the Ardsheal and Onich GCR sites is illustrated in Fig. 18, which shows the five locations discussed above. The later folds and associated crenulation cleavage, which are seen at Ardsheal on the north-west limb of the Appin Syncline, but also on its south-east limb at Onich, show a consistent north-west vergence on both limbs. These folds belong to the regional D2 phase of Treagus and Treagus (1971) and are subsidiary to the major Stob Ban Synform, seen to the east (see the *Tom Meadhoin and Doire Ban* and *Stob Ban* GCR site reports).

The paper by Treagus and Treagus (1971) contradicted the earlier view of Bowes and Wright (1967) that the syncline is a third-phase structure associated with the retrograde metamorphism in the area. Bowes and Wright proposed that 'elongate compositional blebs' and small isoclinal fold noses in semipelites within the Appin Phyllite are evidence of an earlier D1 phase. Mica flakes that cut across these earlier features were said to be attributable to a D2 phase. The lineation resulting

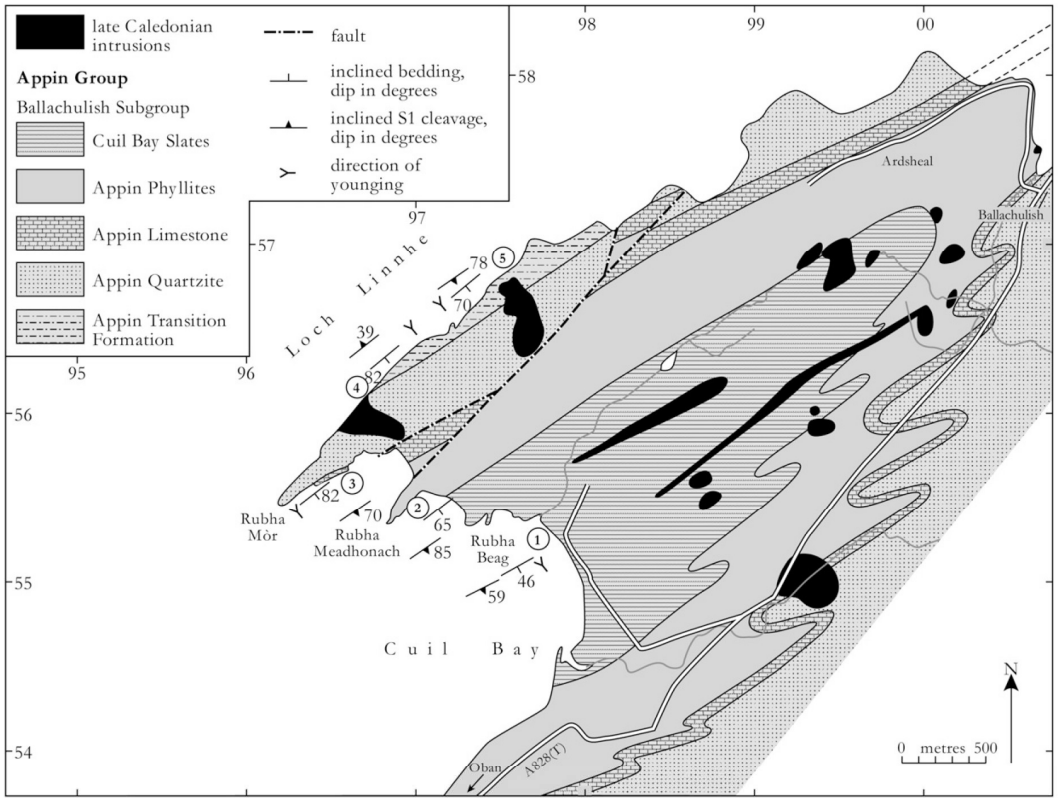



**Fig. 20.** Map of the Appin Syncline on the Ardsheal Peninsula. Numbers relate to localities discussed in the text. After Treagus and Treagus (1971).

from the mica flakes was said to be folded around the Appin Synform, which was therefore interpreted as a D3 phase. No locations nor detailed measurements of these structures were given and no evidence was found by Treagus and Treagus (1971), or has been found since, to support these views. Importantly, thin-section examination of the pelites from the Ardsheal Peninsula GCR site has shown that the penetrative cleavage that is axial planar to the dominant minor folds and to the major syncline was developed as part of the regional, progressive, chlorite-muscovite metamorphism.

### *9.4. Conclusions*

At the Ardsheal Peninsula GCR site it can be demonstrated that the Appin Syncline, one of the major folds in the western part of the Grampian Fold-belt, was produced as part of the earliest deformation of these rocks. The site provides superlative examples of the minor folds produced by this deformation as well as evidence of the effects of a later period of deformation, superimposed on the earlier one. The site also provides excellent clean exposures of five stratigraphical units in the upper part of the Appin Group with clear evidence of the nature of the original sediments; it contains the type locality of one of them – the Cuil Bay Slates. The site is a necessary complement to the GCR sites at Onich to give a complete cross-section of the Appin Syncline. The site is of exceptional importance educationally and is used extensively by undergraduate parties and students undertaking mapping projects.

*Key points:*

$D_1$ produced the main $F_1$ fold which is a synclinal synform plunging about 30°SW, and verging NW. The synform has a penetrative NE-SW striking slaty axial planar cleavage dipping 50-60°SE. This cleavage contains a stretching lineation of elongate chlorite-muscovite porphyroblasts that pitches steeply NE, that is, it is at almost 90° to $F_1$.

$D_2$ is present primarily as a fine-scale crenulation of the first cleavage ($S_2/S_1$) that is particularly visible in the more pelitic lithologies (Appin Phyllites and Cuil Bay Slates). In the Cuil Bay Slates this crenulation is developed into a widely spaced crenulation cleavage ($S_2$) dipping 60-70°NW, and is axial planar to open folds. $F_2$ plunges to SW and verges NW (like $F_1$ here). These structures are subsidiary to the major $D_2$ Stob Bhan synform about 10 km to the east

There are some late kink bands and faults

*If you were to be asked to write a short, illustrated report (2-3 page including figures) of the structural history of the Cuil Bay mapping area to accompany a map and cross-section, it might be arranged something like the following:*

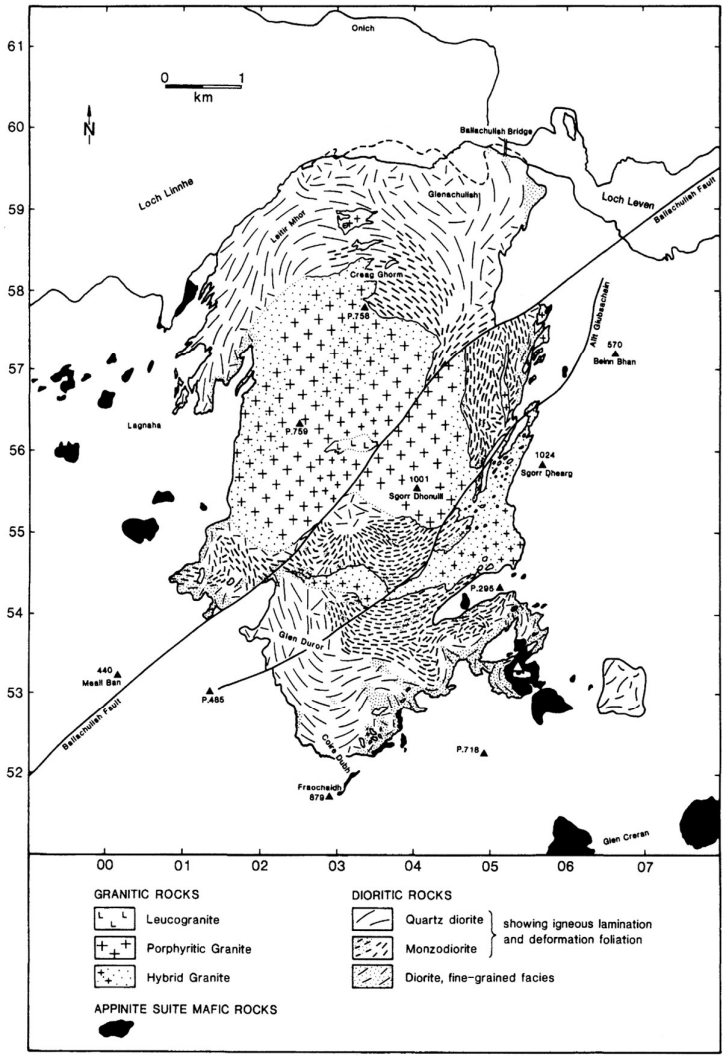
- ***Introductory statement:*** The units at Cuil Bay are exposed within the core and western limb of a 2-3 km wavelength upright $D_1$ synclinal synform that plunges about 30°SW, and verges NW. This has been locally overprinted by minor $D_2$ folds, and subsequently faulted and intruded by three compositionally distinct sets of small igneous bodies
- ***Stratigraphical column:*** with brief lithological descriptions, ideally illustrated by an annotated stratigraphic column. This should include evidence of younging
- ***Geometry of the folding:***
  - $F_1$: quote average $S_1$ and $F_1$ ($S_1/S_0$). Observe that a penetrative $S_1$ cleavage is well-developed in fine-grained lithologies and is evident in all lithologies. Describe $L_1$ and its average orientation. Illustrate all these points with an annotated stereogram
  - $F_2$: quote average $S_2$ and $F_2$ ($S_2/S_0$). Observe that the $F_2$ structures are only weakly developed as minor folds and as a spaced crenulation cleavage in the more pelitic units. Note that $F_2$ is coaxial with $F_1$ but that $F_2$ and $F_1$ are not coplanar. Illustrate with interpreted stereogram

    Illustrate the interrelationships of $F_1$ and $F_2$ using a section (with local stereogram) across the folds in the Cuil Bay Slates at Rudha Beag
- ***Metamorphism:*** describe the mineralogy/textures of the mapping units noting any variations between lithologies → provide an interpretation of the grade and timing of metamorphism [regional lower greenschist facies (chlorite zone), probably syn- or post-$F_2$ because the micas define $S_2$]
- ***Intrusion / faulting / kinking:***
  - ***Faulting:*** faults generally minor (small displacements), and late (cross-cutting $D_1$ and $D_2$ structures). The most significant faults are consistently oriented ~010/V with sinistral strike-slip offsets and unknown dip-slip movement. There are some faults conjugate to this oriented ~080/V with dextral strike-slip offsets
  - ***Intrusions:*** in each case give summary descriptions of mineralogy/textures, and describe the geometry of the igneous bodies
    (a) Appinite body and appinitic/lamprophyric dykes: post-folding (undeformed, contact metamorphism obliterates cleavages) but pre-faulting (intrusions affected by faults)
    (b) Felsitic dykes: unknown but late age
    (c) Doleritic dykes: exploit faults and are therefore later than them
  - ***Kinking:*** conjugate kink bands – describe the geometries, note that these are not congruent with the faults. Illustrate the different orientations of faults and kink bands (map-view will do)
- ***Summary:*** point-by-point outline of the sequence of events

# KENTALLEN

Kentallen Pier lies in the thermal aureole of the Ballachulish Igneous Complex, a few hundred metres west of the contact with the main intrusive body. Here we will examine the contact metamorphic rocks together with a granitic dyke of very similar modal mineralogy to the outer parts of the pluton itself. We will also walk a short distance along the coast to examine a small pluton belonging to the Appinite Suite, a suite of mafic igneous rocks closely associated with the intrusion of the Newer Granites.

**Ballachulish Igneous Complex**




*Geological map of the Ballachulish Igneous Complex (after Pattison & Harte, 1997)*

The Ballachulish Igneous Complex was emplaced 427 ± 1 Ma (U-Pb dating of zircon) making it one of the earliest of the Newer Granites. The complex comprises a zoned monzodiorite-quartz diorite envelope with flow- and deformation-fabrics, that surrounds a core of variably porphyritic granite. The internal contacts between the granite and quartz-diorite are mostly hybrid, and so it is inferred that the granite was emplaced whilst the older quartz-diorite envelope was still partially molten.

The contacts with the surrounding metasediments are mostly sharp, discordant, and steeply outward dipping. The intrusive complex is approximately cylindrical in shape to a depth of ~4 km below the present land surface. The lack of intrusion-related deformation features in the marginal country rocks, together with the presence of large blocks and a high density of xenoliths in the marginal parts of the intrusion, imply that the intrusion was emplaced by stoping and block faulting. From the mineralogical composition of the thermal aureole (see below) the emplacement depth is estimated to be ~10 km.

The major element chemistry identifies the monzodiorites as metaluminous ($Na_2O + K_2O < Al_2O_3 < CaO + Na_2O + K_2O$) I-type granites, the quartz-diorites as transitional to peraluminous ($Al_2O_3 > CaO + Na_2O + K_2O$) crustal-derived magmas, and the granites as S-type granites derived from crustal melting. The rare earth element composition of the monzodiorites and granites is similar implying that the geochemical

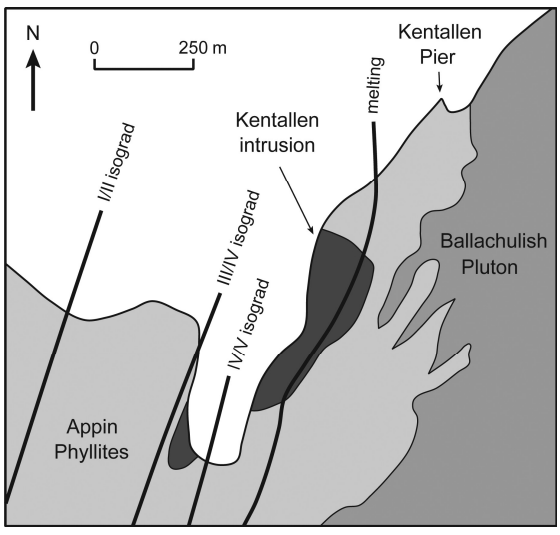
signature of both was imposed by the source region and is not the result of fractional crystallization of one large magma batch – the interpretation is that the magmas contained a significant component derived from melting of lower crustal intermediate granulites or gneisses. The monzodiorites and quartz diorites have a greater range of differentiation index compared with the granites, and this has been interpreted as indicating that the former underwent greater differentiation / fractionation whereas the latter crystallized from a crystal mush with limited separation of cumulate phases or intercumulus liquids.

***Intrusion story***

1. Dry, alkaline, convecting monzodioritic magma was emplaced at a depth of ~ 10 km at ~ 1100°C to form a small elliptical stock. During cooling and crystallization, a sub-vertical crystal mush zone slowly advanced from the margins inwards. It was this phase of the intrusion that was primarily responsible for the contact metamorphism
2. In response to crystal fractionation, flow differentiation, and assimilation of country rock, a quartz-diorite melt evolved from the monzodioritic magma at 1050°- 900°C. Ongoing regional faulting and block stoping may have led to enlargement of the pluton. Continued fractionation and crystallization led to the accumulation of a residual rhyodacitic magma in the centre of the intrusion
3. At 900°- 850°C a granite magma was emplaced. When this reached its buoyancy level, the granite diapir expanded and mixed with the partially consolidated quartz-diorite envelope magmas. With further cooling, hydrous fluids were released at 700°- 680°C. Consolidation proceeded inwards leaving a residual, hydrous leucogranite melt, which crystallized rapidly to fine grain size when it suddenly lost volatiles. The thermal event (time above ~ 550°C) lasted ~ 500 kyr

**Ballachulish thermal aureole**

The intrusion of the Ballachulish igneous complex led to extensive overprinting of the regional mineral assemblages by the contact metamorphic mineral assemblages. In the field the most obvious outermost effect of the metamorphism is the appearance of cordierite as spots in the regional pelitic schists. Texturally, as one moves upgrade from the first appearance of these cordierite spots towards the igneous contact, the pelitic rocks gradually change from fine-grained schists and phyllites to cordierite-rich hornfelses. A sequence of contact metamorphic zones / isograds may be defined, although this is sensitive to the bulk composition of the pelites and, in particular, to the presence or absence of graphite (which influenced the metamorphic fluid composition).




*Position of the isograds near Kentallen (after Pattison & Harte, 1997)*

Zone I ms + qtz + chl ± bt ± gt

I/II isograd ms + chl + qtz = crd + bt + $H_2O$
*(appearance of crd)*

Zone II ms + crd + chl + bt + qtz

II/III isograd elimination of chl by I/II reaction
*(disappearance of chl)*

Zone III ms + crd + bt + qtz

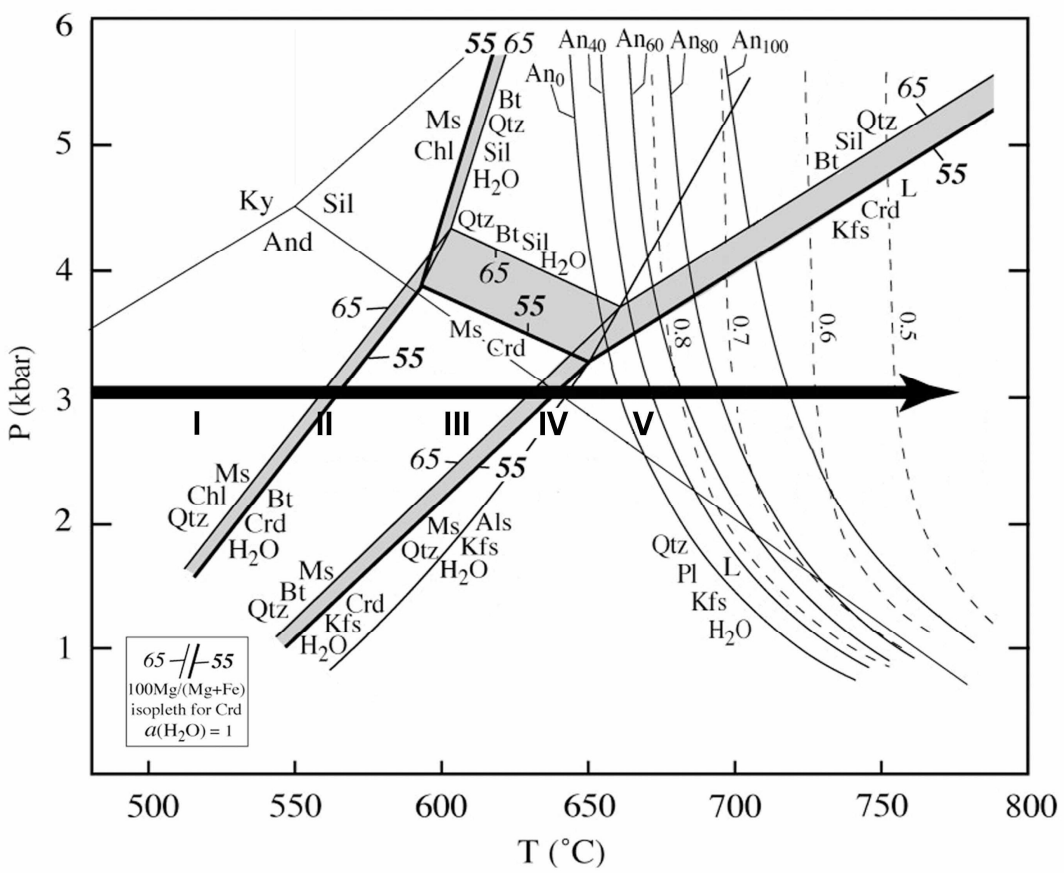
III/IV isograd ms + bt + qtz = crd + ksp + $H_2O$
*(appearance of ksp)*

Zone IV ms + crd + ksp + bt + qtz

IV/V isograd ms + qtz = and + ksp + qtz
*(appearance of and)*

Zone V crd + bt + and + ksp + qtz ± sill

Partial melting migmatitic / quartzofeldspathic veins

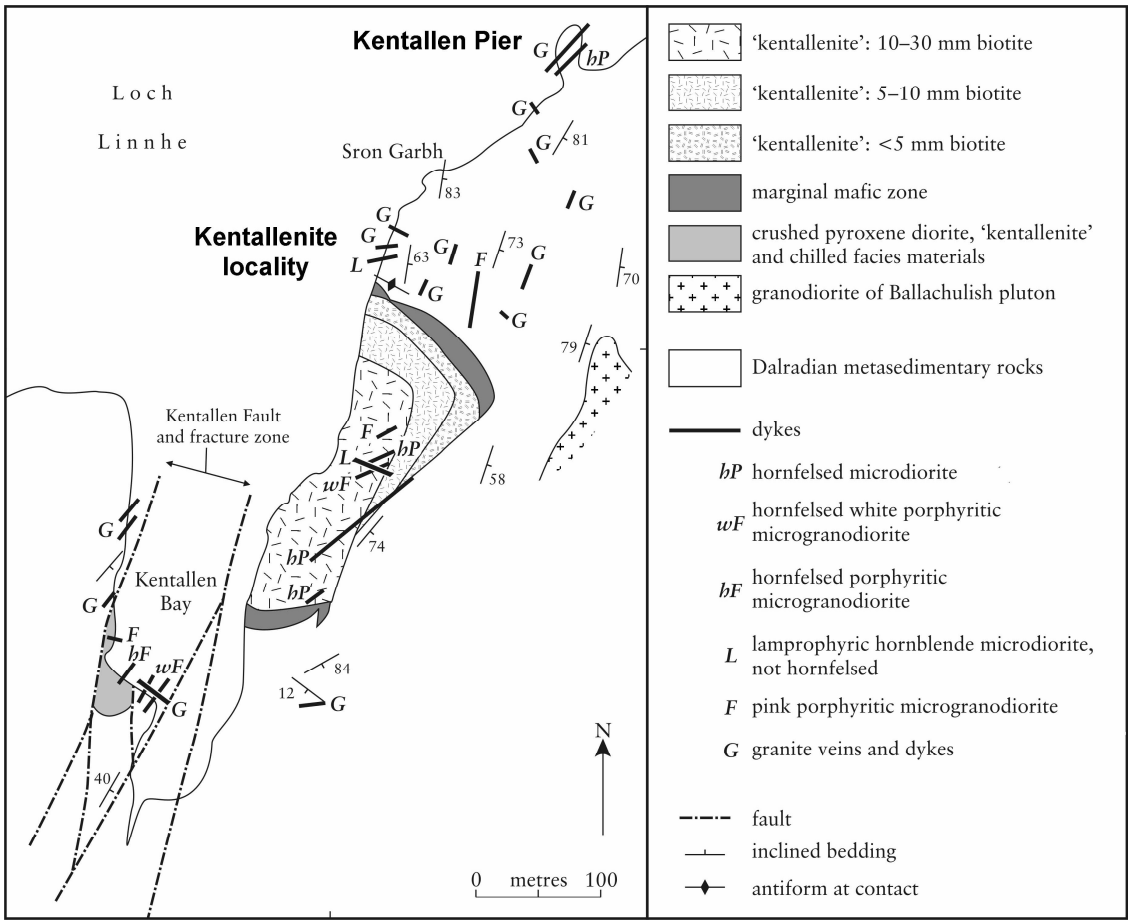



*Petrogenetic grid for the pelites at Kentallen Pier (KFMASH system, $a_{H2O}$ = 1) calculated for $X_{Mg}$ cordierite = 0.55 and 0.65, showing melting for different plagioclase compositions (solid lines) and different $H_2O$ activities. The black arrow shows the pressure of metamorphism implied by the succession of mineral assemblages observed in the aureole at Kentallen. A pressure of 3 kbar = 300 MPa is equivalent to a depth of ~ 10 km. (After Pattison & Harte, 1997)*

**Some tasks for the visit to Kentallen Pier (NN0134 5841)**

1. Draw an oriented sketch map in your notebook showing the location of the pier, igneous intrusion, and position of the more psammitic and more pelitic parts of the Appin Phyllite Formation
2. Examine the psammites, noting the distinct bedding. Look for sedimentary structures that could indicate the younging direction. Briefly describe these units and measure the attitude of the bedding.
3. Examine the pelites, noting their texture. Look for the presence or absence of tectonic fabrics (*e.g.*, cleavage). In addition to biotite, K-feldspar, andalusite (not visible), and quartz, these rocks contain a mineral that appears as ovoid pits on weathered surfaces – what is it? Briefly describe these units. If the tide is low, look for minor folds, note their asymmetry and measure the orientation of an axial plane and fold axis.
4. Note the thin (1 - 3 mm) cross-cutting veinlets of pale quartzo-feldspathic material – what are these, and how did they form?
5. Using the petrogenetic grid above, put limits on the pressure-temperature conditions of contact metamorphism.
6. Note the 1 m thick igneous intrusion. Fully describe its geometry and contact relationships. Identify the lithology using the QAPF classification scheme. Look for xenoliths within it and describe these. Account for the presence of the intrusion at this locality.

**The Kentallen intrusion**




*Map of the Kentallen intrusion (after Platten, 1999, In: Stephenson et al. (editors), Caledonian Igneous Rocks of Great Britain, Geological Conservation Review Series 17, Chapter 8, Site: Kentallen)*

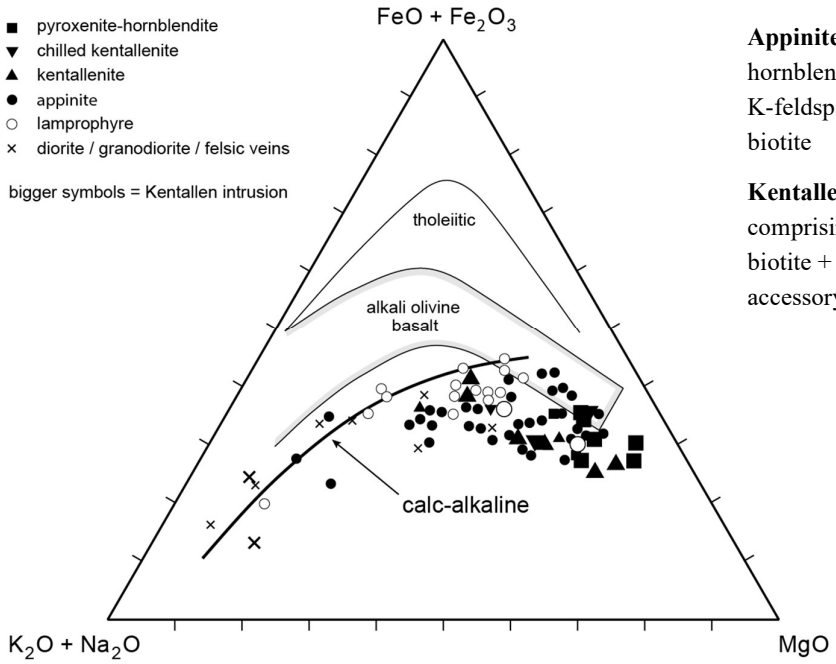
Associated with the Ballachulish Igneous Complex are several small intrusive complexes, collectively known as the Appinite Suite. These comprise ultramafic (peridotite, pyroxenite, hornblendite), mafic to intermediate (olivine monzonite, biotite- and pyroxene-diorite) and subordinate felsic (granodiorite) bodies in association with explosion breccias. The lithologies are texturally and mineralogically highly variable.

Geochemically, the Appinite Suite lithologies have high K and volatile ($H_2O$, $CO_2$, S, Cl) contents, and are enriched in Mg, Ni, Cr, Co, Ba, Sr, and LREE. On chemical variation diagrams, the lithologies clearly lie on a differentiation trend from a primary K-rich, calc-alkaline basaltic magma.

They are interpreted as a suite of plutonic rocks emplaced in sub-volcanic pipes within the upper crust under conditions of high vapour pressures. Their textural variability is attributed to crystallization under highly variable fluid pressures that reflect differences

a) in the fluid pressure required to breach obstacles during the ascent of the magmas,
b) the amount of crystallization that had occurred at the time of breaching,
c) the rate of gas release when breaching occurred

Hence some members of the suite are cumulates, some are the crystallization products of residual liquids, and some are mixtures produced during explosive emplacement.



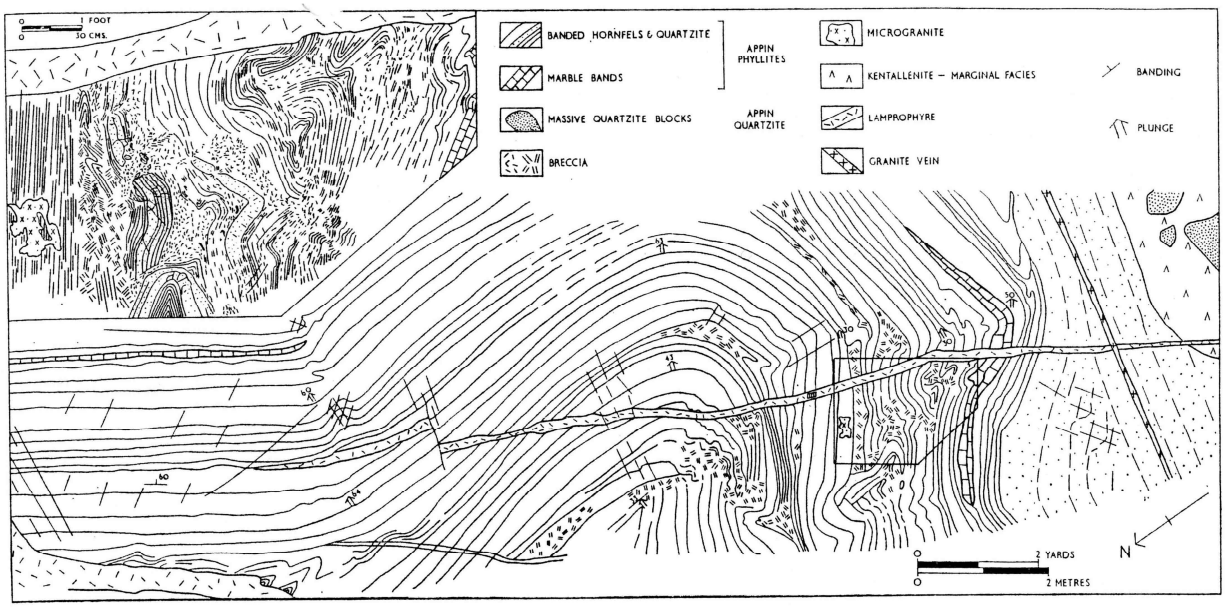

**Appinite** = hornblende diorite comprising hornblende + plagioclase with interstitial K-feldspar and quartz, and accessory biotite

**Kentallenite** = olivine monzonite comprising olivine + clinopyroxene + biotite + plagioclase + K-feldspar, and accessory apatite

*AFM plot of compositions of appinite suite rocks (after Hamidullah & Bowes, 1987, Acta Universitas Carolinae Geologica 4: 295-396)*

At Ballachulish, intrusions of the Appinite Suite pre-date the Ballachulish Igneous Complex, although the presence of a hybrid transitional zone between an appinite body and quartz-diorites on the south-eastern margin of the complex suggests that they were emplaced only shortly before the main igneous complex. The intrusion at Kentallen forms the type-locality of kentallenite, a less hydrous sub-type of appinite which comprises olivine, augite, phlogopite, K-feldspar, and andesine. On the adjacent Ardsheal peninsula (including in the Cuil Bay mapping area), the appinites are hornblende diorites comprising green-brown, euhedral hornblende, strongly zoned, tabular plagioclase, and interstitial K-feldspar and quartz in granophyric intergrowths, with accessory biotite.

**Intrusion related deformation, northern contact of the Kentallen intrusion**




*Intrusion-induced deformation in the country rock (Appin Phyllites) at the northern contact of the Kentallen intrusion (after Bowes & Wright, 1967, Transactions of the Royal Society of Edinburgh 67: 109-143)*

**Textural and mineralogical variability of the intrusion at Kentallen**

***The main body of the intrusion***

The interior of the Kentallen intrusion comprises a melanocratic olivine monzonite ('kentallenite'). The olivine and clinopyroxene crystals are typically 2 - 8 mm but rarely the olivine crystals may be up to 20 mm. The crystals are euhedral to subhedral and form a densely packed, nearly crystal-supported fabric. The olivine gives the impression of being corroded, and it is densely charged with magnetite. Plagioclase occurs as small interstitial laths. Phlogopite (bronze in hand specimen) forms large (10 - 30 mm) crystals that poikilitically enclose the plagioclase and mafics. K-feldspar either overgrows the plagioclase or forms poikilitic crystals. Apatite is a common accessory.

Towards the margin, the phlogopite decreases in grain size (> 10 mm to < 1 mm) and changes from poikilitic to interstitial. The olivine becomes less abundant, the olivine and pyroxene crystals are less densely packed, and the colour index decreases. Some enclaves of a porphyritic kentallenite facies with very fine-grained matrix are present close to the outer margin.

*Kentallenite in thin section: olivine (close-stipple with magnetite), augite (light stipple), phlogopite (showing parallel cleavage), plagioclase (in lighter outline with traces of the twins), K-feldspar (blank), and euhedral apatites (from Hatch et al., 1972, Petrology of the Igneous Rocks (13th edition), Allen & Unwin).*

**Early contact facies** (moving into the main intrusion):

(a) very fine-grained porphyritic kentallenite with olivine and clinopyroxene phenocrysts;

(b) pyroxenite = clinopyroxene-phlogopite-plagioclase rock that is coarser grained (5 - 10 mm pyroxenes) than most of the intrusion;

(c) olivine appears abruptly so that (b) becomes a phlogopite-clinopyroxene bearing peridotite (mica forms a sub-poikilitic framework and shows evidence of cataclasis and kinking)

(d) pyroxene-olivine diorite with millimetre-scale vertical layering at the contact with the chilled margin of the main body

**Marginal intrusions** (between country rock and early contact facies):

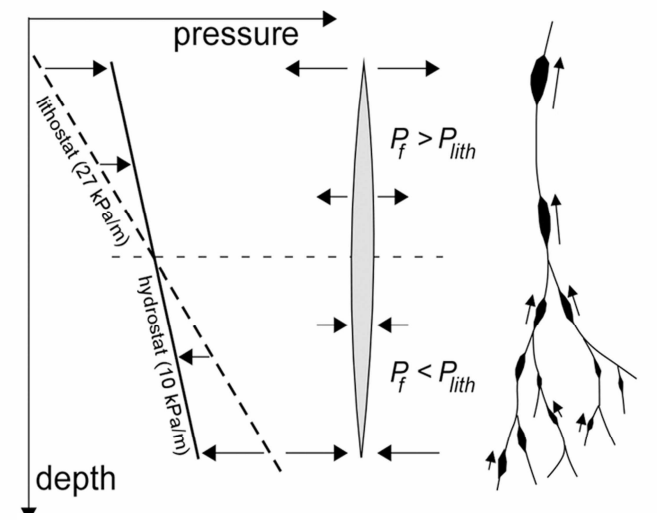
(a) medium-grained, pyroxene-hornblende-biotite diorite with xenoliths of Appin Quartzite

(b) pyroxene-biotite diorite with 5 mm pyroxenes and a rounded xenolith of very fine-grained porphyritic kentallenite

***Intrusion story***

1. Formation of a volatile-rich, high-K, calc-alkaline basaltic magma by small degree partial melting of phlogopite-peridotite (metasomatized subcontinental lithospheric mantle)
2. Crystallization of olivine at ~1800°C and 25 kbar (70 - 80 km) but resorption of these crystals during ascent of the magmas; rapid magma ascent was facilitated by the deeply penetrating Great Glen Fault
3. Explosive emplacement of magmas at high crustal levels:
   → formation of the chilled porphyritic kentallenite of the early contact facies (this porphyritic kentallenite is representative of the primary magma)
   → wall crystallization from the convecting magma produced the pyroxenite and peridotite cumulates of the early contact facies; as the olivine, clinopyroxene and some phlogopite crystallized (1125°- 1000°C), fluid pressures increased leading to crystallization (960°- 770°C) of amphibole and further phlogopite; cataclasis of the phlogopite in these cumulates indicates internal deformation whilst these aggregates contained interstitial fluids and were only supported on one side (the wall)
   → temporary cessation of the convection allowed the crystallization of the pyroxene-olivine diorite of the early contact facies
4. A new batch of more mafic magma was introduced into the pipe, displacing the existing magmas and triggering partial collapse and removal of the wall cumulates and chilled rocks:
   → emplacement of the marginal intrusions may have occurred during the collapse of the wall cumulates
   → the main magma body quenched against the walls of the pipe, trapping enclaves of the porphyritic kentallenite and forming the relatively fine-grained outer parts of the main intrusion
   → settlement and accumulation of olivine and clinopyroxene phenocrysts in the centre of the pipe and crystallization of interstitial liquids led to the formation of typical kentallenite
5. Compaction of the main intrusion expelled some interstitial residual liquids to form biotite-feldspar veins and lamprophyre dykes
6. Subsequent gas streaming led to hydration and carbonation of some of the products of crystallization

**Rapid ascent of fluids/melts (mobile hydrofracturing)**




*Principles behind mobile hydrofracturing (after Bons et al., 2012, Journal of Structural Geology 43: 33-62)*

Rapid ascent of fluids from great depths may be achieved by mobile hydrofracturing. When a fracture is filled with stagnant fluid the vertical fluid pressure gradient ( $p_{fluid} = \rho_{fluid} g z$ ) is ~ 10 kPa/m (for $H_2O$) while the lithostatic pressure gradient in the rock ($p_{lith} = \rho_{rock} g z$ ) is ~ 27 kPa/m. Hence if the fluid and lithostatic pressure are in equilibrium at one level in the fracture then the fluid is over-pressured above this level, and under-pressured below it. Above a critical height (~ 10s of metres) the rock cannot sustain the over-pressure and the crack propagates upwards closing its lower end as it does so – the fracture moves with its containing fluid. The propagation rate is rapid (~ 1 $ms^{-1}$ ~ 11 km $hr^{-1}$). The fractures move preferentially up pre-existing planes of weakness, merge, and then ascend as even bigger and faster fluid batches. In this way ascent of volatile-rich magmas across the full crustal thickness is achieved in a few hours without the need to invoke large fluid fluxes.

**On lamprophyres**

'Lamprophyre' is a field term for hypabyssal rocks of wide-ranging composition but which are usually mesocratic or melanocratic (occasionally ultramafic) and comprise phenocrysts of biotite and/or amphibole in a groundmass of feldspars and/or feldspathoids. Those which are geochemically calc-alkaline (typically, containing no feldspathoids) are classified on the basis of whether they are biotite- or amphibole-bearing and on the relative proportions of plagioclase and K-feldspar:

| | *biotite* | *hornblende* |
|---|---|---|
| *orthoclase > plagioclase* | minette | vogesite |
| *plagioclase > orthoclase* | kersantite | spessartite |

Appinites are therefore essentially plutonic equivalents of vogesites and spessartites.

Post-orogenic (415 - 395 Ma) lamprophyric dykes and sheets occur across Scotland in sufficient numbers for them to account for 10 - 20% of the total Caledonian magmatism. They have two distinct chemistries: those in the Northern Highlands terrane north of the Great Glen Fault are predominantly minettes, and those in the Central Highlands terrane south of the Great Glen Fault are predominantly spessartites.

Calc-alkaline lamprophyres have primitive compositions (*i.e.*, the magmas came to the surface rapidly with little assimilation). The hydrous nature of the rocks reflects volatile retention via crystallization at high fluid pressures. Their rare earth chemistry suggests that the magmas originated at depths of over 100 km, and so the presence of lamprophyres with two different chemistries within a few kilometres on either side of the Great Glen Fault implies that this fault is near vertical and extends deep into the sub-continental lithospheric mantle.

On the field course we may encounter some Permo-Carboniferous lamprophyres. These are typically alkaline lamprophyres (containing feldspathoids and/or alkali-rich glass). Two of the most common lithologies among these lamprophyres are ***camptonites*** (feldspar > feldspathoid with the feldspars being predominantly plagioclase) and ***monchiquites*** (feldspar-free with a groundmass that may be glassy). In both of these types, the ferromagnesian minerals are typically sodic, Ti-rich amphiboles, Ti-augite, olivine, and biotite.

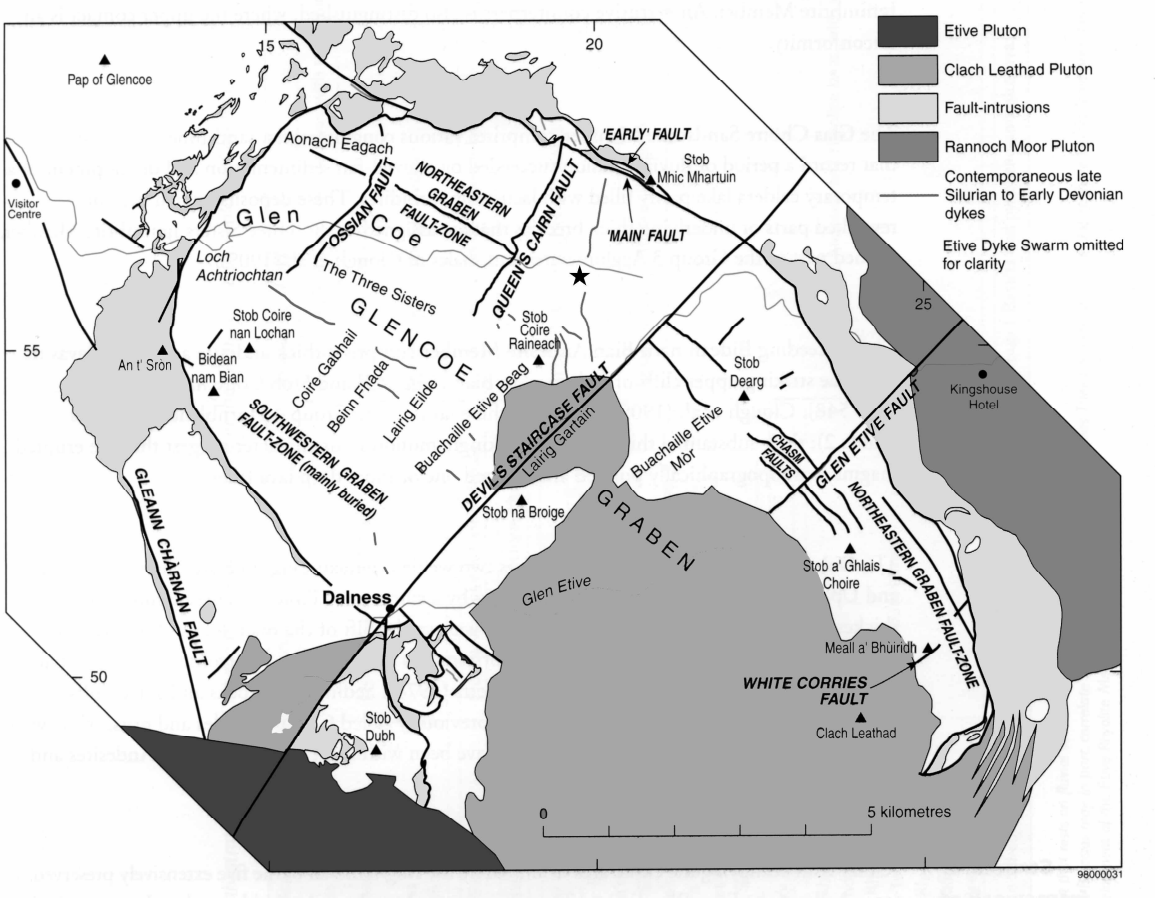
**Some tasks for the visit to the northern contact of the Kentallen intrusion (NN0106 5808)**

1. Examine the country rock lithologies (to the north of the contact). See if you can identify the features illustrated in the sketch on p.39 – the folding, country rock brecciation, and cross-cutting dykes. Seek evidence for timing relationships.
2. Examine the intrusive rocks (to the south of the contact). You should be able to distinguish, on the basis of texture and mineralogy, the various lithologies described above, including "normal" kentallenite, porphyritic kentallenite, fine-grained kentallenite, biotite-pyroxenite, and biotite-gabbro.
3. Draw an oriented, full page sketch map in your notebook of the section, spanning 15 - 20 m either side of the contact, to show the field relations. Include measurements of the strike/dip of bedding in the country rock and the strike/dip of any igneous contacts, and show cross-cutting relationships.
4. Establish whether or not you agree with the following sequence of events:
   (a) explosive brecciation of country rocks; (b) emplacement of mafic-rich, volatile-rich marginal units; (c) intrusion of the main kentallenite body; (d) intrusion of lamprophyres; (e) intrusion of granodioritic offshoots of the Ballachulish pluton and thermal metamorphism

# GLEN COE

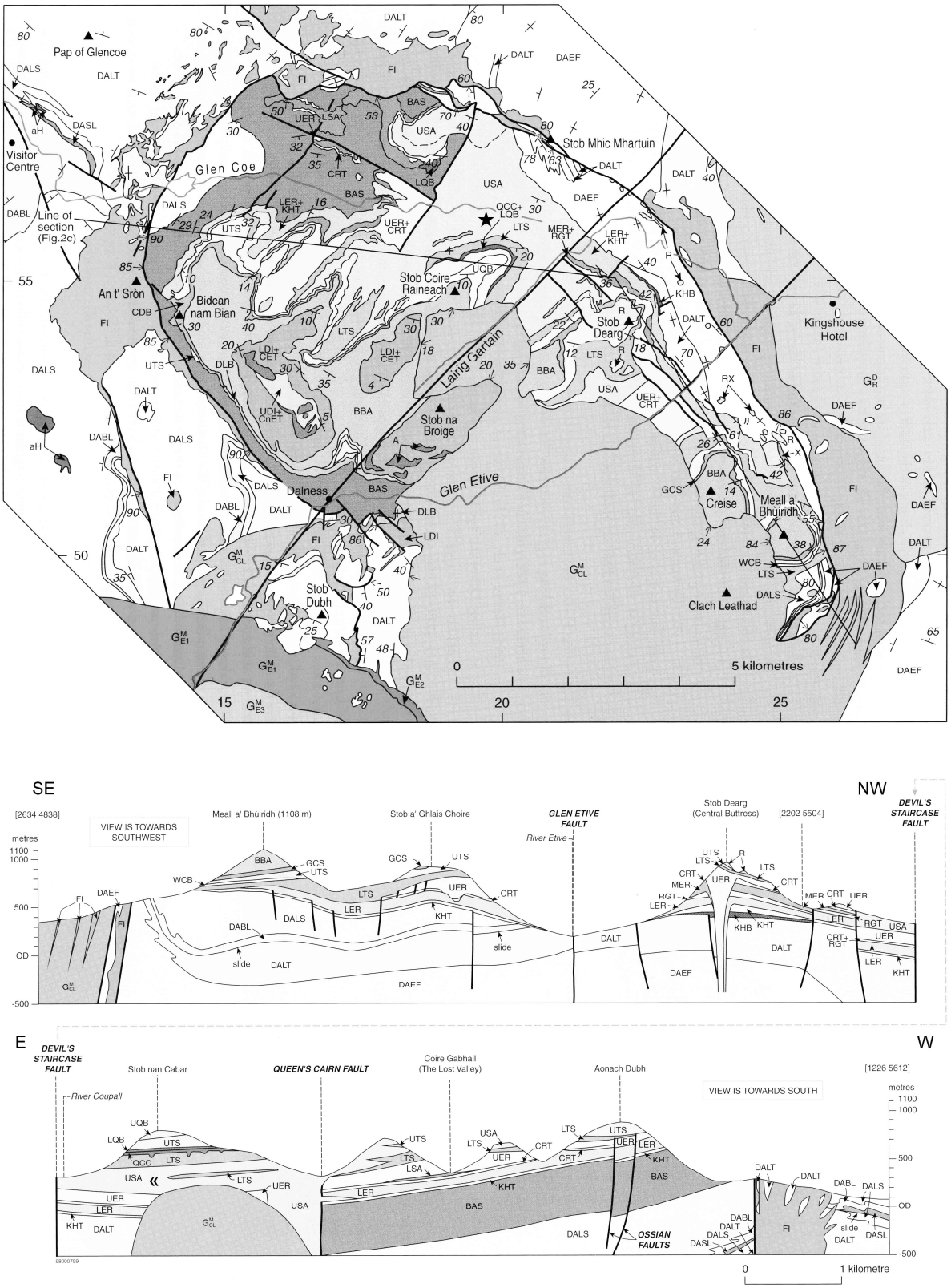
In the Glen Coe area the roots of a large Silurian-Devonian volcano are exposed. The geology is primarily an intra-caldera succession of silicic pyroclastic rocks and intermediate / silicic lavas with intercalated alluvial deposits, which overlies Dalradian metasedimentary rocks and pre-caldera andesitic sills. The succession is intruded by a wide variety of igneous rocks.

Most of the intra-caldera succession comprises seven thick ignimbrites which represent major caldera-forming explosive eruptions that were associated with volcano-tectonic subsidence and caldera collapse. The subsidence was piecemeal, involving movement on numerous cross-cutting faults, many of which tapped the underlying magma chambers. After the early stages of caldera formation, a ring-fault system developed, a large part of the caldera succession foundered, and voluminous silicic magmas (now comprising the Clach Leathad Pluton) were emplaced.

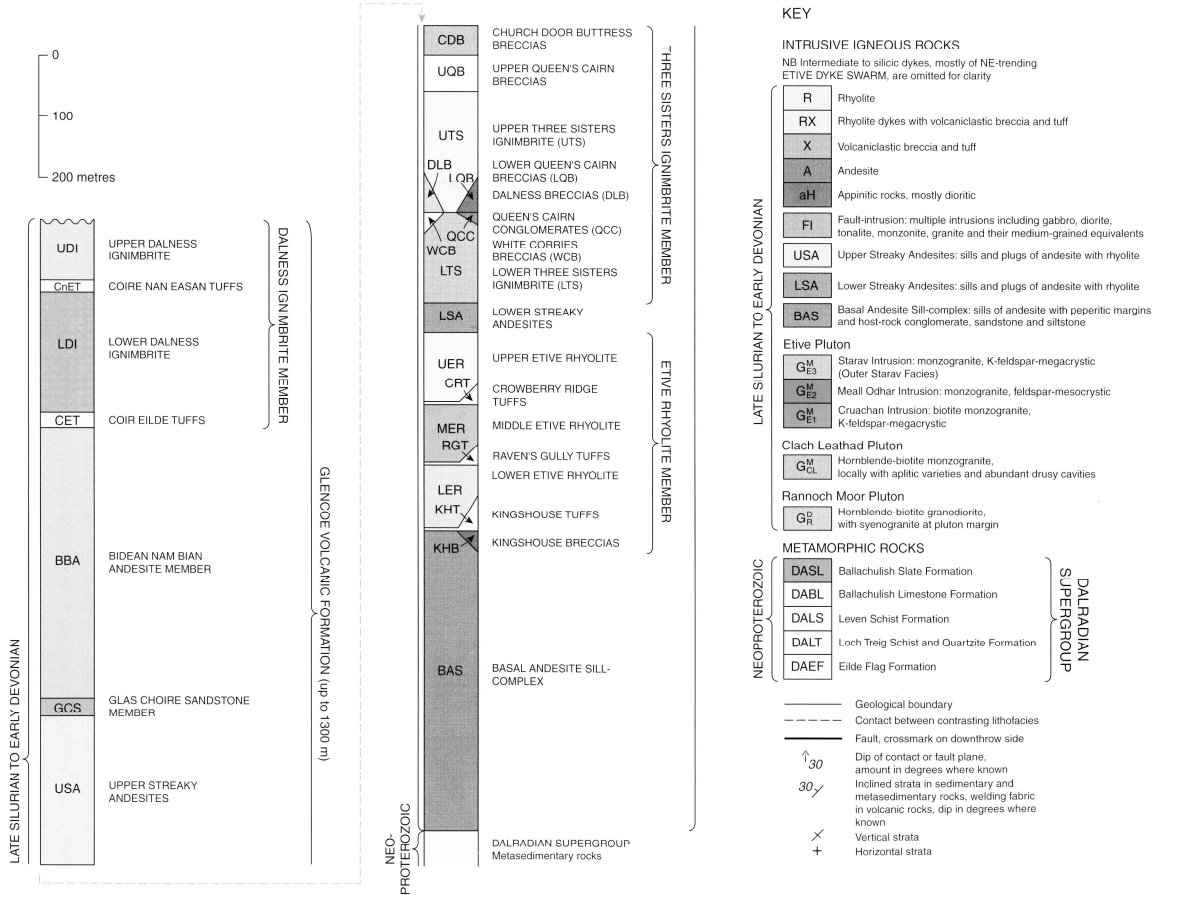



*Principal structural features of the Glen Coe caldera-volcano complex; Etive Dyke swarm not shown (after Kokelaar & Moore, 2006). The star indicates the position of the compass-and-pace exercise*

**The stratigraphy of the Glen Coe succession**




*Geological map (top), cross-section along the indicated line of section (bottom) and stratigraphic column (facing) of the units associated with the Glen Coe caldera (after Kokelaar & Moore, 2006)*




***Basal Andesite Sill Complex:*** thick sequence of basaltic to andesitic sills with intercalated sedimentary strata. Peperites (coarse to fine breccias admixed with vesicular sedimentary rocks) occur along the top surfaces of these sheets indicating that the sills were intruded into unlithified wet deposits. These units, together with a cover of lavas, were extensively eroded by rivers before the onset of the magmatism that formed the caldera.

***Etive Rhyolite Member:*** predominantly comprises three flow-laminated "rhyolites" formed in three discrete explosive eruptions, each associated with caldera subsidence. The lava-like appearance of the rhyolites reflects rapid, near-vent deposition and coalescence of hot, fragmented magma, followed by fluid-lava-like laminar flow. Each rhyolite is underlain by fluvial/alluvial fan sedimentary rocks and silicic tuffs.

***Lower Streaky Andesites:*** lavas and sheets that intrude the Etive rhyolites. The millimetre to centimetre scale streaky texture is the result of incomplete mixing of coexisting rhyolitic and andesitic magmas.

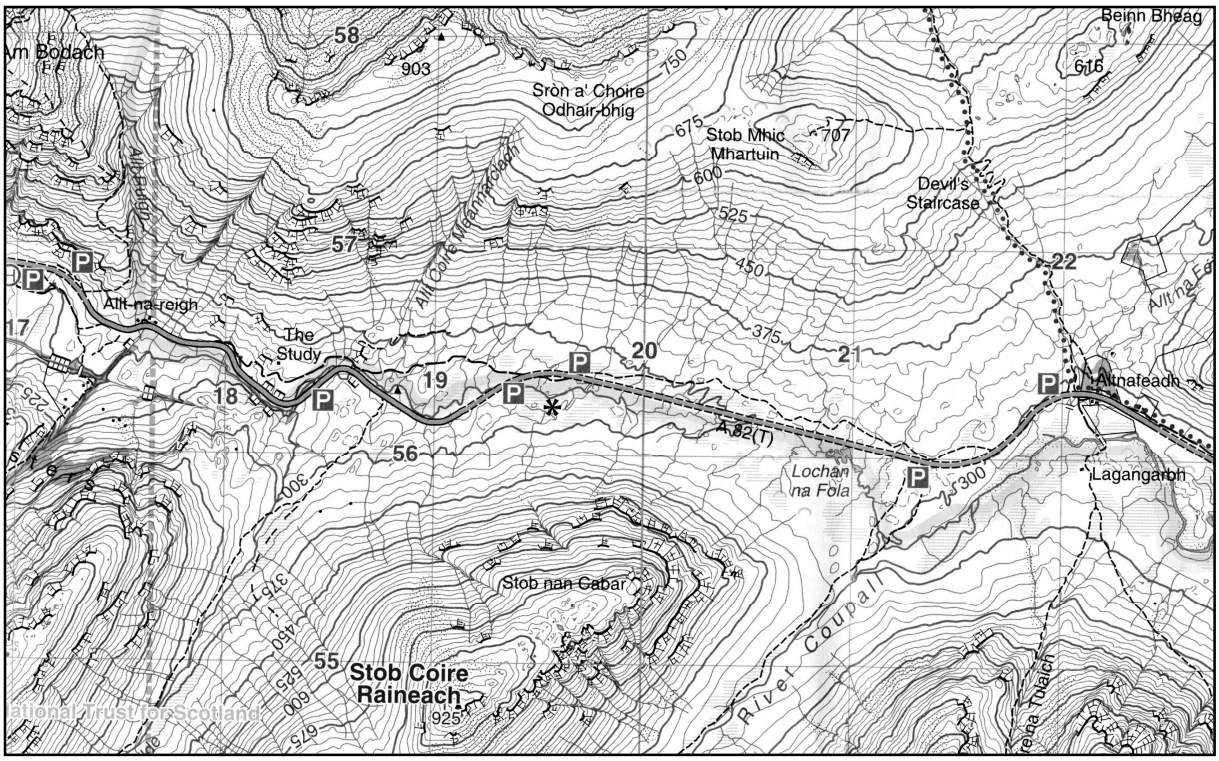
***Three Sisters Ignimbrite Member:*** predominantly comprises two strongly welded ignimbrites recording two major explosive eruptions. Each are underlain and overlain by sedimentary rocks and breccias.

***Upper Streaky Andesites:*** a further phase of activity in which andesitic and rhyolitic magmas were intermingled. The andesites occur in a plug that forms part of a vent infill, and also as numerous sill-like sheets within the Three Sisters Ignimbrite Member.

***Glas Choire Sandstone Member:*** various conglomerates, sandstones, and siltstones recording a period of fluvial incision succeeded by alluvial-fan sedimentation and the development of a caldera lake that was partly filled by lacustrine turbidites.

***Bidean nam Bian Andesite Member:*** thick andesite and dacite lavas. Their thickness and monotonous character suggest that the erupted magma was topographically ponded.

***Dalness Ignimbrite Member:*** two welded ignimbrites, each underlain by a silicic tuff. The ignimbrites are thickest near Dalness, suggesting a south-westward shift of the main locus of intra-caldera deposition.



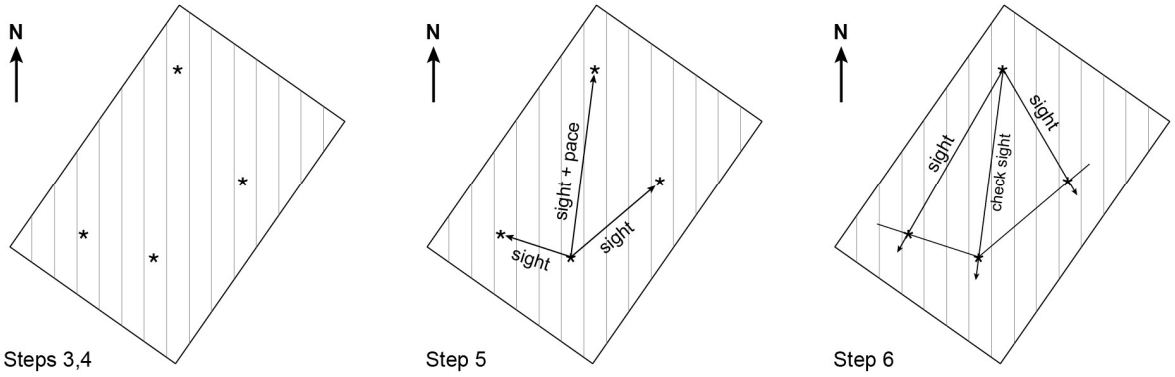

*Location of the compass-and-pace exercise in Glen Coe (asterisk). The grid squares are 1 km². The location is also shown by a star on the geological maps on p.43 and p.44*

**Compass-and-pace exercise: Pass of Glen Coe (NN1953 5626)**

The aim of this exercise is to construct an accurate map of a small area using the method of compass-and-pace. The technique is particularly useful for showing detailed field observations obtained within a small area (*e.g.*, cross-cutting relationships, local deformation features, local variations in sedimentary features). The locality lies within the Upper Streaky Andesites (USA) which are here cut by numerous sub-vertical, NE-SW trending dykes belonging to the "Etive Dyke Swarm". These dykes exhibit a number of small-scale features that provide information about the regional stresses during their emplacement, and your compass-and-pace map will help to document those features.

The following compass-and-pace method is appropriate if you have a long tape measure and are in an area without much topography. You may have to adapt it in other situations. You will need a blank sheet of paper, mapping board, compass-clinometer, ruler and sharp pencil. The exercise will be done in pairs; each pair will produce one map.

1. We will start by walking around the area to see its lateral extent.
2. Then we will place markers (*e.g.*, rucksacks) on four prominent features within the area to be used as reference points. These will define the four corners of an approximately rectangular shape.
3. You will want to draw your map with these four points positioned sensibly on your sheet of paper so that the map is a good size (*i.e.*, small enough to fit on the page but not too small to show the required detail). Therefore, orient your sheet of paper so that the four points will be well distributed across it. Now choose one of the markers as your starting point, and lightly draw a cross at an appropriate position on your sheet of paper. You will want to map outside the shape defined by the four points and so don't place this cross right at the edge of the paper. ***You have now fixed one position on your map.***




4. With your sheet of paper oriented so that the four points will fit nicely on it, use your compass to determine where north is, and lightly rule a set of parallel north lines on the page. ***You have now fixed the direction of north on your map.***
5. Standing at your reference point, take accurate compass bearings to each of the other three marker points, and using the north lines that you have just drawn, lightly draw a line going out from your reference point in the right direction to each of these marker points. Now pace out the distance from your reference point to the point diagonally across from it. Use this to choose a scale for your map, relating mm on the map to paces on the ground ($x$ mm = $y$ paces). Make sure that your 2nd marker point is in a sensible position on the map. You can now mark the 2nd marker point on the map at an appropriate distance along the line from the reference point. ***You have now fixed the scale of your map.***
6. Now standing at your 2nd marker point, take accurate compass bearings to the two marker points you have yet to locate on your paper, and using the north lines, lightly draw lines going out from your 2nd marker point in the right direction to each of these two marker points. The locations of the 3rd and 4th marker points are where corresponding sighting lines drawn from the reference point and the 2nd marker point cross. You might like to check the accuracy of your sighting technique by also sighting the reference point from the 2nd marker point – you should find that it is exactly 180° different to the value that you got when sighting the 2nd marker point from your reference point.
7. ***Checking.*** If you are feeling brave, starting at any of the marker points, sight the next marker point clockwise (or anticlockwise) and pace out the distance to it and see how near you get to where you have plotted that point. Continue going around the rectangle in like fashion until you arrive back at the point you started from. The misclosure that you get when you arrive back provides an indication of the uncertainty on your measurements.
8. ***Mapping.*** Now you can mark other features as follows:
   a) Stand at any feature that you want to mark on your map and sight two of the marker points; the location of the feature is at the intersection point of the two compass bearings drawn from the sighted marker points (since you are sighting from the feature to the marker point don't forget to add 180° to the bearing when drawing a line from the marker point to the feature)
   b) Alternatively (less accurately), take a compass bearing from any point that you have already marked on the map to the feature, and then pace out the distance to that feature and mark it on the map.

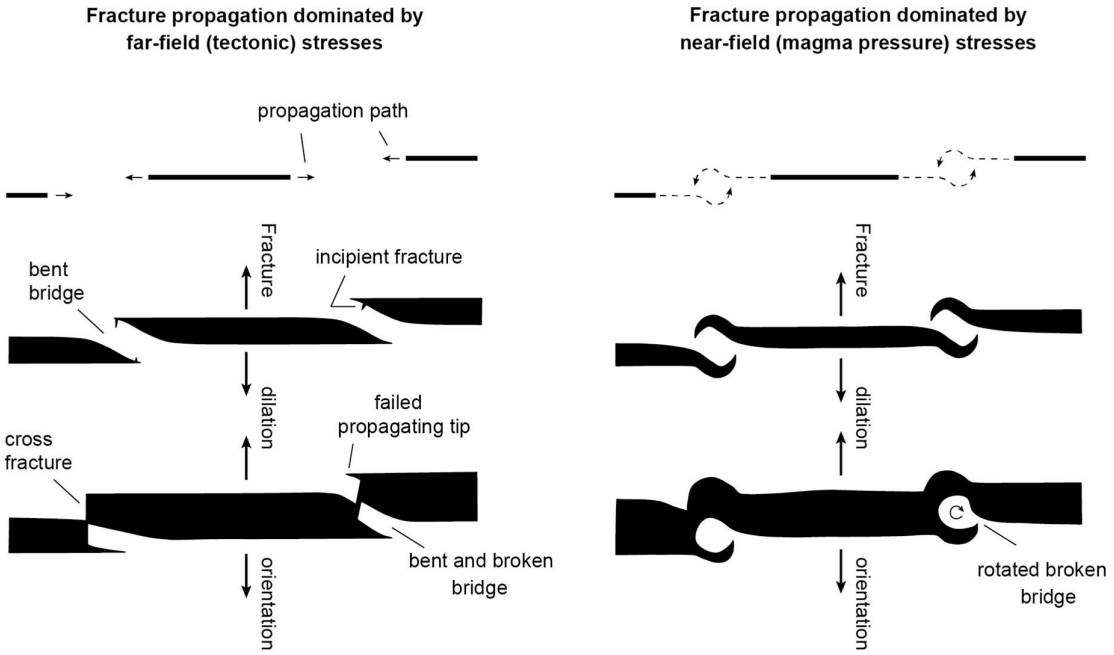
   Locate topographic features and draw them on your map, *e.g.*, the river and the edges of the exposures. Then add the geology in exactly the same way as you would during regular geological mapping. In particular, show the dykes and any prominent fracture sets.

9. Before finishing, measure the length of your pace by pacing out 10 m (a tape will be provided)
10. Record a hand specimen description of both the dykes and the country rock

When you return from the field you should ink and colour in your map in exactly the same way as you would for any other geological map. There should be a key (with scale and north arrow), and the map should be annotated with comments and blow-up sketches of significant features.

***Some interpretative remarks on the form of the tips of the dykes***

The dykes form en échelon arrays, that is, they are parallel but offset from each other. As the tips of individual fractures in an en échelon array propagate towards each other and enter each other's stress field, they curve in a way that depends upon the magnitude and orientation of the regional stresses. In the present case we can say something about whether the fractures filled by the magmas propagated in a way that was controlled by the regional (tectonic) stresses or by the magma pressure:




*Schematic illustration of dilatant en échelon crack growth controlled by (a) tectonic stresses and (b) fluid pressure (from Morris et al., 2008, Geological Society, London, Special Publication 302:149-158). Any comments on which of these cases we have at the Pass of Glen Coe site?*

You will probably conclude that the fractures containing the dykes propagated in a way that was controlled by the tectonic stresses. The dykes are mostly NE-SW trending and so this indicates their emplacement during a period of regional NW-SE extension.

Look carefully to see if there is any evidence of accompanying strike-slip displacement (transtension).

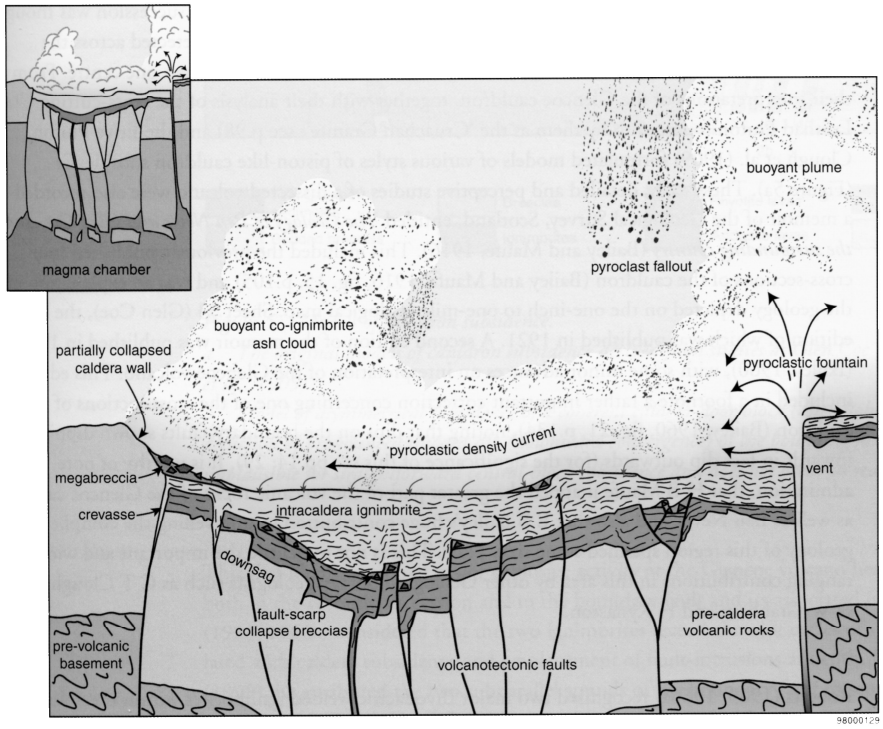
Consider also whether the dyke emplacement was influenced by pre-existing fracture sets.

**Some remarks on calderas**

Catastrophic eruption of tens to thousands of cubic kilometres of magma from a chamber beneath a volcano typically leads to collapse of the chamber roof and subsidence of the overlying volcanic pile. This subsidence forms the surface depression that is described as a caldera.

Most caldera-forming eruptions are explosive, involving sustained jets and gas-rich flows of vitric ash particles, pumice fragments, and crystals with some included rock fragments. Much of the erupted mixture is denser than air and so falls to the ground and then moves away from the vent as a pyroclastic flow. The resulting deposits are described as ignimbrites. Commonly the fragmented magmatic material in ignimbrites retains temperatures above 550° to 600°C and so the glassy constituents deform in ductile manner and become fused together. Fallout, both from the eruptive fountain and the associated pyroclastic flow, produces blanket-like deposits of pumice and ash. The magma that erupts is usually of intermediate to silicic composition, commonly dacite, rhyodacite or rhyolite.

Since the eruptive vents are mostly around the margins of the caldera, pyroclastic material usually only partially infills the topography. The topographic depressions are liable to flooding by water, and so eruptions commonly involve the explosive interaction of water and magma. When the interaction occurs beneath the ground surface in, for example, an aquifer, the resulting phreatomagmatic explosions typically build low profile tuff cones composed of ash and fragments of the aquifer.




*Sketch of the principal features of an eruption within a caldera (after Kokelaar & Moore, 2006)*

# FIELD AND MAPPING SKILLS

*The describing igneous and (meta)sedimentary units in the field comments in the following pages will be developed properly during the Fife part of the field course but are included here so that you can be starting to think about them when looking at the units in NW Scotland*

**Further reading**


- Lisle RJ, Brabham PJ, Barnes JW, 2011, *Basic Geological Mapping* (5th edition), John Wiley & Sons.
- Coe AL (editor), 2010, *Geological Field Techniques*. Wiley-Blackwell.
- McClay KR, 1991, *The Mapping of Geological Structures*, John Wiley & Sons.
- Jerram D, Petford N, 2011, *The Field Description of Igneous Rocks* (2nd edition), John Wiley & Sons.
- Fry N, 1991, *The Field Description of Metamorphic Rocks*, John Wiley & Sons.
- Tucker ME, 2011, *Sedimentary Rocks in the Field* (4th edition), John Wiley & Sons.

# Describing igneous rocks in the field

Like sedimentary units, igneous units need to be described from the outcrop scale, through the field exposure scale, to the hand specimen scale. Focus on observations that have the potential to be interpreted.

## A. Outcrop scale observations

The outcrop scale description is a general synthesis of the observations obtained from different exposures. Apart from reference to the nature (plutonic, hypabyssal, volcanic) and basic geometry of the igneous unit (dyke, sill, lopolith, laccolith, *etc.*), include an indication of its areal extent and any map-scale variations in lithology and fabric (layering, flow-banding, *etc.*). *As you work in the field, think about how your observations from different exposures fit together*.

## B. Exposure scale observations

**1. Plutonic rocks**

(a) Describe the contacts
- – Nature of the contact (*e.g.*, concordant/discordant, sharp/gradational)
- – Shape and orientation of the contact (*e.g.*, planar, non-planar, strike/dip)
- – Fabrics/foliations and xenoliths in the igneous body near the contact
- – State of disturbance of the country rock

*Seek observations that might constrain the depth and mode of emplacement (e.g., stoping, diapirism, forceful vs. passive intrusion)*

(b) Describe any internal compositional or fabric variations within the exposure
- – Nature of the variation (*e.g.*, layering, mixing)
- – Spatial and compositional scale of the variation (compile measured sections when appropriate)

*Seek observations that help to constrain the origin of the variations (e.g., cumulate settling)*

(c) Describe any internal boundaries/contacts that indicate different phases of intrusion (*e.g.*, pegmatite dykes)
- – Describe the contacts (as above)
- – Look for evidence of relative ages (*e.g.*, cross-cutting relationships, chilled margins)

(d) Record any jointing

**2. Hypabyssal rocks**

(a) Describe the contacts
- – Nature of the contact (*e.g.*, concordant, discordant)
- – Shape and orientation of the contact (*e.g.*, planar, non-planar, strike/dip)
- – Thickness of the intrusion
- – Offsets of surfaces in the country rock across the intrusion

*Seek observations that might constrain the mechanism of emplacement*

(b) Describe any internal compositional or fabric variations
- – Presence of chilled margins
- – Presence of flow-differentiation
- – Presence of vesicles/amygdales

(c) Record any jointing (*e.g.*, columnar) and measure its orientation

## 3. Volcanic rocks

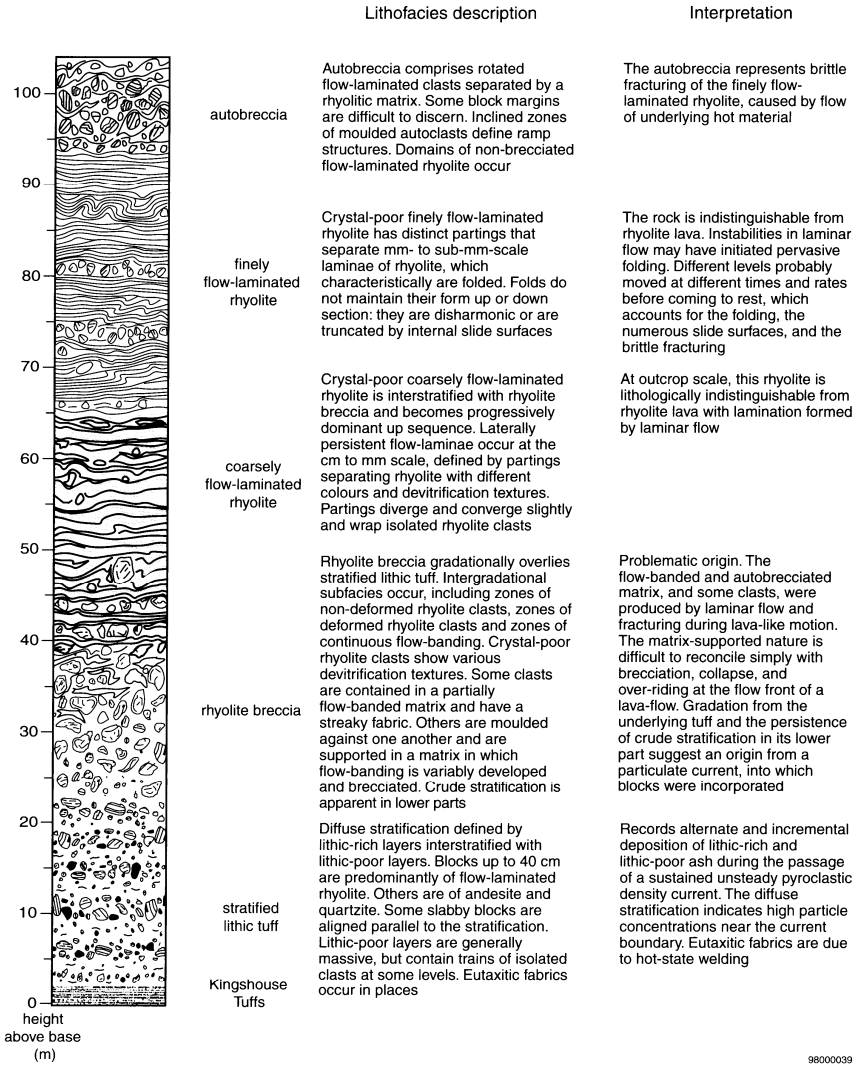
Describe lava and volcaniclastic/pyroclastic sequences in the same way as sedimentary sequences. Record up-stratigraphy variations on a measured section, and lateral variations by constructing more than one measured section through the same part of the stratigraphy in different places.

(a) Describe the contacts
  – Orientation of the contact
  – Nature of the contact (*e.g.*, sharp/gradual, transitional, loaded, palaeosol horizons)

(b) Describe the geometry of the units
  – Shape of unit (*e.g.*, tabular, lenticular, wedge-like)
  – Thickness of unit

(c) Describe the within unit variations
  – Changes in grain size (*e.g.*, normal grading, reverse grading)
  – Changes in fabrics (*e.g.*, flow banding, laminations, cross-laminations)
  – Changes in composition (*e.g.*, proportions of lithics, pumice, fiamme, crystals)

(d) Record any jointing/fractures/veining/mineralization




*Example of a measured section from Glen Coe (after Kokelaar & Moore, 2006, Classical Areas of British Geology: Glencoe Caldera volcano, Scotland). Note that in this case grain size variations have not been formally described.*

## C. Hand specimen scale observations

Hand specimen scale observations lead to the point where an igneous rock may be named/classified. Note that naming comes *after* making these observations and not before. The following is a recommended strategy for hand specimen inspection.

1. ***Weathering.*** Examine a weathered surface noting its colour and any textural features. Sometimes weathered surfaces show up features not easily visible in fresh surfaces.

2. ***Colour.*** Colour is a preliminary guide to the approximate chemical and mineral composition of an igneous rock. Inspect a fresh surface of the rock and assign its colour index based on an estimate of the total percentage of mafic minerals present:

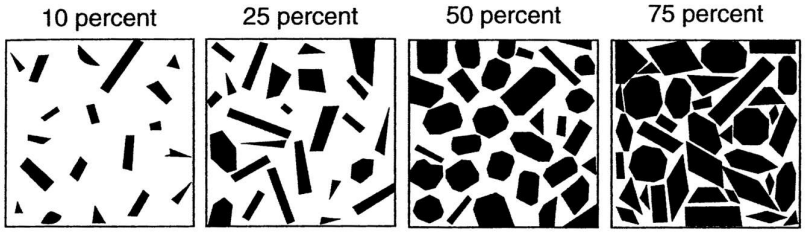
   (a) *Felsic/leucocratic*: 0 - 33% mafic minerals and therefore light-coloured
   (b) *Intermediate/mesocratic*: 34 - 66% mafic minerals
   (c) *Mafic/melanocratic*: 67 - 100% mafic minerals and therefore dark-coloured

   Colour index estimates are more reliable for coarse-grained samples.

3. ***Texture of the sample.*** Texture provides information about the physical conditions under which the magma crystallized. Examine the grains (ideally with a hand lens):

   (a) *for aphyric samples*:
   - record grain size: fine-grained (< 0.25 mm), medium-grained, coarse-grained (> 2 mm)
   - note any other textural features (*e.g.*, if it is equigranular)

   (b) *for porphyritic samples*:
   - record grain size of phenocrysts and groundmass
   - record relative proportions of phenocrysts and groundmass




4. ***Relative proportions and textural relationships of the minerals present***

   (a) *Identify the minerals present*
   (b) *Estimate the proportions of each mineral*
   (c) *For each mineral record*:
   - the basis for the identification (*e.g.*, colour, cleavage, habit, hardness)
   - crystal size and crystal shape (euhedral, subhedral, anhedral)
   - the textural relationship with other phases (*e.g.*, groundmass phase, phenocryst phase, poikilitic)

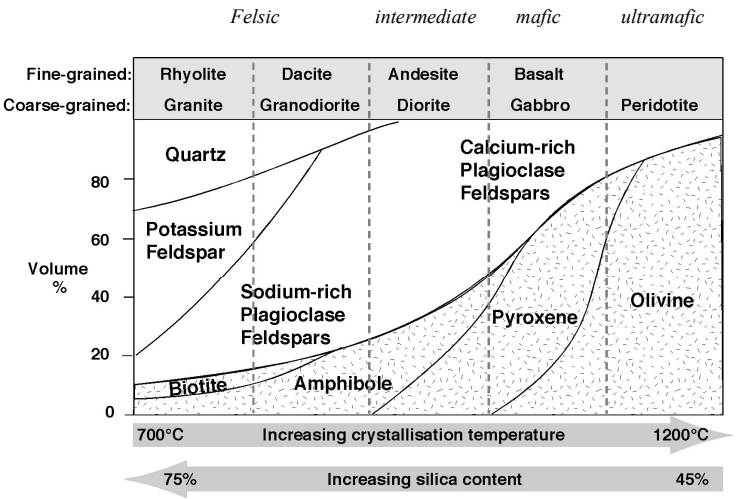
   (d) *Use mineral associations* to predict the possible presence of other minerals that you may have missed (*e.g.*, see the diagram on the facing page of typical minerals expected in igneous rocks associated with subduction zones)

5. ***Degree of compositional and textural homogeneity***

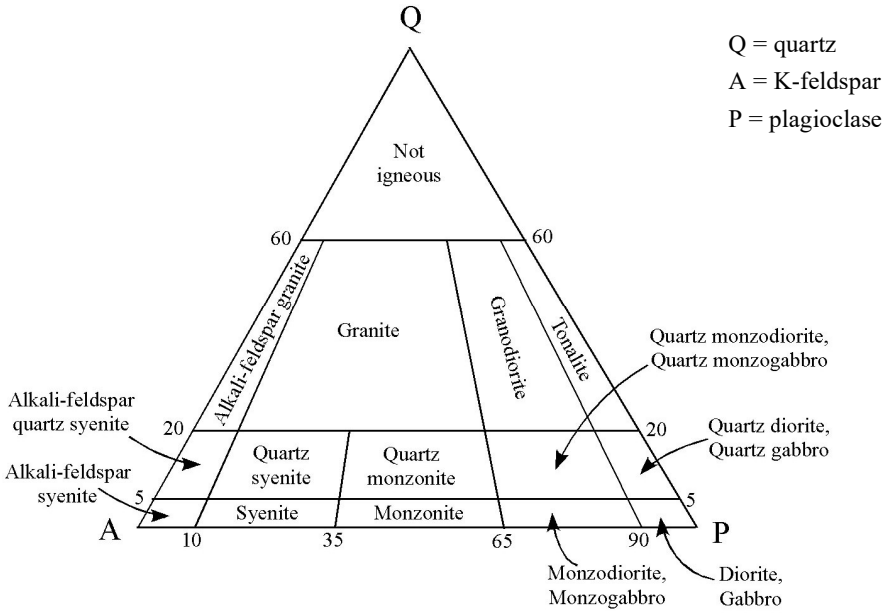
   Record hand specimen scale layering, laminations, vesicles, inclusions, xenoliths and other notable textural characteristics

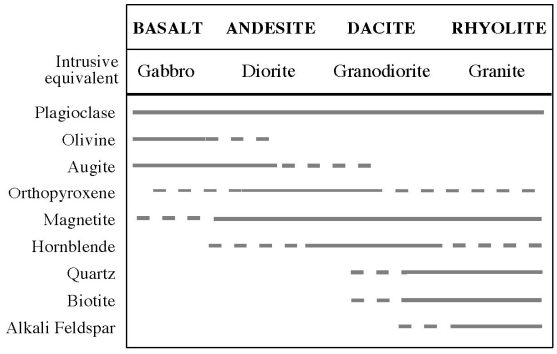
## 6. *Give the rock a field name*

(a) In many cases it may only be possible to use the preliminary classification scheme below




(b) For coarse-grained rocks, however, it ought to be possible to use the formally ratified QAPF scheme, the relevant part of which for this field course (*i.e.*, no feldspathoids) is reproduced below.




*Mineral associations in igneous rocks associated with subduction zone magmatism*

## Describing metasedimentary rocks in the field

The metasedimentary units that you will see on this field course are of sufficiently low metamorphic grade for you to describe them at the outcrop and exposure scales in the same way as you would a sedimentary unit. Bed geometries have been modified by the deformation but remain recognizable, and many of the sedimentary structures (*e.g.*, cross-bedding, graded bedding, erosive contacts) have been preserved. However, at the hand specimen scale, deformation, recrystallization, and new mineral growth have heavily overprinted the sedimentary textures and so the observations at this scale need to reflect this.

An accurately drawn stratigraphic column showing up-stratigraphy variations in lithology, unit thicknesses, the nature of contacts, and sedimentary structures should be drawn, and if this is combined with stratigraphic columns drawn in other parts of the area, lateral variations in the stratigraphy can be depicted.

Locally, where the variability of the lithologies warrants it (*e.g.*, particularly near contacts between the mapping units), short lengths of properly measured section are the best method of communicating your lithological observations. Such measured sections ought to be primarily diagrammatic with annotations and comments reserved to convey points not easily communicated in a diagram. They should cover the following matters:

(a) Contacts
  – Orientation of the contact
  – Nature of the contact (*e.g.*, sharp / gradual, transitional, faulted)

(c) Geometry of the units
  – Shape of unit (*e.g.*, tabular, lenticular, wedge-like)
  – Thickness of unit

(c) Within unit variations in sedimentary structures
  – Changes in grain size (*e.g.*, pebbly bases, any hints of graded bedding)
  – Indicate the present and scale of sedimentary structures (*e.g.*, cross-bedding, slumping)
  – Show any sedimentary features associated with the contacts (*e.g.*, erosive bases, loading)

(d) Hand specimen scale features
  – Mineral assemblages with criteria used to identify the minerals (*e.g.*, colour, cleavage, habit, hardness)
  – Comment on metamorphic textures: grain size, porphyroblasts, porphyroclasts, presence of any segregations
  – Note the nature of the tectonic fabrics providing orientations of cleavages and lineations in the area covered by the measured section, and with observations relevant for inferring the timing relationships of deformation and mineral growth when possible

(e) Details of any later minor structures and their orientations (*e.g.*, fractures, veins)

# Writing descriptions of formations

The content of a written (and illustrated) description of a geological formation depends upon the nature of that formation, for example, whether it is igneous (plutonic, hypabyssal, volcanic), metamorphic (low grade / high grade, *i.e.*, the extent to which features in its protolith are recognizable), or sedimentary (*e.g.*, siliciclastic, carbonate, evaporitic). For igneous and metasedimentary units the key ingredients are summarized above.

Although the details of the content may vary, any good description should (a) be heavily focused on observations, (b) keep observation and interpretation separate as far as is practicable, and (c) be organized in something like the following way:

1. ***Formation name*** with a short (perhaps only one-line) description and indication of stratigraphic context to serve as an introduction
2. ***Outcrop scale observations:*** detailing the location and areal extent of the outcrop with a comment on the nature/quality of the exposure; describing any subdivisions (if any) and their locations and areal extents; indicating the total apparent thickness and apparent thicknesses of any subdivisions
3. ***Exposure-scale observations:*** primarily conveying information (quantitative whenever appropriate) on the nature of contacts, the geometry of units, and internal features (*e.g.*, layering, sedimentary structures)
4. ***Hand specimen-scale observations:*** primarily conveying information on mineralogy/composition and on textural features
5. ***Microscopic observations (if thin section work has been carried out):*** amplifying the hand specimen scale compositional and textural observations with those features that require a microscope to be seen
6. ***Interpretation:*** a concise interpretative synthesis of the preceding observations, typically incorporating explicitly made deductions about the processes by which the unit was formed and was subsequently modified.

In general, point 2 above will be ***illustrated*** with a simplified map showing the outcrop and any contacts or features within it, point 3 will almost always be illustrated by a measured section to show up-stratigraphy and lateral variations, and points 4 / 5 will be illustrated by synoptic sketches illustrating textural relationships.

# Field Notebook

On this field course your notebook is not being marked. However, be aware that the quality of your field notebook is generally a good indicator of how you are approaching the task – if it is well-organized and tidy, what you are doing probably illustrates those qualities too. In order to keep it well-organized and tidy, you have to think about what you are doing, and the major benefit of this is that it helps you to come up with more effective and efficient mapping strategies. Keeping a good notebook takes effort but after a few days it becomes progressively easier.

**Some preliminary remarks**

The field notebooks of professional geologists belong to their employer and not to the geologist who compiles them. Hence, they must be fully intelligible to a third party, and contain all your data and any other relevant information. In principle, it should be possible to reconstruct your map and the observational part of your report using your notebook alone.

A field notebook that falls apart is incredibly depressing to the one who is compiling it. Hence it is worth getting a good one. It should be hard-backed, robust (*i.e.*, reasonably weather-proof), and of manageable size (*e.g.*, A5). The purist takes the view that by recording your observations systematically in the field, only one version of the notebook is needed; in practice, some might like to keep a fair-copy notebook that is written up ***every*** evening. If you adopt the latter strategy, your fair copy notebook should be with you at all times in the field even if you are actually writing into another notebook when actually collecting your observations.

Notes should be written using a good quality pencil, with a good quality pencil sharpener and a good quality eraser to hand. There is no particular need to ink-in unless there is a risk of the writing becoming smudged or faint through rubbing. Colouring pencils can be used to advantage on sketches. Above all, be neat, and make effective use of terminology to keep your text concise.

**Layout**

You should write your name and permanent (or University) address inside the front cover as well as your local address. Provide information about the map grid and the magnetic declination used (see p.67). Before going into the field, many find it convenient to compile a few pages of notes about their mapping area that they have obtained from the literature. This may include a stratigraphical column with some details of the mapping unit lithologies, published maps, notes on key features to look out for, details of the deformation history, *etc*. – essentially a set of quick reference facts that might come in handy while in the field.

Use a systematic layout for the notebook. Begin the entry for each day with the date, and with brief remarks about the weather and anything else that might affect the quality of that day's observations (*e.g.*, if you feel that you have illness coming on). This should be followed by a short statement of the day aims.

You are then ready to describe each locality as you visit it. The exact layout of how you record locality observations is a matter of personal taste, and will depend upon the nature of the data being collected. A commonly used layout has two 1 cm margins on each page with the central portion used for the main body of notes, the left-hand margin for locality numbers, grid references, readings, sample numbers, photo numbers, *etc.*, and the right-hand margin for notes made at a later date (*e.g.*, minerals and fossils identified back in the laboratory). However, it is not necessary to follow that layout just as long as the layout that you do use is consistent from locality to locality and well organized.

At the end of each day (or at least after every two or three days), you should give a summary of your current working hypotheses, so that the development of your ideas can be followed. It does not matter if you later retract those working hypotheses. Editing of notes and writing summaries are most appropriately done in the evening.

**Content**

You should record field *observations* (*e.g.*, detailed descriptions of lithologies, field sketches, structural readings, *etc.*) and *interpretations* (*e.g.*, hypotheses about the features that you have seen). The observations and interpretations must be kept separate from each other and be clearly distinguished.

Notes should be accompanied by informative diagrams such as sketch maps and sketch sections showing critical geological features and relationships. Every diagram should be clearly annotated and marked with an orientation and scale, and you should state whether the sketch is a map, section, or view.

The following is a generic list of things that you should aim to record at each locality.

1. ***Locations.*** Always use an 8-figure grid reference and a locality number. Describe the position of the locality with respect to topographical / map features (*e.g.*, gullies, bends in road, fence intersections, buildings, *etc.*). Record compass bearings if you have used them to pin-point the locality. Describe the size and nature of the exposure at the locality.

2. ***Exposure scale observations (overview of geological structure).*** Summary of key exposure scale features, position of the locality with respect to any recognized major structure in the area, brief qualitative description of the principal structural (folds, faults) and stratigraphical (lithologies) features visible in the exposure. A sketch section and/or map may be important here.

3. ***Lithology.*** Descriptive details of the rock-types present (see the sections above on how to describe metasedimentary and igneous units in the field). Pay particular attention to observations that can be interpreted, for example, the nature of bedding surfaces (*e.g.*, sharp, erosive, gradational), sedimentary structures (*e.g.*, cross-bedding, graded bedding, *etc.*, and direction of younging implied), other primary structures (*e.g.*, igneous xenoliths, layering, *etc.*), grain sizes and shapes, fossils, mineralogy, *etc.*, *etc*. Sub-headings should be used to organize these notes.

4. ***Geological structure.*** Formal description of any structures (folds, faults) and structurally-induced fabrics present (foliations, lineations) including their orientation (strike/dip for planar features; plunge/plunge direction for linear features); note the angular relations of bedding and foliations ('Z' or 'S' asymmetry or vergence); describe any kinematic indicators (*e.g.*, shear sense indicators) and minor fractures.
   - Remember the checklists for what it takes to fully describe a fold and a fault
   - All orientation readings must be correctly and fully specified, *e.g.*, $S_0$ 045/25SE, $S_1/S_0$ 24/230

5. ***Evidence for age relationships.*** If the locality contains the contact between two or more mapping units, pay particular attention to describing the nature of that contact (*e.g.*, gradational, sharp, unconformable, faulted), and its orientation (strike/dip, map trend); outline any deductions about age relationships between units, and the timing of any deformation features.

## Assessment of notebooks

*As a guide to the above and to the importance attached to good notebook skills, below are the assessment criteria that we generally consider when marking field notebooks.*

**Presentation and organization (intelligibility to a third party)**
General comments on neatness/condition of the notebook
General comments on organization of the notebook
Do the first few pages of the notebook provide adequate comment on the overall field trip dates/objectives/*etc*?
Day objectives stated at the beginning of each day?
Day summaries present, including the development of working hypotheses?
Is the quality of the notebook maintained throughout the trip?

**Localities**
Locality information (size and nature of the exposure) provided?
Are the localities given grid references (generally 8 figure)?
Do the grid references tally with the localities as marked on the map?

**Lithological descriptions**
Are exposure scale features (bedding, *etc.*) and hand specimen scale features (petrography) discriminated?
Are the petrographic (sedimentary/metamorphic/igneous) descriptions comprehensive?
Are there any interpretations of the lithological observations?

**Field sketches**
Are there sufficient field sketches?
Are the field sketches oriented and given a scale?
Are the field sketches well drawn?
Are the field sketches informative?
Are the field sketches interpreted (labelled)?

**Structural observations**
Are the orientation measurements correctly and fully specified?
– planar features as strike and dip as, *e.g.*, 045/25SE
– the nature of the planar reading noted, *e.g.*, bedding ($S_0$), cleavage ($S_1$), *etc.*
– linear features as plunge/plunge direction, *e.g.*, 24/230
– the nature of the linear reading noted, *e.g.*, intersection lineation ($S_1/S_0$ with asymmetry noted)
Is the geometry of the structure at each locality clearly reported and interpreted?
Is there a clear sense of the map-scale geometry between localities (*e.g.*, sketch cross-sections)?

## Key Features of a fold

- Scale (wavelength, amplitude), antiform/synform, anticline/syncline, axial plane, fold axis, limbs, interlimb angle, bisector (or comment on limb thickness)
- Fold asymmetry (if a fold pair), hinge roundness, nature of axial planar fabric, nature of any linear features

## Key features of a fault

- Fault plane, slip vector orientation, sense of strike-slip movement, sense of dip-slip movement, indicators of slip magnitude
- Description of fault-related fabrics (*e.g.*, S-C fabrics, Riedel shears, foliations, lineations)

## Key geometrical features of an intrusion

- Shape of intrusion, orientation of contacts (strike/dip, concordant or discordant), size of intrusion
- Emplacement processes (chilled margins, baked contacts, flow banding, xenoliths, country rock deformation, opening direction of dykes and sills)

# Field Map

You are producing an ***exposure map***, that is, one that shows the exposures as well as the readings. In contrast, an ***outcrop map*** shows the readings but does not show the outlines of the exposures. Arguably, marking exposures is unnecessary now that GPS systems make it easy and quick to locate oneself accurately at all times without sighting other exposures. However, showing exposure does indicate how much of it there is.

In general, the more geological information that you can put on your map without confusion the better; geological information is most easily assimilated and interpreted in its correct geographical position.

**Precision**

The base maps that you will be using are at a scale of 1:10000, that is, 1 mm on the map = 10 m on the ground. Thus a 0.2 mm thick line on the map has a 'ground' thickness of 2 m. This represents the limit of precision of your mapping and it is the precision that you should be aiming for when you locate features on the map. In order to achieve this precision, you must work as neatly and accurately as possible. Use fine lines when marking information on your map, and take care when locating yourself.

**Putting information on your map**

In the field, mark information on your map in pencil. Make sure that your pencil is well sharpened. It will take a day or two before you establish how much data can be accommodated on the map. The following should be recorded on the map while 'on location': locality number, mapping unit (using colour to show the extent of exposure), all structural readings, bedding trends (form-lines), and contacts. Most (including me) are sparing in how many readings we actually put on the map in the field but you ***absolutely must shade the extent of the exposure in the field*** – you cannot do this accurately in the evening. Normally, you should show topographic features that help to constrain contacts but this is not really necessary at Kinlochleven.

In the evenings, you should ink- and colour-in ***all the data*** collected in exposed areas. It is best to leave unexposed boundaries in pencil until you have all your data. Take care when inking in: (1) to use the recommended symbols and line colours, and (2) not to obscure any areas on the map that you have not yet visited. Lettering and symbols should be neat and small, but still legible.

**Data to be recorded**

Data recorded on the map should include things actually *observed* in exposed rocks as well as those *inferred* therefrom in unexposed areas.

1. ***Outlines of exposures at clearly defined numbered localities.*** When finally inked in, locality numbers should be neat and legible but unobtrusive. Use a green pen so that the numbers are not confused with readings. Exposures should be outlined with a thin green line. Outlines should be accurate; stylized circles or ellipses will be marked down.

2. ***Lithologies.*** The mapping units are **formations** not lithologies. Formations are defined on the basis of stratigraphical position as well as lithology. A formation is a mappable unit characterized by a particular lithology or group of lithologies. Where a sedimentary or meta-sedimentary unit comprises two or more lithologies, these may be inter-layered on a scale too small to be mapped individually; thus, for mapping purposes, the formation has to be defined on the basis of the *proportions* of different

lithologies present and the *scale* on which they are inter-layered. A formation is also defined by its stratigraphical position – two different units, each containing the same proportion of lithologies inter-layered on the same scale but in different parts of the stratigraphical column, are assigned to different formations. Within a formation, an abnormally thick layer (> 5 m) of one of the constituent lithologies, or a thick layer of an uncharacteristic lithology, should be mapped as a separate member.
Each exposure on the map should be coloured to indicate the formation or member present. A lighter tone of the same colour can be applied to those unexposed areas of the map where the same unit is inferred. Colours should be selected so that there is clear contrast across contacts. Bearing in mind the limits to precision, the thinnest unit that you can colour in accurately is about 4 m thick.

It is an internationally agreed convention that formations should be named on the basis of geography, not lithology (*e.g.*, "Mam Beag Formation", not "Mam Beag Schist Formation").

3. ***Geological boundaries.*** Primary contacts and faults should be mapped with three categories of confidence. Within exposures, boundaries can be mapped with certainty. In unexposed ground, boundaries are either located with confidence (*e.g.*, where the two units straddling the boundary are both exposed, or if there is a tell-tale break in slope), or else are speculative (*e.g.*, if correlations are uncertain). When estimating the trend of an inferred boundary, remember that it will be influenced by both the topography and the dip of the contact. Wherever possible, trends should be estimated from topography and dip using stratum contours.

4. ***Form-lines.*** Within thick sedimentary (or metasedimentary) units, the trends of bedding should be shown as form-lines. A form-line can be a powerful constraint on the position of an unexposed concordant sedimentary contact as the two features should be parallel. (A concordant sedimentary contact can be considered as a special kind of form-line that separates two formations.) Form-lines should be drawn wherever there is a *large* exposure within which bedding has variable attitude. Form-lines may be expected to bend at changes of slope; however, on a flat exposed surface, strongly curving form-lines pin-point fold hinges.

5. ***Structural data.*** Symbols to show the strike/dip of planar structures (bedding, first cleavage, second cleavage, fold axial planes, *etc.*), the plunge/plunge direction of linear structures (first fold axes, second fold axes, bedding-cleavage intersection lineations, stretching lineations, *etc.*), the axial traces of synclines, anticlines, synforms and antiform, the direction of younging.

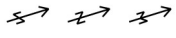
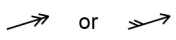
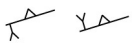
   The strike line on a strike/dip symbol should be about 6 mm long, as should be a lineation arrow. Symbols should be plotted accurately using a protractor. The symbol for bedding or the dominant cleavage should be centred at the exact spot where the measurement was made. Other symbols (*e.g.*, lineations) should be plotted as close as possible, preferably along strike from their true position.

6. ***Annotations and micro-sketches.*** Your maps should be annotated with important field information, *e.g.*, fossil locality, mineral locality, good examples of key structural relationships, well-exposed contacts, *etc*. Small sketches of key features should also be added, making use of 'spare space' in map margins and unexposed areas.

7. ***Topographical features related to underlying geological structures.*** Spring lines, scarps, gullies, *etc*. Such features are of greatest use in poorly exposed areas (not an issue at Kinlochleven).

8. ***Superficial deposits.*** In general, the nature and boundaries of superficial deposits (excluding soils) should be noted, *e.g.*, peat, till, beach sand, talus, *etc*., but we will not worry about this at Kinlochleven.

**9.** ***Key.*** Space should be set aside to draw up a key to formations mapped (coloured boxes) and the main symbols used. ***Do not forget*** to include a scale bar and a north arrow. Recommended symbols are:

| | | | |
|---|---|---|---|
| Planar Structures: | Bedding: strike/dip | | black |
| | 1st generation cleavage: strike/dip | | black / red |
| | 2nd generation cleavage: strike/dip | | black / red |
| Linear Structures: | Palaeocurrent [evidence: cb = cross-bedding] | | black / blue |
| | Mineral lineation [evidence: SL = stretching lineation] | | black / red |
| | Fold axis | | black / red |
| | Bedding/cleavage intersection | | black / red |
| | Fold (or intersection lineation) asymmetry: S/Z/M |  | black / red |
| | or tick towards antiform | | |
| | 2nd generation fold axis (or intersection lineation) |   | black / red |
| Contacts: | Observed contact |  | black |
| | Inferred contact |  | black |
| | Speculative contact |   | black |
| | Normal fault (tick= dip direction; tooth = downthrown side) |  | red |
| | High angle reverse fault |  | red |
| | Thrust fault |  | red |
| | Strike-slip fault (with sense of movement) | | red |
| Other lines: | Exposure boundary | | green |
| | Axial trace of antiform |  | red |
| | Axial trace of synform |  | red |
| | Axial trace of anticline |  | red |
| | Axial trace of syncline |  | red |
| Point symbols: | Locality number | | green |
| | Younging [evidence: gb = graded bedding] | | black / blue |
| | right way up , overturned |  | |
| | Fossil locality | * | black / blue |
| | Mineral locality |   | black / blue |
| Topographic features: | Break in slope (top of incline) |  | blue |
| | Break in slope (bottom of incline) |  | blue |
| | Ridge feature | | blue |
| Drift: | Unclassified drift | | blue |
| | Peat |  | blue |
| | Boulder clay |  | blue |
| | Landslip | | blue |

Add a sensible letter-key to each mapping unit so that a colour-blind person can distinguish the units.

## Assessment of field slips

*As a guide to the above and to the importance attached to good field slip skills, below are the assessment criteria that we generally consider when marking field slips.*

**Presentation**

General condition of the field slips
Good choice of colours for lithological units
Quality of drafting (fineness of line)
Quality of lettering (neatness and legibility)
Are the symbols of good size and proportion?
Is it easy to discriminate the different symbols?
If strikes are stated, are they standardized to a number between 0 and 180° and specified with three digits?

**Key**

Is the key comprehensive (containing all formations/lithologies and all symbols)?
Are all the planar and linear measurements correctly specified?
Is there a simplified stratigraphical column?

**Data coverage**

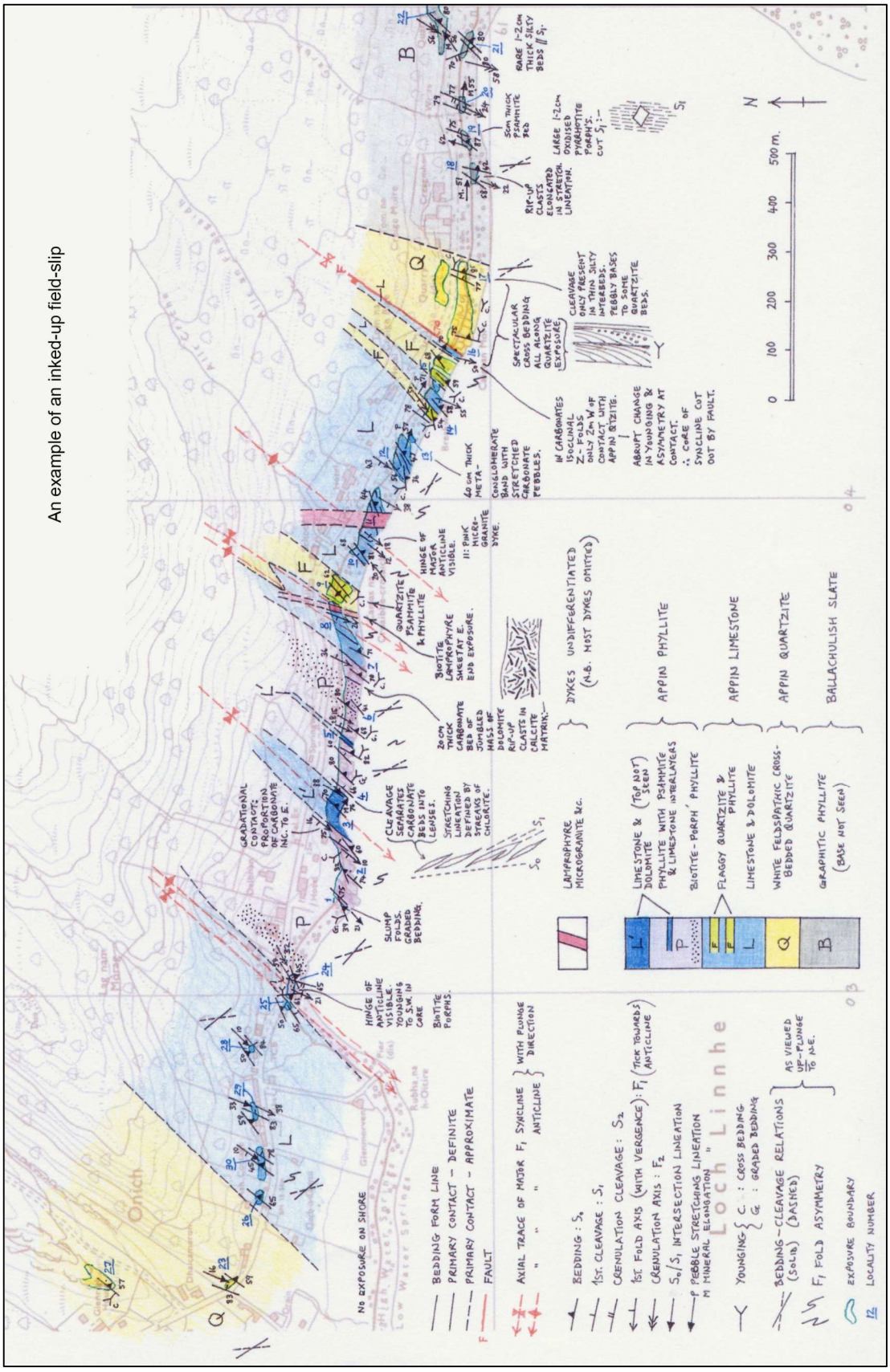
Are the boundaries of the exposure clearly shown?
Is there a difference in shading intensity between exposed and non-exposed areas?
Are the locality numbers unobtrusive?
Is the data coverage evenly distributed?
Are there sufficient bedding/cleavage/lineation readings, younging symbols?

**Contacts**

Is the distinction between observed and inferred contacts made?
Are tectonic contacts clearly distinguished as such?

**Interpretation**

Are the mapping unit contacts shown accurately and are they in agreement with the readings?
Are the field-slips fully shaded to indicate outcrop in areas of no exposure?
Is the extent of extrapolation appropriate?
Are the fold axial traces indicated and located accurately?
Are the faults drawn accurately with appropriate offsets of contacts across them?
Are the dip of the fault and the sense of dip-slip and strike-slip movement indicated?
Is good use made of form-lines?
Are written annotations/sketches shown, and if so are they useful?




*Example of an inked-in field slip (actually the same synform as at Cuil Bay but ~8 km up-plunge of Cuil Bay). In this case the exposure lay in a strip (mostly along the shoreline). Note that exposed rocks are shaded more darkly and green-lined, and use has been made of the sea to incorporate annotations*

# Constructing Cross-Sections

Whereas maps illustrate the geometry of the geological structures in plan view, that is, on a nearly horizontal cut through the structures, cross-sections illustrate those structures on a more optimally oriented surface. In the Kinlochleven section drawing exercise you will be drawing a profile plane section to illustrate the geometry of the $F_1$ folds along the coastline in the mapping area.

**The data to be shown on the cross-section**

- Ideally, the cross-section should be treated just like a field map: all the data used in its construction should be recorded on the section, that is, all the planar orientation measurements (bedding, cleavages) should be presented as apparent dip arrows, and younging information shown where it was observed (note that the younging symbol should be at right angles to the apparent dip arrow). Ideally, the exposure should also be shown projected on to the plane of section but we won't ask you to do that for this exercise.
- The cross-section should be completed with the interpretation, that is, the positions of the mapping unit boundaries, fold axial traces, and any faults should be shown using the same solid/dashed line conventions as on the map. Show form-lines of bedding and cleavage.
- As this is not a vertical section, it is not necessary to show topography.

**The orientation of the cross-section**

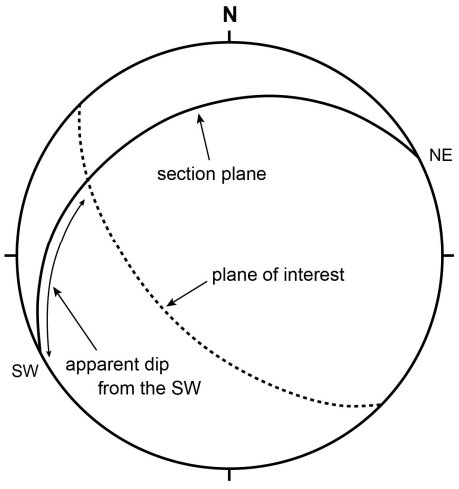
- In an area of folded rocks, the ideal orientation for a cross-section is the fold profile plane, that is, a section drawn such that one is looking along the fold axis. The section may be drawn looking up-plunge or down-plunge, whichever is the more convenient.
- Hence the first step is to plot a stereogram to determine the strike/dip of the fold profile plane. It is good practice to present a fully interpreted stereogram illustrating the geometry of the folds (complete with simplified profile plane sketch section) to accompany the section.

**Constructing the section**

Once you have chosen a cross-section plane orientation, putting the data on to the section involves two steps:
(a) the apparent dip of each planar reading (bedding, cleavages) on the cross-section needs to be determined;
(b) the projected position of each planar reading on the section needs to be determined.

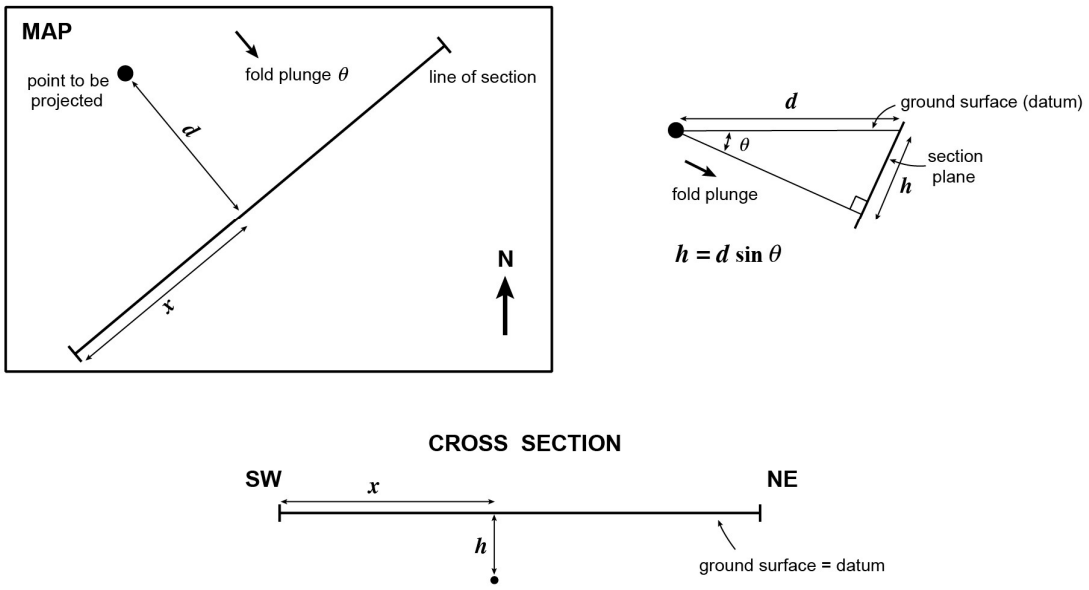
***Evaluating apparent dips***

The easiest way to determine apparent dips is by using a stereonet. Note that for planes with a strike close to perpendicular to the line of section, the difference between true dip and apparent dip is small.




***Projecting (flat topography case)***

In an area of folded rocks, measurements that lie off the line of section should be projected parallel to the fold axis on to the line of section. In areas of flat topography this may be done using the simple trigonometrical procedure illustrated on the facing page:

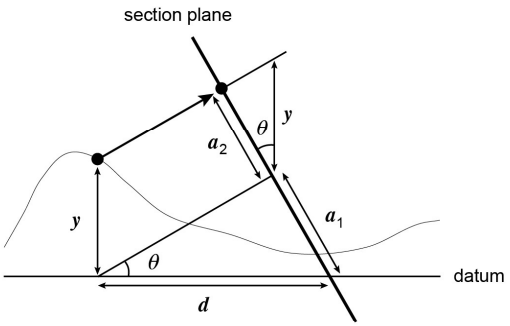



*Note that if we have a contact on the map, we can do this correction at several points along the contact. The resulting shape of the contact on the cross section has the correct apparent dip in the section plane*

***Projecting (rugged topography case)***

When the topography is not flat the trigonometrical correction is more complicated because the elevation of the point of interest above datum must be accommodated:

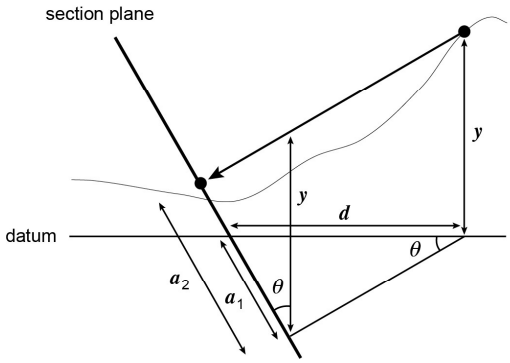
**Readings that project up onto the section**




*For a reading at elevation y above datum*

$h = a_1 + a_2 = d \sin\theta + y \cos\theta$

**Readings that project down onto the section**




*For a reading at elevation y above datum*

$h = a_2 - a_1 = y \cos\theta - d \sin\theta$

You have a topography of up to about 40 m. How many millimetres is the correction (= $a_2$) on your cross-section for an increase in topographical elevation of 10 m?

**Presentation**

- Don't forget a title with the strike/dip of the section plane included.
- Don't forget to put orientation indicators on each end of the section. Include a scale bar for both the horizontal and vertical scales.
- Include a comprehensive key to the symbols and a simple stratigraphical column.

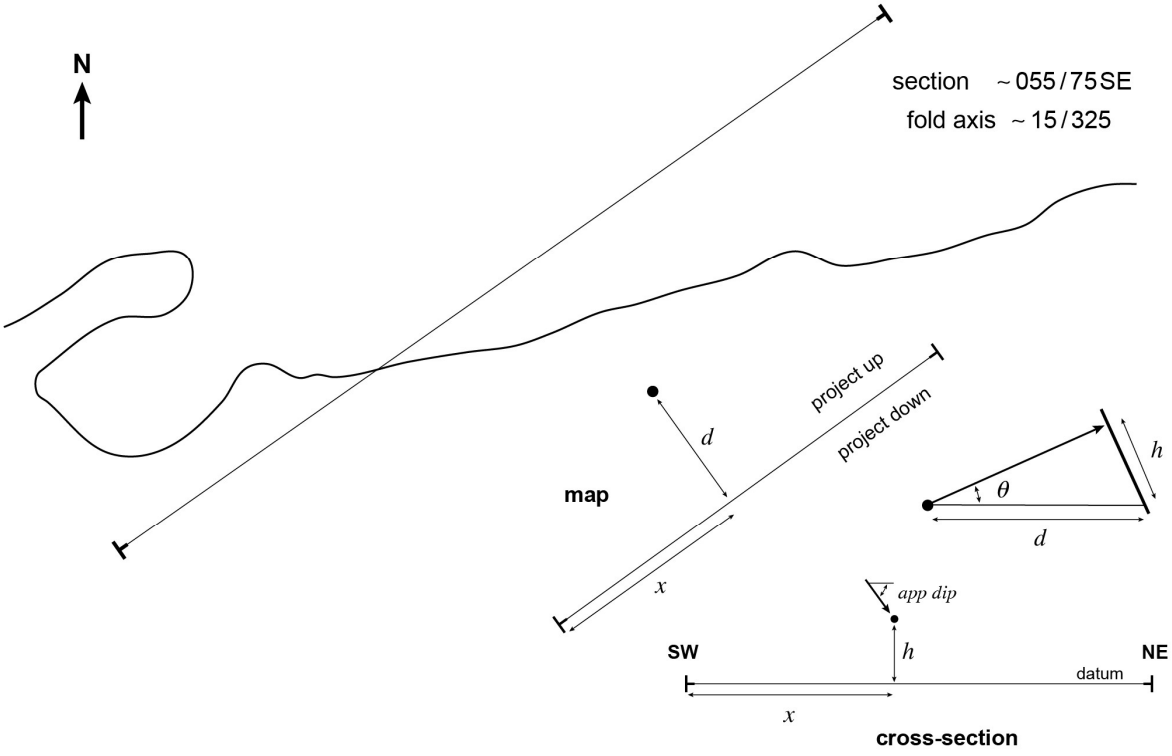
## Assessment of cross-sections

*As a guide to the above and to the importance attached to cross-sections, below are the assessment criteria that we generally consider when marking cross-sections.*

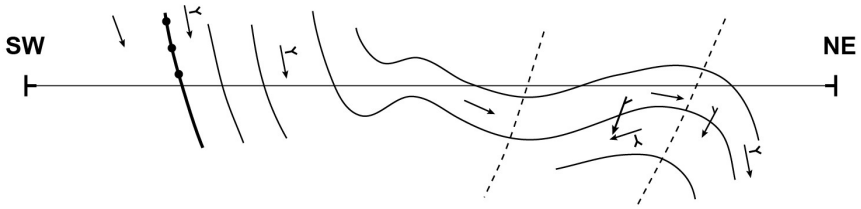
**Presentation**
General neatness
Are the length, orientation, and location of the cross-section well chosen?
Is the line of cross-section indicated on the map?
Vertical and horizontal scale noted?

**Accuracy**
(When it is appropriate to show topography) is the topography accurately drawn?
Does the distribution of lithologies correspond with that shown on the map/stratigraphical column?
Are units off the line of section correctly projected on to it?
Are the apparent dips of the planar readings (bedding/cleavage) in the cross-section plane shown?
Is younging information included?

**Interpretation**
Is the geometry of structures accurately shown?
Are fold axial traces accurately located and labelled?
Is there an appropriate amount of extrapolation above and below datum?
Are the orientations of the faults accurately drawn and the movement sense indicated?




*Draw axial traces (parallel to A.P.) before sketching form-lines*




*Don't forget all the presentational elements*

# APPENDICES

# Loch Leven – Glen Coe sample list

Over several years the Department has amassed a large number of samples from this area, many with thin sections. These are available in the Department for inspection. The following lists only the samples from the localities we will visit on the 2026 field course. Many of the Kinlochleven ones are from further up the hill from where you will work. Note that although we have several samples of the appinite body at Cuil Bay (labelled here as Ardsheal), we have none from the country rock there.

H: hand-specimen available T: thin section available * nearby locality not actually visited n: not *in situ*

**Dalradian metasediments from the Kinlochleven mapping area**

| *Sample* | *Formation* | *Locality* | | *Lithology* | *Features* | |
|---|---|---|---|---|---|---|
| Lev 21 | Binnein Schist | NN 1721 6206 | | bt musc schist | Penetrative $S_2$. Differentiated $F_3$ cren's | HT |
| Lev 41 | Binnein Quartzite | NN 159 629 | | Quartzite | Truncated cross-bedding | H |
| Lev 42 | Binnein Quartzite | NN 1587 6211 | | Spotted quartzite | Spots of diopside, tremolite & calcite | HT |
| Lev 104 | Binnein Schist | NN 1721 6204 | | Graphitic mica schist | Bedding, oblique $S_2$, $S_3$ cren. cleavage | HT |
| Lev 105 | Binnein Schist | NN 1721 6204 | | Mica schist | Bedding, oblique $S_2$, $S_3$ cren. cleavage | HT |
| Lev 106 | Binnein Schist | NN 1734 6306 | | Graphitic mica schist | Bedding & oblique $S_2$ cleavage | H |
| Lev 107 | Eilde Schist | NN 1651 6319 | | Mica schist | Psammite beds. $S_2$ crenulated by $F_3$ | HT |
| Lev 108 | Eilde Schist | NN 1651 6319 | | Psammite | Bedding. $S_2$ cleavage | H |
| Lev 127 | Binnein Schist | NN 1721 6204 | | Interbanded quartzite & schist | Distinct bedding. Odd sed. structures | H |
| Lev 128 | Binnein Schist | NN 1750 6381 | | Banded graphitic qtz-mica schist | Bedding at high angle to $S_1/S_2$ | HT |
| Lev 149 | Eilde Schist | NN 1897 6285 | | Graphitic gt mica schist | $S_1$ cleavage, 2 sets crens. gt p/b | HT |
| Lev 150 | Eilde Schist | NN 1906 6396 | | Garnet mica schist | Spaced $S_2$ cleavage. gt p/b | HT |
| Lev 154 | Binnein Schist | NN 1645 6294 | | Mica schist with psammite beds | Oriented sample. $F_2$ folds ax.pl. $S_2$ | HT |

**Intrusive rocks from the Kinlochleven mapping area**

| *Sample* | *Formation* | *Locality* | *Lithology* | *Features* | |
|---|---|---|---|---|---|
| Lev 103 | Dyke | NN 1795 6223 | hbl lamprophyre (dioritic) | plag & abundant euhedral hbl | HT |
| Lev 155 | Dyke | NN 1703 6361 | hbl lamprophyre | abundant euhedral hbl | H |
| Lev L | Dyke | NN 1773 6216 | hbl lamprophyre (dioritic) | abundant euhedral hbl. Poikilitic fsp | HT |

**Dalradian metasediments from Kentallen**

| *Sample* | *Formation* | *Locality* | | *Lithology* | *Features* | |
|---|---|---|---|---|---|---|
| Lev 35 | Appin Phyllite | NN 0115 5824 | | bt crd ksp hornfels | crd poikiloblasts. Bedding. No fabric | H |
| Lev 36 | Appin Phyllite | NN 0115 5824 | | bt crd ksp hornfels | crd poikiloblasts. Bedding. No fabric | HT |
| Lev 37x | Appin Phyllite | NN 0115 5824 | | bt crd ksp hornfels | crd poikiloblasts. Bedding. No fabric | HT |
| Lev 85a | Appin Phyllite | NN 0146 5834 | * | Migm. and sil crn crd ksp hornfels | crd poiks. ksp-rich leucosomes | HT |
| Lev 86 | Appin Phyllite | NN 0134 5840 | | bt crd ksp hornfels | crd poikiloblasts. Bedding. No fabric | HT |
| Lev 94 | Appin Phyllite | NN 0146 5834 | * | Migmatitic hornfels | crd poikiloblasts. ksp-rich leucosomes | H |
| Lev 95 | Appin Phyllite | NN 0146 5834 | * | Semi-pelitic hornfels | Fine grained. Granoblastic texture | H |
| Lev 118 | Appin Phyllite | NN 0146 5834 | * | Semi-pelitic hornfels | crd poikiloblasts. ksp-rich leucosomes | H |
| Lev 119 | Appin Phyllite | NN 0146 5834 | * | Qtzite & semi-pelitic hornfels | ksp-rich veins from qtzite to semi-pelite | HT |
| Lev 135 | Appin Phyllite | NN 0134 5840 | | bt crd ksp hornfels | crd poikiloblasts. ksp-rich vein | H |
| Lev 137 | Appin Phyllite | NN 0134 5840 | | Finely interlayered psammitic & pelitic hornfels. No fabric | | H |
| Lev 164 | Appin Phyllite | NN 0115 5824 | | Semi-pelitic hornfels | crd poikiloblasts. ksp-rich leucosomes | H |
| K4 | Appin Phyllite | Kentallen | | phlog diop hornfels | Bedding. Granoblastic diopside | T |
| K96 | Appin Phyllite | ¼ mi SE Kentallen | | and sil crd ksp hornfels | Subhedral and. Granoblastic ksp segr. | T |
| K202 | Appin Phyllite | Kentallen | | bt crd ksp hornfels | Granoblastic. Sub-poikiloblastic crd | T |
| Lev 85c | Appin Phyllite xenolith | NN 0146 5834 | * | Cordierite K-feldspar hornfels | Granoblastic texture. crd poikiloblasts | H |
| Lev 93 | Appin Phyllite xenolith | NN 0146 5834 | * | Biotite cordierite hornfels | crd poikiloblasts | H |

**Intrusive rocks of the Ballachulish pluton**

| *Sample* | *Formation* | *Locality* | | *Lithology* | *Features* | |
|---|---|---|---|---|---|---|
| Δ234 | Balla. Quartz Diorite | n | * | Granodiorite | Xenoliths of pelite & quartzite | HT |
| Lev 30 | Balla. Quartz Diorite | NN 022 590 | * | hbl bt granodiorite | Xenolith of psammite | HT |
| Lev 83 | Balla. Quartz Diorite | NN 0534 5961 | * | Tonalite | Igneous xenolith | HT |
| Lev 85b | Balla. Marginal Aplite | NN 0146 5834 | * | Pink microgranite ('aplite') | Granular texture. High modal % qtz | HT |
| Lev 85c | Balla. Marginal Aplite | NN 0146 5834 | * | Intrusive breccia (hornfels xenoliths in pink microgranite) | | H |
| Lev 92 | Balla. Quartz Diorite | NN 0146 5834 | * | Granodiorite | Xenolith of psammite | HT |
| Lev 133 | Balla. Quartz Diorite | NN 0146 5834 | * | qtz monzodiorite | Much biotite. Small clots of hbl | HT |
| K137 | Balla. Quartz Diorite | 2mi NE Kentallen | * | hbl bt granodiorite | Microdiorite xenolith | T |

**Intrusive rocks of the Appinite Suite (kentallenite *etc*.)**

| *Sample* | *Formation* | *Locality* | *Lithology* | *Features* | |
|---|---|---|---|---|---|
| Lev 31 | Kentallen intrusion | NN 0105 5780 | bt ol monzonite ('kentallenite') | ol & aug phenocrysts. Poikilitic bt | HT |
| Lev 32 | Kentallen (dyke phase) | NN 0107 5808 | Microkentallenite | ol & aug phenocrysts | HT |

| | | | | | | |
|---|---|---|---|---|---|---|
| Lev 33 | W Ardsheal intrusion | NM 9667 5612 | * | bt aug diorite (‘appinite’) | aug rimmed by hbl | HT |
| Lev 34 | W Ardsheal (N mass) | NM 9675 5620 | * | bt aug quartz monzonite | aug & brown hbl phenocrysts | HT |
| Lev 43 | Kentallen (dyke phase) | NN 0107 5808 | | Microkentallenite | ol & aug phenocrysts | HT |
| *Sample* | *Formation* | *Locality* | | *Lithology* | *Features* | |
| Lev 44 | Kentallen intrusion | NN 0106 5805 | | bt ol monzonite (kentallenite) | ol & aug phenocrysts. Poikilitic bt | HT |
| Lev 45 | Kentallen (dyke phase) | NN 0107 5808 | | Microkentallenite | ol & aug phenocrysts | H |
| Lev 87 | Kentallen (margin) | NN 0107 5808 | | bt gabbro | Euhedral aug. 3m S of Qtzite contact | HT |
| Lev 96 | Kentallen intrusion | NN 0107 5806 | | bt ol monzonite (kentallenite) | Like Lev 44 | HT |
| Lev 120 | Kentallen intrusion | n | | bt ol monzonite (kentallenite) | Like Lev 31. Middle of intrusion | H |
| Lev 121 | Kentallen intrusion | NN 0107 5806 | | bt ol monzonite (kentallenite) | Like Lev 44. 20m S of Qtzite contact | HT |
| Lev 122 | Kentallen intrusion | NN 0107 5806 | | bt ol monzonite (kentallenite) | Like Lev 44. 15m S of Qtzite contact | HT |
| Lev 123 | Kentallen (margin) | NN 0107 5807 | | bt clinopyroxenite | cpx altered. 10m S of Qtzite contact | H |
| Lev 125 | W Ardsheal (N mass) | NM 9675 5619 | * | bt aug qtz monzonite | Similar to Lev 34 but finer-grained | HT |
| Lev 126 | W Ardsheal (N mass) | NM 9674 5619 | * | bt hbl clinopyroxenite | Slightly altered. Minimal feldspar | HT |
| Lev 136 | Kentallen (margin) | NN 0107 5807 | | bt clinopyroxenite | cpx fresh. 10m S of Qtzite contact | HT |
| Lev 162 | W Ardsheal (N mass) | NM 9673 5618 | * | bt hbl clinopyroxenite | Like Lev 126 | H |
| Lev 163 | W Ardsheal (N mass) | NM 9670 5614 | * | bt aug qtz monzonite | Like Lev 34 | H |
| Lev 166 | W Ardsheal (margin) | NM 9691 5591 | * | bt diorite (‘appinite’) | Euhedral hbl & bt | H |
| Δ674 | ? C Ardsheal intrusion | n | * | Pegmatitic qtz bt diorite | Long euhedral hbl. Micrographic | HT |
| Δ774 | Kentallen intrusion | NN ~ 013 582 | | bt ol monzonite (kentallenite) | ol & aug phenocrysts | HT |
| 9751 | Kentallen intrusion | ‘New Quarry’ | * | bt ol monzonite (kentallenite) | ol & aug phenocrysts. Poikilitic bt | HT |
| K34 | Kentallen intrusion | NN 0106 5805 | | bt ol monzonite (kentallenite) | Similar to Lev 31 | T |
| K37 | Kentallen (dyke phase) | NN 0106 5805 | | Microkentallenite | Similar to Lev 43 & Lev 45 | T |
| K43 | Kentallen intrusion | NN 0106 5805 | | bt ol monzonite (kentallenite) | Similar to Lev 31 | T |
| K116 | C Ardsheal intrusion | NM ~ 995 569 | * | bt hbl qtz gabbro | Hornblendite xenolith. Like K201 | T |
| K200 | Kentallen intrusion | NN 0106 5805 | | bt ol monzonite (kentallenite) | Like Lev 31 | T |
| K201 | C Ardsheal intrusion | NM ~ 995 569 | * | bt hbl qtz gabbro | Zoned plag, euhedral aug, interstit qtz, cc | T |

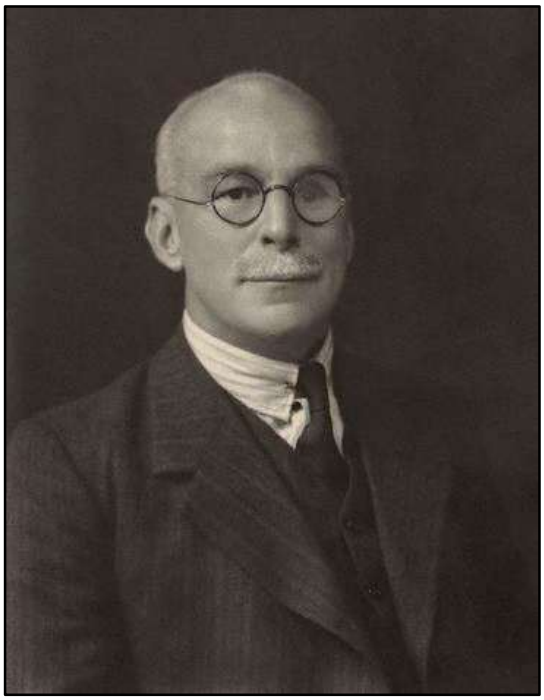
**Volcanic and sedimentary rocks of the Glen Coe Volcanic Complex**

| *Sample* | *Formation* | *Locality* | *Lithology* | *Features* | |
|---|---|---|---|---|---|
| Lev 64 | Three Sisters Ignimbrite | n | Rhyolitic ignimbrite | Flattened glass shards | H |
| Lev 65 | Etive Rhyolite | n | Flow-banded rhyolite | Folded flow banding. fsp phenocrysts | H |
| Lev 66 | Etive Rhyolite | n | Flow-banded rhyolite | Folded flow banding. fsp phenocrysts | H |
| Lev 67 | Three Sisters Ignimbrite | n | Volcanic breccia | Large variety of clast types | H |
| Lev 68 | Basal Andesite Sills | n | Peperite | Vesicular ‘clasts’ | H |
| Lev 69 | Three Sisters Ignimbrite | n | Rhyolitic ignimbrite | Flattened glass shards | H |
| Lev 73a | “ORS sediment” | NN 2022 5750 | Volcanic breccia | Angular clasts. Green matrix | H |
| Lev 73b | “ORS sediment” | NN 2021 5749 | Volcanic ‘conglomerate’ | Polymictic. Rounded volcanic pebbles | HT |
| Lev 74 | Basal Andesite Sills | NN 2031 5660 | Andesite | Dark grey, fine-grained, aphyric | H |
| Lev 75a | Lr Streaky Andesites | NN 2132 5598 | Rhyolite | Flow banding | H |
| Lev 75b | Lr Streaky Andesites | NN 2132 5598 | Andesite / Dacite | Dark, massive | H |
| Lev 76a | Etive Rhyolite | NN 1822 5632 | Rhyolite | Flow banding. fsp phenocrysts | H |
| Lev 76b | Etive Rhyolite | NN 1822 5632 | Rhyolite | Flow banding. plag phenocrysts | HT |
| Lev 77 | Basal Andesite Sills | NN 1800 5651 | Andesite | Dark grey, fine-grained, aphyric | H |
| Lev 81 | Basal Andesite Sills | NN 1415 5583 | Andesite | Aphyric. Fresh cpx. Altered opx | HT |
| AT8 | Etive Rhyolite | NN ~ 213 560 | Rhyolite | Flow banding. plag phenocrysts | T |
| Lev 82 | Old Red Sandstone | NN 1409 5591 | Red siltstone | Clasts of qtz and musc | HT |

*The Loch Leven area was originally mapped for the British Geological Survey by Edward Bailey. He was one of the greats of British Geological mapping but had somewhat eccentric ways . . .*

## Sir Edward Bailey (1881-1965)

The following is taken (with minor editing) from A. Maltman (1998) *Geological Maps: An Introduction* (2nd edition). For a more detailed biography see C.J. Stubblefield (1965), Edward Battersby Bailey, 1881-1965, *Biographical Memoirs of Fellows of the Royal Society* 11:1-21.



Edward Bailey was born 1 July 1881 at Marden in Kent. At school he was picked on by bullies, and took some cruel beatings, but his reaction to this was to 'toughen himself up'. As an undergraduate student he would go for marathon walks at night, took up boxing, and would sleep all year with his bedroom window wide open and only a cotton sheet for a cover. He graduated from Cambridge University in 1902 and applied to join the Geological Survey. Only two people were appointed that year, but Bailey was one of them – despite having broken the nose of the Director's nephew in an undergraduate boxing match.

Bailey was assigned to the West Highland unit. He joined a team comprising three of the finest geologists the BGS has ever employed. Ben Peach and John Horne were fresh from their brilliant unravelling of the Moine Thrust region of northwest Scotland, and the field maps of the equally gifted C.T. Clough have often been cited as the standard against which others should be judged. The mapping training that Bailey received, and his sudden exposure to 'a jostling crowd of problems awaiting solution', exhilarated him. By his twenty-first birthday he was mapping independently for the BGS and went on to be involved with many of the superb geological maps of the Scottish western seaboard.

Yet for Bailey the maps alone were not sufficient. The accompanying memoirs were vital channels for him to expound his interpretations of the geological histories of the surveyed areas. The memoir accompanying the map of Glencoe, for example, allowed the first detailed three-dimensional analysis of a volcanic caldera. It was a characteristic of Bailey's writing that terms were used exactly, and if no suitable terms existed, he would invent them – antiform and synform, for example. In addition, Bailey began to publish the results of geological studies carried out in his own time – evenings, weekends, holidays – in journals other than official Survey publications. The liberty to do this was to become to Bailey a fundamental principle.

All was not straightforward in these early days – Bailey's eccentricities were raising eyebrows. Why was some of his fieldwork being done barefoot? And was he really conducting official Survey work wearing shorts? The hierarchy reeled, and informed Bailey that he was not wearing 'sufficiently formal attire'. Hearing this, Bailey promptly resigned, which rattled his senior supervisors even more. A special dispensation was produced which allowed him to continue his work in shorts.

At the start of World War I, Bailey volunteered for duty. He was posted to an island off Plymouth, but this did not stop him doing geology for he immediately began surveying the island, the results of which he published after the war. He was eventually sent to the front line and was badly wounded during the Battle of the Somme and later at Ypres. He sustained a badly damaged arm and lost an eye. He was later knighted for gallantry in battle.

After the war, Bailey was able to return to the survey of western Scotland and was put in charge. He was thereby enabled to complete a project that he had been involved with off and on since 1907 – the mapping of the island of Mull and nearby areas. The completed one-inch map appeared in 1924, and the following year the accompanying two-volume explanatory memoir was published. The map has been called a 'landmark in the history of geological cartography', 'one of the most wonderful chapters of the geology of Britain', and when the map was published it was said that 'new lustre had been added to the Geological Survey'. The precision with which the intricate geology is portrayed, including over 40 kinds of igneous rocks, is remarkable. Bailey was clearly entranced, both with the routine surveying, and the gradual deciphering of the volcanic story that was hidden in the rugged mountains. His commitment to geology and oblivion of danger are notable. On one occasion of prolonged heavy rain, the survey team was camped in a meadow which became flooded. The team was forced to leave during the night, only to realise the next morning that Bailey had not joined their evacuation. One of them waded back to Bailey's tent, to find the water lapping just inches below the level of the canvas camp bed. On the bed lay Bailey, happily reading a textbook on physics.

Nevertheless, despite this success, Bailey was continuing to annoy his superiors with his machismo activities. He would soak his boots at the start of a day in the field, so that further wettings would go unnoticed. He was renowned for eating nothing more than a chocolate bar for lunch, or, if he had been supplied with more, eating it first thing to 'get it over with'. On one occasion Bailey received an official reprimand for leading his new Assistant Director through a chin-deep river, when he knew there was a perfectly good bridge just out of sight upstream. But much more significantly than all of this, in 1920 a director was appointed to the BGS who felt that all his staff's scientific writings, whether or not they were related to survey work, had to be vetted by him. This was completely contrary to the principle of scientific liberty cherished by Bailey. He became increasingly uneasy with the new regime. In turn, his director grew paranoid about enforcing his new rule. Things eventually came to a head when Bailey was banned from looking at any of the Survey's field maps, even though they were available for consultation by the general public. Bailey resigned, taking up the life of a university professor, at Glasgow.

At Glasgow Bailey clearly enjoyed himself, travelling far and wide, coming up with innovative interpretations, and letting his geological imagination roam. He was one of the first in the UK to embrace a wild new theory which involved the continents actually drifting apart. He seized on the significance in rocks of 'way-up' indicators, was one of the first to recognize submarine slumping in rocks and anticipated the concept of growth faulting. Bailey had an unrivalled grasp of the three-dimensional aspects of map work. He began to apply his skills to geometrically difficult regions such as the Swiss Alps. The convoluted forms of the sections he constructed became nicknamed 'Bailey's bicycles'. He, himself, enjoyed a play on his first name in his paper entitled 'Eddies in mountain structure'. He formulated some of the methods for dealing with the outcrop patterns of structurally complex areas, and, following a visit to the Appalachians, helped promote the 'down-structure' viewing technique.

However, despite the scientific liberty which a university offered, it seems that the Geological Survey was still Bailey's first love. In 1936 he was invited to return – as Director. Bailey accepted, but on his own conditions regarding research. As it transpired, he had little time to put his ideals into action before once again he and his Survey were plunged into supporting the national war effort. In World War II, BGS geologists were not involved in the fighting, but rather in all kinds of strategic support work. Geological surveying was required for underground vaults, new airbases, temporary water supplies, *etc.*, as well as for sources of raw materials which could no longer be imported. These ranged from coal and iron to less obvious but vital needs – sand for sandbags, mica for electrical equipment, silica for optical glass, even sapphires for aircraft compass pivots. Bailey threw himself into his new responsibilities. He was later to be

proud of the BGS war effort and was eager to point out that its duties had been greatly facilitated by the fact that high-standard geological maps of the UK already existed.

During the war, the BGS office was evacuated, becoming the Civil Defence Headquarters. The geological maps and materials were shipped to North Wales for safety; the geologists, too, were moved. Except Bailey. He found space sufficient for a temporary office on the topmost gallery of the Survey building and worked there throughout the London blitz. True to form, he seemed oblivious of the danger, even after he was 'temporarily buried by a $V_1$ or Doodlebug'.

At the end of the war Bailey saw the need for the Survey to have a fresh look. Reorganization was necessary, and Bailey felt that it was time for him to make way for new blood. He retired, although until his death 20 years later he continued actively in geology, publishing research papers, and attending geological meetings – invariably wearing shorts.

In 1948, there came to fruition one of his pet projects. Back in 1942 Bailey had been appointed to look after the geological aspects of a 'National Atlas'. He had arranged for the BGS to produce a map at the innovative scale of 1:625000, or 'about ten miles to an inch'. The scale was selected in order that the land mass could be accommodated on 'two not over-large sheets'. The map was to carry a new kind of 10 km grid. This was to help unify the two sheets and give easy correlation with larger-scale maps which from then on were to carry a kilometre grid. This 'National Grid' would also provide a convenient and precise reference system for locations 'much more easy to employ than that given by longitudes and latitudes'. Most of the detailed survey data were already available on the one-inch sheets; it was largely a matter of compilation and reorganization to fit the new scale. Bailey took the project to heart. The north sheet covered the Scottish Highlands and Islands, and he saw this part, especially, as his own.

On its publication, the map was hailed a great success. Since then, it has remained the standard geological map of Great Britain, and a model for many other national maps. Without a great deal of alteration, a second edition was published in 1957, and a third in 1979. The north and south sheets are widely available today for purchase. Also still sold are the one-inch maps of the Scottish western seaboard which Bailey helped produce. They remain the definitive maps of those areas (some now photographically enlarged to 1:50000), especially the celebrated Mull map. The legacy to geological maps of Sir Edward Bailey remains very tangible.

Bailey produced the first comprehensive geological map of the Loch Leven area, and while the interpretation of the structures has evolved since he did the work, his map remains a fine statement of what is there on the ground.

# Health and Safety / Behaviour

## Nationally agreed Code of Safety and Behaviour (relevant parts for this course)

**To: All students attending a geological field course**

- It is important that we know if you have any disability which may influence your work, *e.g*., diabetes, epilepsy, *etc*., so that appropriate action can be taken should you be affected while on a field course. You should therefore have filled in the Medical Questionnaire provided prior to the field course. This will be kept by the field course leader.
- ***All information will be treated with strict confidence.***
- If for any reason you are prevented from attending a prescribed field course, or have to withdraw during such a course, please inform the course leader at the earliest opportunity. *If the reasons are medical, you will be expected to produce a doctor's certificate.*

**I. General**

Geological fieldwork is an activity involving some inherent special risks and hazards, *e.g*., exploring coast exposures, quarries, and mountains. Severe or dangerous weather conditions may also be encountered at any season, especially on mountains or the coast. In accordance with the Health and Safety at Work Act, leaders will have been advised by the School to take certain safety precautions and every reasonable care concerning the safety of members of their parties. *It is imperative that students should cooperate by behaving reasonably in order to reduce the risk of accidents*. ***Each individual is responsible for his or her own safety.***

**II. In particular you must:**

1. Observe all safety instructions given by party leaders or field advisers. Anyone not conforming to the standards required may be dismissed from the field course (with financial and examination liabilities as a result).
2. Stay with the party, except by clear arrangement with the leaders. Assemble *where* requested in order to receive specific instructions regarding likely hazards. Observe instructions for reporting back after completion of work.
3. Report any injury or illness to the field course leader as soon as possible.
4. Carry identification of yourself and the School.

**III. Equipment**

Note: leaders are advised by the School to refuse to allow ill-equipped students on their field courses.

1. Wear adequate clothing and footwear for the type of weather and terrain likely to be encountered. Shirt, loose-fitting trousers, warm sweater, and brightly coloured rain-proof jacket with hood, are desirable in Scotland. A woollen hat (in addition to a coat hood) is useful on high ground. Waterproof over-trousers are necessary for wet weather. Jeans generally do not give sufficient protection when wet and subject to cold wind but can suffice if waterproof over-trousers are worn as well. Walking boots with rubber mountaineering soles are normally essential. Sports shoes/trainers, etc., are unsuitable for mountains, quarries or rough/wet country. Wellington boots are generally best reserved for streams, peat bogs, *etc*., but are inadequate and usually uncomfortable on rocky ground.
2. Always wear a safety helmet, preferably with chin strap, where there is a risk from falling objects.
3. Wear safety goggles when hammering rocks or using chisels. ***Do not use*** a geological hammer as a chisel while hitting on it with another hammer; use only a soft steel chisel, preferably with a plastic guard. Avoid hammering near another person. Avoid looking towards another person hammering.
4. Carry at all times a survival bag, emergency food supplies, whistle, small first aid kit.

**IV. Behaviour**

1. Be aware of traffic when examining road cuttings. While there, avoid hammering, and in any event do not leave rock debris on the roadway or verges.
2. Take great care when walking or climbing over slippery rocks on rock shores, particularly below high water mark. More accidents to geologists, including fatalities, occur along rocky shorelines than anywhere else.

3. Do not climb cliffs, rock faces or crags unless this had been approved ***in writing*** as an essential part of the work.
4. Take special care near the edges of cliffs and quarries, or any other steep or sheer faces, particularly in gusting winds.
5. Ensure that rocks above are safe before venturing below. Avoid working under steep overhangs. Avoid loosening rocks on steep slopes and *never* roll rocks down slope for amusement. Beware of landslides and mudflows occurring on clay cliffs and in clay pits, or rockfalls from any cliffs.

Leaders of parties will follow the general guidance contained in: *A Code for Geological Field Work* issued by the Geologists' Association; *Mountain Safety – basic precautions* published by Climber and Rambler; *Guidelines for visits to quarries* – laid down by the British Quarrying and Slag Federation.

**V. To: all students undertaking geological fieldwork alone, in pairs, or small groups (section relevant for this field course):**

1. All members of a mapping group or pair in one area must:
   a. Conduct a team familiarization of each mapping area and its particular risks/hazards before mapping.
   b. Be familiar with local rescue/medical facilities.
   c. Agree a formal day plan and rendezvous scheme with your 'buddy' which includes:
      - listing areas/routes to be mapped that day.
      - definition of meeting locations and times – an end of day rendezvous is required.
   d. Never carelessly break arrangements to meet your 'buddy'; it is better to be early for a rendezvous.
   e. Agree a plan of action for bad weather or illness.
   f. Agree a scheme for rescue in the event of an accident.

## Behaviour on field courses

*A reminder of the Code of Behaviour form that you have signed*

Ordinance XIV on the admission, conduct and discipline of students, paragraph 4 states:
"Every student shall maintain at all times and in all places an acceptable standard of conduct and shall comply with such Regulations relating thereto as shall have been duly made by the University or by the authorities of any hall of residence or affiliated institution: without prejudice to the generality of the foregoing every student shall be liable to disciplinary action in respect of conduct which:

(i) is discreditable to the University or detrimental to the discharge of the University's obligations under the Charter;
(ii) disrupts the teaching, research or administration of the University; damages University property or obstructs or endangers the safety of officers, employees or students of the University or visitors to the University;
(iii) involves the use or attempted use of unauthorized or unfair means in connection with any examination."

I have read and understood Ordinance XIV.4. of the University of Manchester and fully realize that any unacceptable behaviour at any time or place while I am a member of the field course may be liable to the disciplinary action outlined therein. I also understand that financial support may be withdrawn in the event of unsatisfactory behaviour. I acknowledge that in registering as a student of the University I have undertaken to obey the regulations contained in its Charter, Statues, Ordinances and Regulations.

I also agree to use safety belts in minibuses where fitted.

## Specific advice for this field course

- You must have read the material on the preceding two pages.
- Carry a small first aid kit with you in the field.
- Students must notify staff if they have any health problems that may affect their safety and/or that of others.
- You may not to smoke in the bus or in the field (except during breaks, such as lunch-time).

**Qualitative Assessment of Risk**

- Student misbehaviour is one of the most significant sources of risk in the field. Act sensibly, and take care not to put yourself or people near you at risk. Do not fool around, and do not throw stones.
- The weather in Scotland in April can be hot and sunny but is more commonly, cool, windy and wet. If it is cold there is a real risk of exposure, particularly on high ground. Even if the weather does not seem to be very cold, windy or wet in the morning, be prepared by packing spare warm and waterproof (preferably brightly coloured) clothes. Even if you doubt that it will be needed, pack sun-screen to guard against sun-burn.
- The mapping area involves walking across featureless moorland or hilly terrain in which one can easily get lost. Take care not to get separated from the rest of the group. Guard against this by ensuring that you have the mobile phone numbers of fellow students but be alert to the possibility that mobile phone reception may be poor.
- Shore / river platforms and boulders are often extremely slippery when wet, particularly when covered in green algae and seaweed. A bad fall can result in a broken arm, leg or collar-bone or a fractured skull.
- Some shore exposures have irregular surfaces on which it is easy to twist an ankle.
- Grassy / mossy hillsides can be very slippery in the rain. Even in dry weather, the ground can be saturated in water, and hence treacherous under-foot, especially if there has been a lot of rain in recent weeks.
- The tidal range in Loch Leven is small, and there are no localities on the itinerary where one can get cut off by the tide. Even so, you should always be aware of tidal conditions when working along the coast.
- Road cuttings/crags present a risk of falling rocks. Be alert to this.
- The main road through the mapping area carries fast-moving traffic. You must always walk along the pavement or verge in single file.

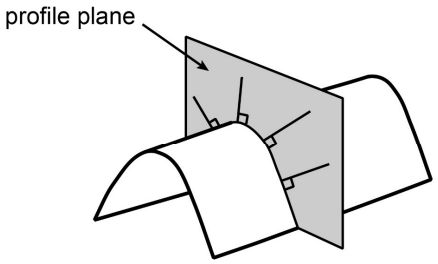
**Site-Specific Hazards**

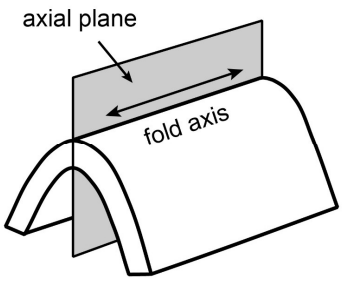
A. *Shore at Loch Leven:* Rock surfaces are slippery when wet and extra care must be taken at low tide on seaweed-covered areas. Take care not to trip or twist your ankle on irregular rock surfaces. Be aware of the tide at all times.
B. *Upland areas at Kinlochleven:* Be prepared for bad weather at all times. Take care to guard against getting exposure. Take care not to get separated from the rest of the group or your mapping partner(s).

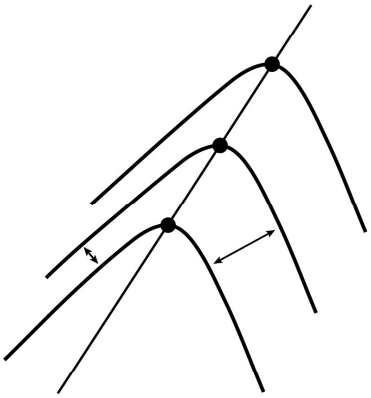
**Accommodation**

- All appliances must be switched off before leaving each day.
- All appliances and utensils must only be used for their designated purposes.
- Accommodation must be kept clean – the owners will inspect the accommodation during the trip and on the morning of departure. Should there be any damage or extra cleaning necessary, the costs will be charged to those involved.
- The use of intoxicating drugs, including alcohol and tobacco, must not interfere with the well-being of other students. Personal consequences and liability arising from intoxication will be the sole responsibility of those who indulge.

# Some helpful figures (fold terms, cross-bedding)



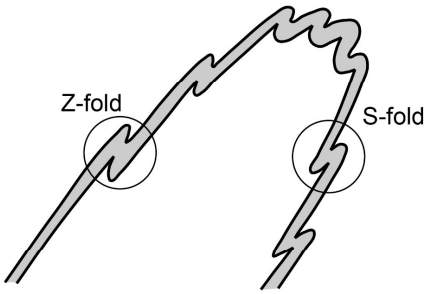



axial plane
(axial trace)

***axial plane*:** the plane that passes through the points of maximum curvature in a nested set of folded surfaces, and which also contains the fold axis

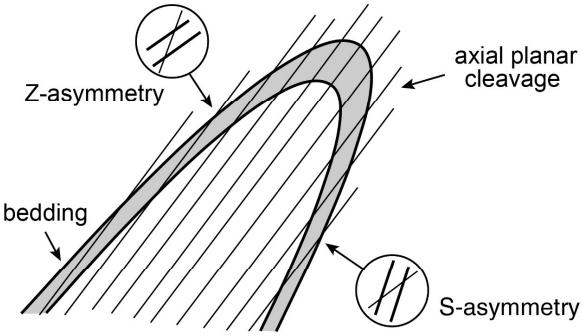
***axial trace*:** the intersection of the axial plane with any surface of interest, *e.g.*, ground surface, cross-section plane

**Change in minor fold asymmetry**

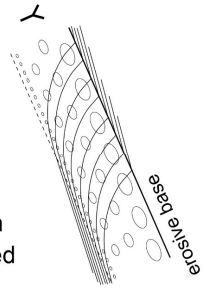



vergence = direction towards antiform

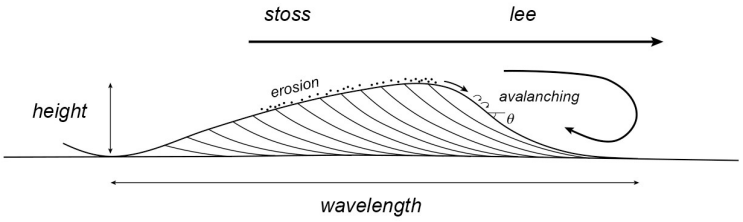
**Change in cleavage / bedding asymmetry**



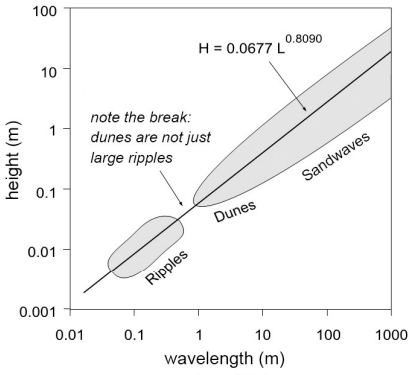

along bedding - along cleavage - along bedding
(making acute angles)



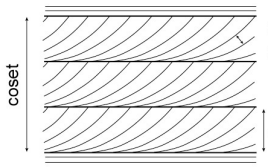

cleavage refraction
through a graded bed




$\theta$ = angle of repose
≈ 25°-30° subaqueous

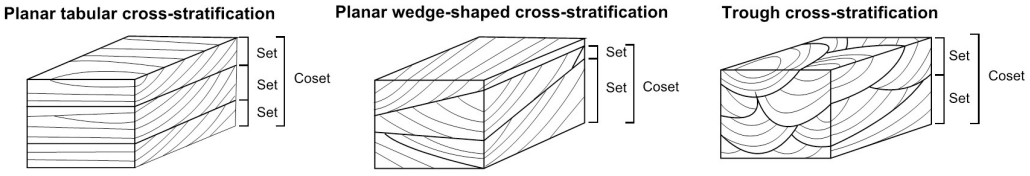


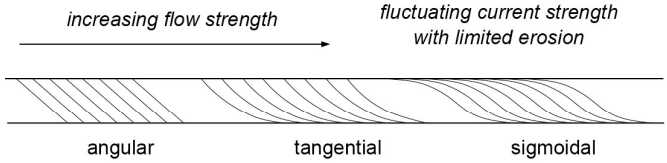

| | | subaqueous | | |
|---|---|---|---|---|
| | | wave ripple | dune | sandwave |
| wavelength / height | = | 0.01 - 0.6 m / 0.003 - 0.25 m | 1 - 10 m / 0.1 - 1 m | 5 - 500+ m / 0.5 - 5+ m |




cross-bed if > 6 mm
cross lamination if < 6 mm

## *set geometry*



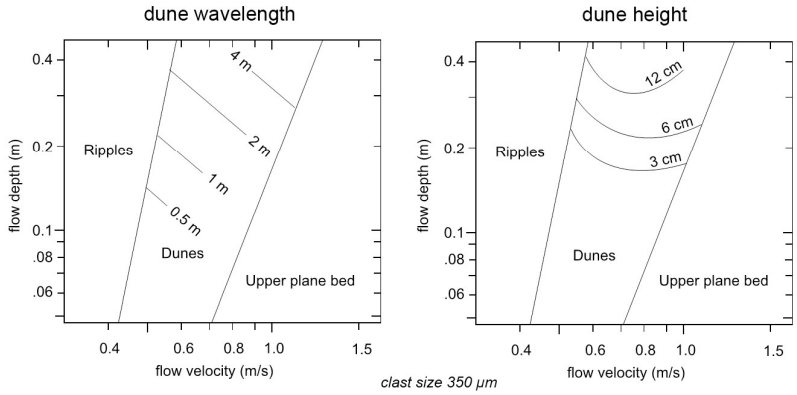

***bedform size controlled by:***

*clast size* ↕ ripples

*flow velocity* ↕ dunes

*flow depth*

*For subaqueous dunes:*
*flow depth ~ 6 × dune height*
*dune height ~ 3 × set thickness*



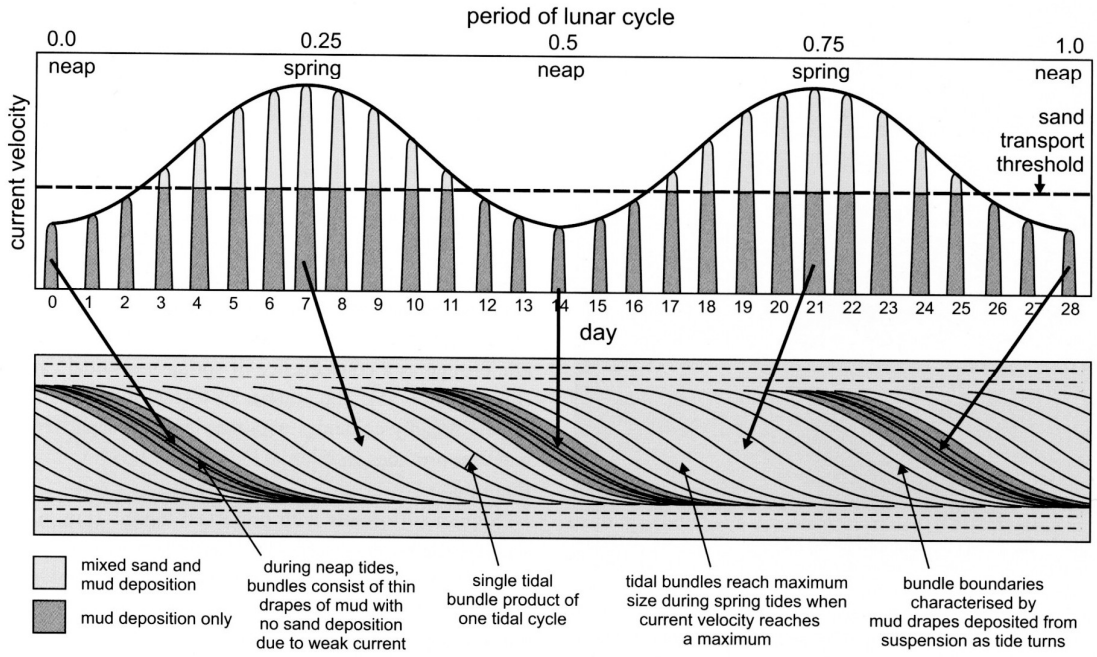

*clast size 350 μm*

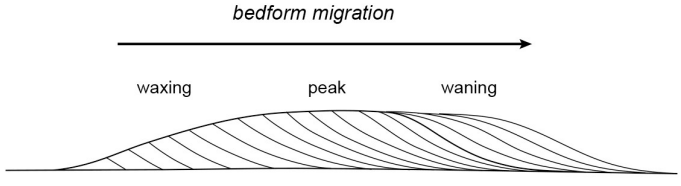
*Tidal bedforms*

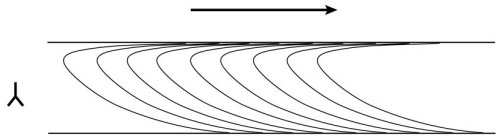
*herringbone cross-stratification*

*tidal bundles*

*mud drapes*



"overturned" cross-bedding

(formed before lithification)